

Family I. **Pasteurellaceae** Pohl 1981b, 382^{VP} (Effective publication: Pohl 1979, 81)

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Pas.teu.rel.la'ce.ae. M.L. fem. n. *Pasteurella* type genus of the family; *-aceae* suffix to denote a family; M.L. fem. pl. n. *Pasteurellaceae* the *Pasteurella* family.

Cells are straight, rigid, **coccoid to rod-shaped**, usually **0.2–0.4 × 0.4–2.0 μm**, nonsporeforming, **Gram negative**, and **nonmotile**. Pleomorphism with cell swelling and formation of filaments can be seen. Demethylmenaquinones, menaquinones, and ubiquinones may or may not be produced. Organisms are **aerobic, with varying degrees of microaerophilia, or facultatively anaerobic**. Both **respiratory and fermentative forms of metabolism** can be found among these organisms, which are **chemoorganotrophic**. Optimal growth temperature is 37°C. Acid is produced through fermentation of D-glucose, other carbohydrates, sugar alcohols, or glycosides. Conventional fermentation test media may not detect accumulation of acid fermentation products from some of the most fastidious species. Usually anaerogenic, but gas-producing species do occur. Fumarate is usually used as a terminal electron acceptor in demethylmenaquinone-mediated anaerobic respiration. Characteristically, **oxidase, catalase, and alkaline phosphatase reactions are positive**, but negative reactions can be detected in some species. **Nitrates are reduced to nitrites**. For primary isolation, complex media supplemented with yeast extract, serum, or whole blood lysate are required. Furthermore, organic nitrogen sources, amino acids, B vitamins, nicotinamide adenine nucleotides, hematin, or protoporphyrin may be needed. Members of the family are **obligate parasites in vertebrates (primarily mammals and birds)**. They inhabit mucous membranes of the alimentary, genital, or respiratory tracts and are often pathogenic in humans, other mammals, birds, and reptiles. Species are generally susceptible to benzylpenicillin and other β-lactam antibiotics, inhibitors of 70S ribosomal protein biosynthesis, as well as sulfonamides, trimethoprim, erythromycin, and colistin. The genome size is 1.2–2.2 × 10⁹ Da.

The mol% G + C of the DNA is: 37–45.

Type genus: ***Pasteurella*** Trevisan 1887, 94 (Nom. Cons. Opin. 13, Jud. Comm. 1954b, 153.)

Circumscription, position and rank Sneath and Johnson (1973), who made numerical phenetic analyses on a restricted number of species, were probably the first authors to indicate that the genera *Actinobacillus*, *Haemophilus*, and *Pasteurella* could form a family. Mannheim et al. (1980) suggested from genetic studies that the genera *Actinobacillus*, *Haemophilus*, and *Pasteurella* should rank as a family, although some species were misclassified and should be removed. In 1979, Pohl proposed the name *Pasteurellaceae* for the family and the maintenance of the three genera as distinct entities (Pohl, 1979). From the results of DNA reassociation experiments, Pohl (1981a) suggested that some rearrangement of the species in the various genera was indicated. Single-linkage clustering of the DNA hybridization data revealed complex intra- and intergeneric relations at or above the 30% DNA–DNA reassociation level. Furthermore, phylogenetic studies based on 16S rDNA sequencing concluded that taxonomic division of the family into phylogenetically and phenotypically coherent genera was problematic (Dewhirst et al., 1992, 1993, 1995; Paster et al., 1995). Nevertheless, such studies contributed to clarification of the taxonomic position of several taxa within the *Pasteurellaceae* (Holmes, 1998).

The genus *Lonepinella* was effectively published (Osawa et al., 1995b) and validated (Osawa et al., 1996) as a fourth genus of the family. This genus was described to encompass isolates from the feces of koala that degrade a tannin–protein complex. So far, the genus consists of only one species, *Lonepinella koalarum*. 16S rDNA sequence analyses have clearly demonstrated that the genus *Lonepinella* clusters within the family *Pasteurellaceae*.

The genus *Mannheimia*, previously the *P. haemolytica* complex, has been validly published as a fifth genus (Angen et al., 1999a). Seven species were identified, five of which were formally named.

The genus *Phocoenobacter* has been validly published as a sixth genus in the family *Pasteurellaceae* and contains the single species *Phocoenobacter uteri* (Foster et al., 2000).

rRNA–DNA hybridization studies showed that the family *Pasteurellaceae* forms one cluster that is linked with both branches of the genus *Alteromonas*, the family *Enterobacteriaceae*, the family *Vibrionaceae*, and the family *Aeromonadaceae* (De Ley et al., 1990). All these clusters constitute superfamily I *sensu* De Ley which together with superfamily II corresponds to the *Gammaproteobacteria*.

16S rDNA sequencing of 54 strains (Dewhirst et al., 1992) and subsequently of 70 strains (Dewhirst et al., 1993) confirmed that the family *Pasteurellaceae* is located in the *Gammaproteobacteria* and demonstrated that the family *Pasteurellaceae* forms a compact phylogenetic group that is distinct from neighboring families. The sequence signatures reported previously for the family (Dewhirst et al., 1992) are still useful, particularly 667 A, 739 U, 831 G, and 855 C. However, if a full sequence is available, it is much better to place the sequence in a phylogenetic tree with appropriate outgroups than to examine only a few signature bases. The DNA probe 5'-ACCAACTACCTAATCCCACTTGGG-3' hybridizes with DNA (at ~80%) from most of the members of the family, but not with members of other taxa, so the specificity is 100% (Dewhirst et al., 1992).

The phenotypic traits that distinguish the *Pasteurellaceae* from other families and genera are presented in Table BXII.γ.286. The dearth of clearly distinguishing phenotypic traits is a reason that phylogenetic placement at the genus and higher level must rely heavily on molecular sequence data. Nonmotile and nutritionally fastidious strains of *Enterobacteriaceae*, *Bacillus*, or aerotolerant clostridia, as well as nonfermentative organisms, can be mistaken for species of *Pasteurellaceae* (Mannheim, 1984).

The cellular fatty acid content of *Pasteurellaceae* is uniform, with only minor variations. Separation from the *Neisseriaceae* and from *Moraxella* is possible (Mutters et al., 1993). In addition, the distribution of phospholipids is uniform within *Pasteurellaceae*. Cellular carbohydrates show a common pattern within all members of the family, but characteristic profiles discriminate between groups, often to the species level (Mutters et al., 1993).

Respiratory quinones had been previously considered useful in defining the family. When selected members of the family *Pasteurellaceae* were investigated for their lipoquinone contents using high-performance liquid chromatography (Kroppenstedt and Mannheim, 1989), menaquinones (MK-7 or MK-8 or both), demethylmenaquinones (DMK-7 or DMK-8 or both) and ubi-

quinones (Q-7 or Q-8 or both) were detected in varying proportions. Similar results were obtained in another study concerning strains of *Pasteurella* that were grown under strictly aerobic conditions (Kainz et al., 2000). The view that *Pasteurellaceae* can be distinguished from the *Enterobacteriaceae* and other fermenting Gram-negative bacteria by lack of menaquinones can therefore not be supported. Although the situation is more complex than had previously been recognized, the distribution patterns of lipoquinone structural types and their isoprenologue analogs remain a reliable chemotaxonomic tool for these bacteria.

Genera and Taxonomic Problems There are six genera in the family *Pasteurellaceae*: *Pasteurella*, *Actinobacillus*, *Haemophilus*, *Lonepinella*, *Mannheimia* and *Phococobacter*. Phenotypic traits that are useful for distinguishing the genera are listed in Table BXII.γ.287. As can be seen by the distressingly large number of traits that are variable, there is little phenotypic consistency

within genera, even when the genera are restricted to monophyletic groups based on DNA–DNA hybridization or 16S rDNA sequence analysis. The phylogeny and taxonomy of the family are clearly problematic. Several approaches to elucidating taxonomy have been used, including numerical taxonomy (Sneath and Johnson, 1973; Olsen, 1993), DNA–DNA hybridization (Mannheim and Pohl, 1980; Mannheim et al., 1980; Mannheim, 1983; Escande et al., 1984), rRNA–DNA hybridization (De Ley et al., 1990), genetic transformation (Albritton et al., 1986), chemotaxonomy (Olsen, 1993), and content of cellular fatty acids (Mutters et al., 1993), quinones (Mannheim, 1981; Kroppenstedt and Mannheim, 1989); Engelhard et al., 1991, 1992; Mutters et al., 1993; Kainz et al., 2000), polyamines (Busse et al., 1997), and cellular carbohydrates (Mutters et al., 1993). While these methods have clearly advanced our understanding of the taxonomy of the *Pasteurellaceae* (Bisgaard, 1995), it is now recognized that 16S sequencing is the most useful method for determining phylogeny and unraveling taxonomy at the genus and higher levels

TABLE BXII.γ.286. Differentiation of *Pasteurellaceae* from other families^a

Characteristic	<i>Pasteurellaceae</i>	<i>Aeromonadaceae</i>	<i>Alteromonadaceae</i>	<i>Enterobacteriaceae</i>	<i>Vibrionaceae</i>
Motility	—	+	+	D	[+]
Na ⁺ required or stimulates growth	—	—	+	—	+
Parasitic in vertebrates	+	+	—	D	[—]
V-factor requirement	D	—	—	—	—
Organic nitrogen sources required	+	—	—	[—]	[—]
Mol% G + C of DNA	38–47	57–63	38–50	52–60	38–51

^aFor symbols see standard definitions; also [—], more than 70% negative; [+], more than 70% positive.

TABLE BXII.γ.287. Differential characteristics for members of the family *Pasteurellaceae*^a

Characteristics	<i>Haemophilus</i> ^b	<i>Actinobacillus</i> ^b	<i>Lonepinella</i> ^c	<i>Mannheimia</i> ^d	<i>Pasteurella</i> ^b	<i>Phococobacter</i> ^e
Sticky colonies	—	d	—	—	—	—
Hemolysis on SBA	d	d	—	d	—	—
V-factor dependency	+	— ^f	—	—	—	—
X-factor dependency	+	—	—	—	—	—
Urease	d	+	—	—	— ^g	—
Ornithine decarboxylase	— ^h	—	—	d	d	—
Indole	d	—	—	d	+	—
α-Fucosidase	—	d	nd	d	—	—
Acid production from:						
D-Glucose	+	+	+	+	+	+
D-Mannitol	—	+ ⁱ	—	+	d	—
D-Mannose	—	d	nd	—	+	—
Melibiose	—	d	+	d	—	—
<i>l</i> -Inositol	—	d	—	d	—	—
D-Sorbitol	—	d	—	d	d	—
Trehalose	—	d	nd	—	d	—

^aFor symbols see standard definitions; nd, not determined.

^b*Sensu stricto* species only as shown in Fig. BXII.γ.215.

^cOsawa (1992).

^dAngen et al. (1999a).

^eFoster et al. (2000).

^fBio-type 1 strains of *A. pleuropneumoniae*, *A. indolicus*, *A. minor*, and *A. porcineus* are positive.

^g*P. dagmatis* is positive.

^hSome strains of *H. influenzae* are positive.

ⁱ*A. suis* is negative.

(Olsen and Woese, 1993), while DNA–DNA hybridization is the most useful at the species level.

Organisms of the family *Pasteurellaceae* infect most vertebrate species. In almost every vertebrate species examined carefully, multiple, often unique, *Pasteurellaceae* species have been recovered. Only a subset of these taxa has been formally named, but a large number of strains have found their way into culture collections, and many have been characterized and placed in tentative taxonomic groups. With the advent of 16S rDNA sequence analysis, it has been possible to rapidly examine strains representing these taxa and place them in an overall *Pasteurellaceae* phylogenetic tree. 16S rRNA-based phylogenies have previously been presented by a number of investigators (Dewhirst et al., 1992, 1993) but substantial new data exist, which are included in Fig. BXII.γ.215. At this time, over 230 16S rDNA sequences have been deposited in GenBank, and 114 nonredundant sequences were selected for inclusion in this tree. The neighbor-joining method (Saitou and Nei, 1987) was used for tree construction. Bootstrap values greater than 40% are shown. It can be seen from the tree that species named *Haemophilus*, *Actinobacillus*, and *Pasteurella* are intermixed throughout the tree. If the family is to be divided into phenotypically consistent monophyletic groups, then there will need to be about 20 genera in this family. The genus *Mannheimia* is probably the most distinct and well-defined cluster in the family. The delineation of almost every other cluster is somewhat subjective and in almost every case will require additional investigation before the cluster can be proposed as a genus. Because the taxonomy of the *Pasteurellaceae* is in the process of major revision, we felt it useful to present a phylogeny representing as much of the diversity of the family as possible, regardless of whether the taxa have been formally named.

Named genera

Actinobacillus sensu stricto. The genus *Actinobacillus sensu stricto* should be comprised of the following named species: *A. lignieresii*, *A. pleuropneumoniae*, *A. ureae*, *A. hominis*, *A. equuli*, and *A. suis*. *A. arthritidis* (Bisgaard Taxon 9), Bisgaard Taxon 8, and taxa represented by strains Kunstyr 570 (MCCM 00149), Bisgaard F64 (CCUG 28015), and SSI P585 (CCUG 19799) should be included. This is in general agreement with the proposal of Escande et al. (1984) and Mutters et al. (1989) that the genus *Actinobacillus* should be limited to *A. lignieresii*, *A. equuli*, *A. capsulatus*, *A. suis*, *A. hominis*, *A. pleuropneumoniae*, *A. arthritidis*, and Bisgaard's taxa 5 and 11. 16S rDNA sequencing data suggest addition of *H. parahaemolyticus* and *A. hominis* (Dewhirst et al., 1992, 1993). The type strain of *A. capsulatus* does not cluster with true actinobacilli (Dewhirst et al., 1992, 1993). See discussion under *Capsulatus* Cluster, below. Bisgaard Taxon 5 falls elsewhere in the tree.

Mannheimia. This genus was recently created from strains that had previously been classified as *Pasteurella haemolytica* (Angen et al., 1999a). Five species have been named: *Mannheimia haemolytica*, *M. glucosida*, *M. granulomatis*, *M. ruminantis*, and *M. varigena*. There are strains representing at least two additional *Mannheimia* species.

Lonepinella. A species isolated from koalas has been given the name *Lonepinella koalarum*.

Pasteurella sensu stricto. The genus *Pasteurella sensu stricto* should be limited to the species *P. multocida*, *P. dagmatis*, *P. canis*, and *P. stomatis*. The three *P. multocida* subspecies are nearly identical by 16S rDNA sequences. Avian species that were previously included by DNA–DNA hybridization studies appear to fall into a separate genus level cluster (see Avian Cluster, below). Bisgaard Taxon 13 strain CCUG 16497 appears to be *P. multocida*. A strain identified as *P. pneumotropica* (NCTC 10827) is not related to the type strain (Rodent Cluster) and appears to represent a novel species in the genus. *Pasteurella* species B branches substantially deeper than do other members of the genus but may warrant inclusion in the genus.

The quinone content of strains of the genus *Pasteurella*, including *P. avium*, *P. canis*, *P. dagmatis*, *P. gallinarum*, *P. langaa*, *P. multocida*, *P. stomatis*, *P. volantium*, "*Pasteurella leonis*", *Pasteurella* species A, and *Pasteurella* species B, has been reinvestigated (Kainz et al., 2000), with cells grown in liquid PPLO broth under strictly aerobic conditions. Based on the quinone composition, the species could be grouped into two clusters. Cluster I, consisting of the species *P. avium*, *P. gallinarum*, *P. volantium*, and *Pasteurella* species A, was characterized by the two predominant compounds ubiquinone Q-7 and Q-8. These species correspond to the Avian Cluster in Fig. BXII.γ.215. Cluster II, consisting of the species *P. canis*, *P. dagmatis*, *P. langaa*, *P. multocida*, *P. stomatis*, and *Pasteurella* species B, contained only ubiquinone Q-8 as the major compound and corresponds to *Pasteurella sensu stricto* of Fig. BXII.γ.215 (Mutters et al., 1993). These two clusters may be also distinguished by the presence of *sym*-norspermidine in *Pasteurella sensu stricto* (Busse et al., 1997), whereas this triamine has not been detected in representatives of the Avian Cluster in Fig. BXII.γ.215.

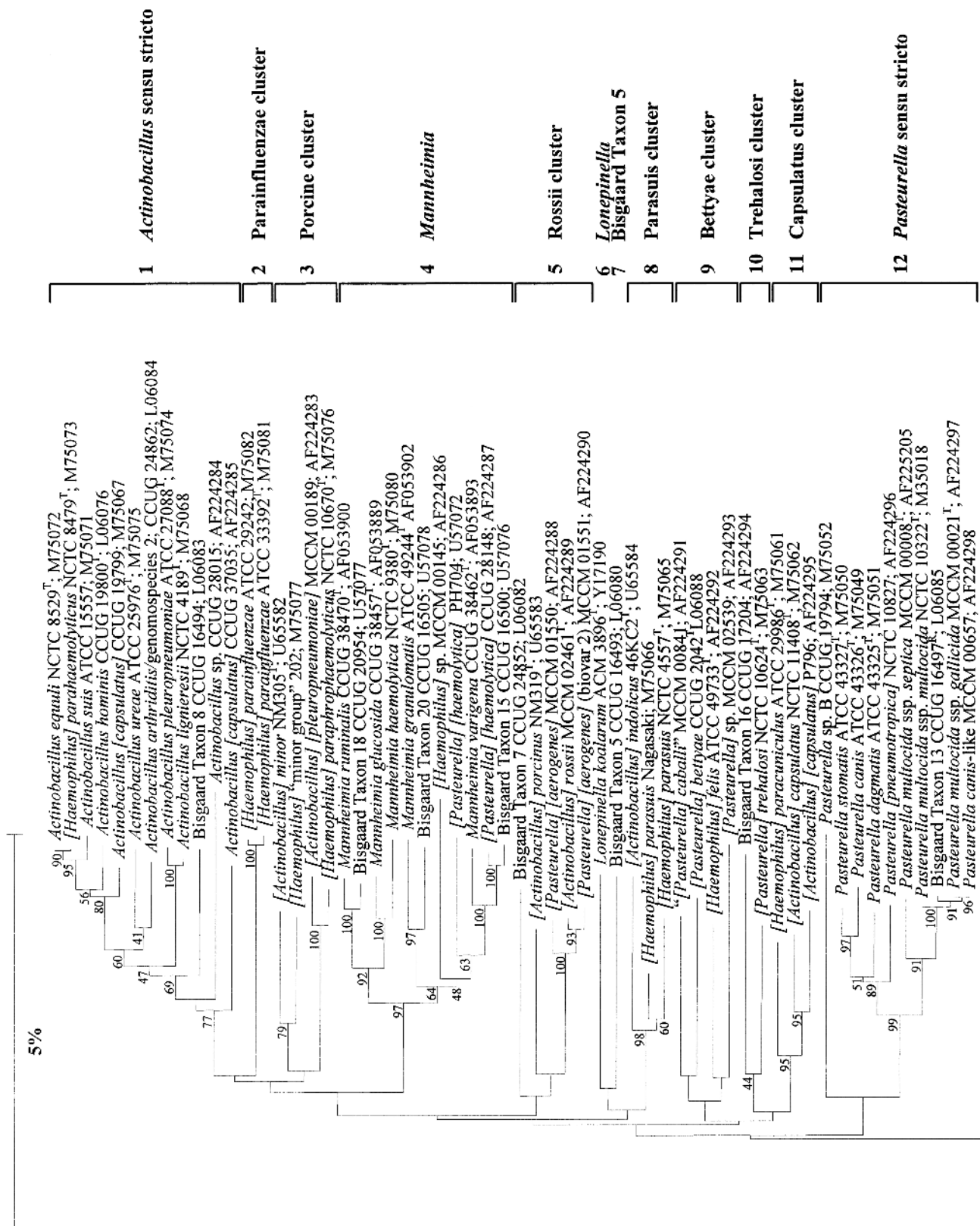
Haemophilus sensu stricto. In the previous edition of *Bergey's Manual*, a primary characteristic of the genus *Haemophilus* was X or V-factor requirement. It is now recognized that factor-requiring species are spread throughout the family. Based primarily on the DNA–DNA hybridization data of Burbach (1987), Mutters et al. (1989) proposed restricting the genus *Haemophilus sensu stricto* to include *H. influenzae*, *H. aegyptius*, *H. haemolyticus*, "*H. intermedius*", and *H. parainfluenzae*. The 16S rRNA tree suggests exclusion of *H. parainfluenzae*. As shown in Fig. BXII.γ.215, strains representing serotype e and f of *H. influenzae* appear distinct from the other serotypes of *H. influenzae*. The tree shows the positions of "*H. intermedius* subsp. *intermedius*", "*H. intermedius* subsp. *gazogenes*", and "*H. quentini*".

Phocoenobacter. A bacterium isolated from a porpoise has been given the name *Phocoenobacter uteri*.

Clusters that represent potential additional genera

Parainfluenzae Cluster. *H. parainfluenzae* does not belong in the genus *Haemophilus*, but rather forms its own cluster. The species is known to be heterogeneous and may represent multiple species.

Porcine Cluster. This cluster contains four taxa in two subclusters: *Actinobacillus minor*, *Haemophilus* "minor group" 202, *A. pleuropneumoniae*, and *H. paraphrohaemolyticus*.



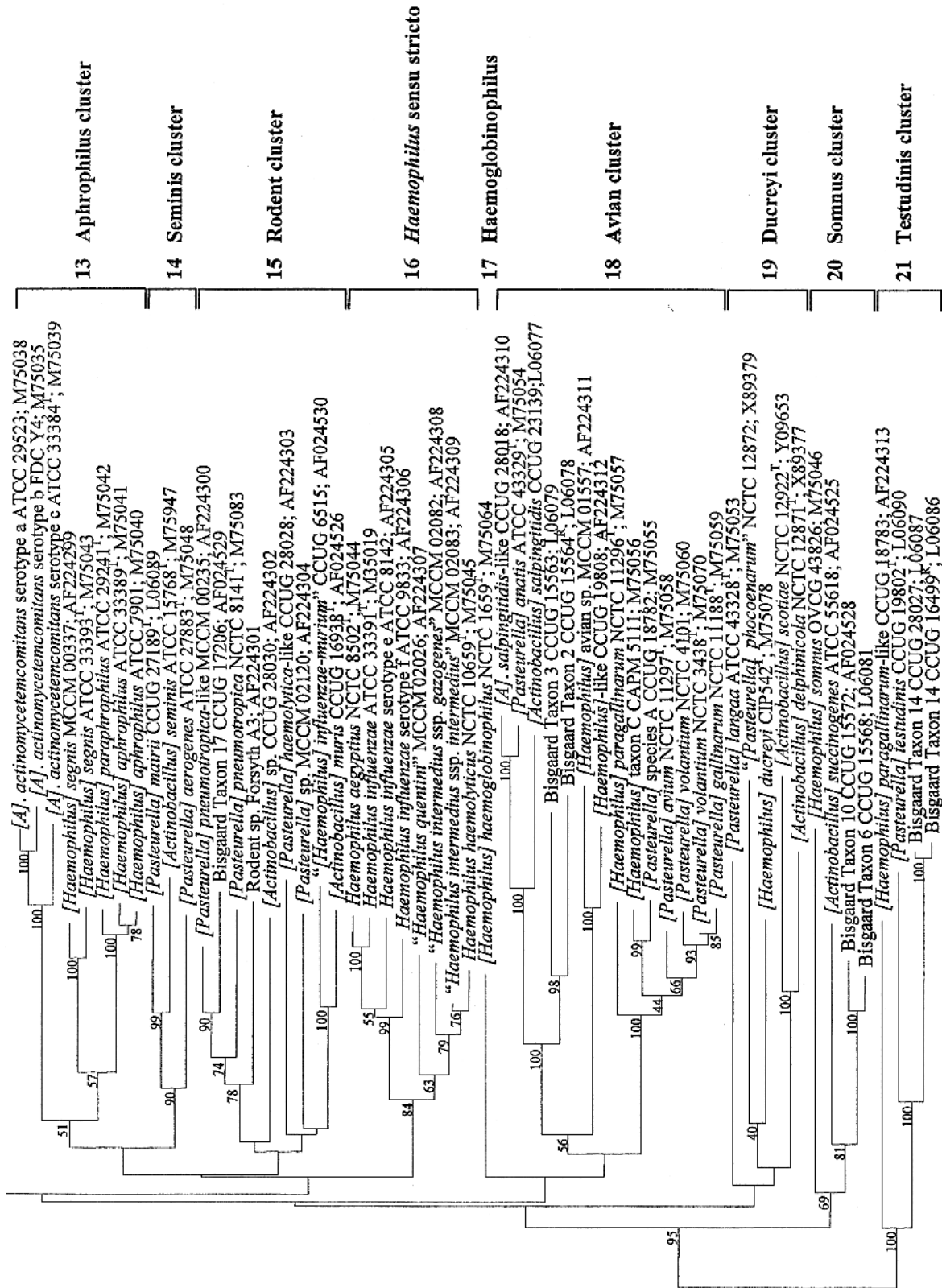


FIGURE BXII.7.215. Phylogeny of the *Pasteurellaceae*. The family *Pasteurellaceae* forms a coherent phylogenetic subgroup within the *Gammaproteobacteria* that is divided into 21 phylogenetic clusters. Five of the clusters are represented by the named genera *Actinobacillus sensu stricto*, *Haemophilus sensu stricto*, *Pasteurella sensu stricto*, *Mannheimia*, and *Lonepinella*. "*Sensu stricto*" is defined herein as a distinct phylogenetic clade. The remaining clusters contain generically misclassified species of *Actinobacillus*, *Haemophilus*, and *Pasteurella*, as well as unnamed species of *Pasteurellaceae* from a variety of culture collections. A bracket around the genus or specific epithet denotes phylogenetic misclassification. The strain number and GenBank accession number of the 16S rRNA sequence are shown. Abbreviations for culture collections are as follows: ATCC, American Type Culture Collection, Manassas, VA; NCTC, National Collection of Type Cultures, London, England; MCCM, Medical Culture Collection of Microorganisms, Marburg, Germany; CCUG, Culture Collection, University of Göteborg, Göteborg, Sweden; CAPM, Collection of Animal Pathogenic Microorganisms, Brno, Czech Republic; Forsyth, The Forsyth Institute, Boston, MA; CIP, Collection de l'Institut Pasteur, Paris, France; ACM, Australian Collection of Microorganisms, Brisbane, Australia. Bar = 5% difference in nucleotide sequences. 200 bootstrap confidence levels (shown as percentages at the nodes) were determined.

Rossii Cluster. This cluster includes *Actinobacillus rossii*, *A. porcini*, and strains that are considered part of the *Pasteurella aerogenes* complex, but do not fall with the type strain, which is in the Seminis Cluster. The Rossii Cluster may also include the more deeply branching Bisgaard Taxon 7, but this taxon is somewhat unstable in its branching position within the tree. See discussion for Bisgaard Taxon 5.

Bisgaard Taxon 5. The phylogenetic position of this taxon is problematic in that its position in the tree is unstable. Changes of taxa included in the analysis cause Bisgaard Taxon 5 to jump from place to place in the tree. Molecular data from other genes may be required to define its true position.

Parasuis Cluster. This cluster includes *H. parasuis*, *A. indolicus*, and a third taxon represented by *H. parasuis* strain Nagasaki.

Bettyae Cluster. This cluster includes *P. bettyae*, "*P. caballii*", *H. felis*, and guinea pig conjunctivitis strain Stenzel T 391 (HIM 1028-4).

Trehalosi Cluster. *P. trehalosi* forms a cluster with Bisgaard Taxon 16. It may be possible to join the Trehalosi and Capsulatus Clusters.

Capsulatus Cluster. This cluster includes *A. capsulatus*, *H. paracuniculus*, and a third taxon represented by *A. capsulatus* strain SSI P796. The taxonomic position of *A. capsulatus* has been controversial because DNA–DNA hybridization placed it with *Actinobacillus sensu stricto* (Escande et al., 1984; Mutters et al., 1989). However, sequences for five isolates (SSI P243^T, P244, P572, P662, and P796) all fall in the Capsulatus cluster. The horse isolate, SSI P585 (CCUG 19799), falls in the middle of the *Actinobacillus sensu stricto* cluster, and if DNA from this strain was inadvertently used in place of that of the type strain (SSI P 243), it could explain the older DNA–DNA hybridization results. A hamster "*A. capsulatus*" strain, MCCM 149, branches with a duck isolate, Bisgaard F64 (CCUG 28015), to form the deepest branch of the *Actinobacillus sensu stricto* cluster. It would appear that *A. capsulatus*-like strains from species other than rabbit represent multiple unique species.

Aphrophilus Cluster. This cluster includes *H. segnis*, *H. aphrophilus*, *H. paraphrophilus*, and an *H. segnis*-like taxon represented by strain MCCM 00337. Based on rRNA–DNA hybridizations, De Ley et al. (1990) suggested that the Aphrophilus cluster be split off from the *Haemophilus sensu stricto*. We would suggest including *A. actinomycetemcomitans* in this cluster to be consistent with the DNA–DNA hybridization data of Potts et al. (1985).

Seminis Cluster. This cluster includes *A. seminis*, *P. mairii*, and *P. aerogenes*. Other strains of the *P. aerogenes* complex, such as the Rossii Cluster, fall elsewhere in the tree.

Rodent Cluster. This genus level cluster includes nine species, all isolated from rodents or guinea pigs. Named species include *P. pneumotropica*, *A. muris*, and "*H. influenzae-murium*". Also included are Bisgaard Taxon 17 (rat), strain Kunstyr 507 (MCCM 02120 [hamster]), Mannheim Michael A (CCUG 28028 [guinea pig]), Kunstyr 246 (CCUG 28030 [hamster]), Mannheim A/5a (MCCM 00235 [rat]), and Forsyth A3

(mouse). This large cluster clearly warrants further investigation for promotion to genus status.

Haemoglobinophilus Cluster. This cluster contains the single species *H. haemoglobinophilus*.

Avian Cluster. *P. volantium*, *P. gallinarum*, and *P. avium* were placed in the *Pasteurella sensu stricto* based on DNA–DNA hybridization (Pohl, 1981a), but by 16S rDNA sequence analysis, they fall into a large cluster of avian species quite distant from the *Pasteurella sensu stricto*. This avian cluster also includes "*Actinobacillus salpingitidis*", *P. anatis*, *H. paragallinarum*, Bisgaard Taxon 2, Bisgaard Taxon 3, *Haemophilus* taxon C, *Pasteurella* species A, and strains representing additional species such as Wuthe DS 1801 (MCCM 01557), "*A. salpingitidis*" CCUG 28018, and *Haemophilus* sp. CCUG 19808. Members of the subcluster consisting of *P. avium*, *P. gallinarum*, *P. volantium*, *Pasteurella* species A, and *H. paragallinarum* contain a polyamine pattern with the predominant compounds 1,3-diaminopropane and spermidine. Strains of Bisgaard Taxon 2 and Taxon 3 are characterized by polyamine patterns with the major compounds putrescine and spermidine (Taxon 2) and putrescine, cadaverine, and spermidine (Taxon 3). The third subcluster, consisting of two strains of "*A. salpingitidis*" and *P. anatis*, contain either the predominant compounds 1,3-diaminopropane and spermidine or additionally putrescine (Busse et al., 1997).

Ducreyi Cluster. This cluster includes *H. ducreyi*, *P. langaa*, and the three cetacean species *A. delphinicola*, *A. scotiae*, and "*P. phocoenarum*" (now *Phocoenobacter uteri*).

Somnus Cluster. This cluster includes "*Haemophilus somnus*", *A. succinogenes*, Bisgaard Taxon 6, and Bisgaard Taxon 10.

Testudinis Cluster. This cluster is the most deeply branching of the family *Pasteurellaceae*. It contains *P. testudinis*, Bisgaard Taxon 14, and the *H. paragallinarum*-like strain CCUG 18783. This cluster is distinct from the rest of the *Pasteurellaceae* in having the common *Proteobacteria* signature 665 A rather than the *Pasteurellaceae* signature 665 G.

The resolution of 16S rDNA sequence analysis may be inadequate to delineate the deeper phylogenetic branching in the *Pasteurellaceae*. Multi-gene sequence analysis has been used effectively to type species (Maiden et al., 1998). Similar methods can also be used to examine species- and genus-level relationships. With the complete genome of *H. influenzae* as well as extensive information for the unfinished genomes of *A. actinomycetemcomitans* and *P. multocida* available, it is reasonably easy to design primers to amplify and sequence essentially any chosen gene. Additional phylogenetic studies using genes such as *recA*, *dnaf*, or *fur* may provide sufficient information to resolve the deeper branches in the family *Pasteurellaceae*. Phylogenetic analysis of genes other than 16S rRNA would also provide a valuable check on whether the 16S tree represents a consensus species tree, or only an evolutionary tree for one specific gene. Examination of local gene order may also be useful for defining genera. While molecular data will no doubt be the primary basis for resolving taxonomic issues, a polyphasic approach to describing the genera is essential. Promising phenotypic traits that could aid taxonomic analysis are polyamine patterns, quinone composition, and carbohydrate profiles.

Genus I. Pasteurella Trevisan 1887, 94^{AL} Nom. Cons. Opin. 13, Jud. Comm. 1954b, 153

REINIER MUTTERS, HENRIK CHRISTENSEN AND MAGNE BISGAARD

Pas.teu.rel la. M.L. dim. fem. n. *Pasteurella* named after Louis Pasteur.

Coccobacilli or rods, generally, $0.3\text{--}1.0 \times 1.0\text{--}2.0 \mu\text{m}$. Depending on the growth stage, cells occur singly, in pairs, or less frequently in short chains. Sometimes threads or filaments are formed resulting in marked pleomorphism. Pleomorphism occurs mainly in old cultures. **Gram negative**, although **bipolar staining often can be observed**. In tissues, *P. multocida* often shows bipolar staining with Giemsa or Wright's stain. Not acid fast. **Endospores are not formed**. **Nonmotile**. **Aerobic to microaerophilic or facultatively anaerobic**. Chemoorganotrophic with both oxidative and fermentative types of metabolism. Electron transport system is cytochrome-based with oxygen, nitrate, or fumarate as the terminal electron acceptor. **Nitrate reductase is produced**. **Oxidase positive, alkaline phosphatase positive**, and almost always **catalase positive**. **Most species are V-factor and X-factor independent, but V-factor-requiring strains do occur**. Even after specific growth factors have been provided, **complex media may be required to obtain better growth**. Optimum growth temperature $35\text{--}37^\circ\text{C}$. Arginine dihydrolase and lysine decarboxylase negative. Gelatin is not liquefied within 48 h. No growth occurs in Simmons citrate medium. Acid is produced from D-glucose, D-galactose, D-fructose, D-mannose, and sucrose. Acid is not produced from L-sorbose, L-rhamnose, *m*-inositol, and adonitol. Parasitic in vertebrates, particularly mammals and birds. Genome molecular weights range from 1.4×10^9 to 1.9×10^9 .

The mol% G + C of the DNA is: 37.7–45.9.

Type species: *Pasteurella multocida* (Lehmann and Neumann 1899) Rosenbusch and Merchant 1939, 85 (*Bacterium multocidum multocidum* (sic) Lehmann and Neumann 1899, 196; *Pasteurella gallicida* (Burrill 1883) Buchanan 1925, 414.)

FURTHER DESCRIPTIVE INFORMATION

Phylogeny Based on 16S rDNA sequence comparison of selected strains, the genus was separated into two phylogenetic groups (Dewhirst et al., 1992, 1993). The core group around the type species included the other species from mammalian hosts: *P. canis*, *P. stomatis*, *P. dagmatis*, and *Pasteurella* sp. B. The avian species *P. avium*, *P. gallinarum*, *P. volantium*, and *Pasteurella* sp. A form a separate cluster together with the type strain of *Haemophilus paragallinarum*. Based on 16S rRNA sequences *P. langaa* and *P. anatis* were not phylogenetically related to the other *Pasteurella* species at the genus level.

Cell wall composition The cell wall and the lipopolysaccharide (LPS) composition of *P. multocida* were comparable to most Gram-negative bacteria and contained lipid A, 2-keto-3-deoxyoctonate, L-glycero-D-mannoheptose, glucose, and glucosamine, and might contain galactose, rhamnose, D-glycero-D-mannoheptose, and galactosamine (Rimmler and Rhoades, 1989). Polyamine analysis showed members of the genus *Pasteurella* to possess 1,3-diaminopropane, putrescine, cadaverine, spermidine, and spermine, and for some species *sym*-norspermidine in the cell wall. The unusual triamine *sym*-norspermidine was found only in *P. multocida*, *P. canis*, *P. dagmatis*, *P. stomatis*, and *Pasteurella* sp. B, but not in other *Pasteurellaceae*. This character is specific for the 16S rRNA subcluster 12 of Olsen et al. (see Fig. BXII.γ.215 of the chapter describing the family *Pasteurellaceae*, this *Manual*). Species of *Pasteurella* in 16S rRNA cluster 18 contained 1,3-diaminopropane and spermidine. In accordance with the 16S rRNA

analysis, the polyamine pattern of the remaining species of *Pasteurella* diverged from these two groups. *P. anatis* diverged from species of 16S rRNA cluster 18 by high putrescine content (Busse et al., 1997). The isoprenoid quinone composition of true members of the genus *Pasteurella* exhibits high amounts of quinones with the chain length eight: ubiquinone Q-8, menaquinone MK-8, and demethylmenaquinone DMK-8 (Mutters et al., 1993).

The capsule of *P. multocida* is composed of carbohydrates and is very hydrophilic (Rimmler and Rhoades, 1989). The biosynthesis and genetic background of capsule formation has been described recently (Boyce and Adler, 2000).

Hyaluronic acid has been reported in the capsular material and the biosynthesis pathway for hyaluronan has been described for *P. multocida* (DeAngelis, 1999).

Outer membrane proteins (OMPs) OMP analysis by SDS-PAGE has been reported for *P. multocida*. Specific OMP profiles of the 16 different serotypes were not given and it was found that *in vivo* propagation led to the expression of additional OMPs (Choi et al., 1989a). The expression of OMP by *P. multocida* was found to be influenced by iron and, in particular, molecules of 76, 84, and 94 kDa were expressed under iron-restricted conditions (Choi-Kim et al., 1991; Zhao et al., 1995). A 37.5-kDa porin of *P. multocida* affected bovine neutrophils (Galdiero et al., 1998). In *P. testudinis*, iron regulated OMPs were also identified (Snipes et al., 1995). Cross-reaction was found between iron-regulated OMPs in pigs and fowl (Zhao et al., 1995). Strains of *P. trehalosi* representing the four serotypes showed only four OMP profiles by SDS-PAGE electrophoresis; however, substantial diversity was observed between OMP profiles of *M. haemolytica* and *P. trehalosi* (Davies and Quirie, 1996), confirming the reclassification of the two taxa at the genus level (see *Mannheimia* chapter).

Monoclonal antibodies raised against different OMPs of *P. multocida* were able to specifically detect strains of capsular type D of *P. multocida* or of the species *P. stomatis*, *P. gallinarum*, *P. bettyae*, *Pasteurella* sp. B, and *P. canis* but not other Gram-negative bacteria, suggesting variable levels of epitope specificity (Marandi and Mittal, 1995, 1996). A relationship between *P. multocida*, *P. gallinarum*, *Haemophilus paragallinarum*, *P. volantium*, and *P. avium* was inferred through the binding of a polyclonal antibody raised against a putative porin of *P. multocida*. The N-terminal amino acid sequence confirmed relationships to other Gram-negative bacteria and to members of *Pasteurellaceae* in particular (Lubke et al., 1994; Hartmann et al., 1996).

Outer membrane proteins and capsular material including hyaluronic acid of serotype A have been characterized and are probably associated with virulence by interacting with host immune system (Rimmler and Rhoades, 1989; Boyce and Adler, 2000; Hunt et al., 2000).

Lipopolysaccharides (LPS) LPS of 13 Heddlestone serotypes of *P. multocida* all contained glucose, 2-keto-3-deoxyoctonate, and heptose. Two isomers of heptose were found in serotypes 2 and 5. Rhamnose was identified with LPS of serotype 9 and galactose found in all serotypes except serotype 11 (Rimmler et al., 1984; Conrad et al., 1996). A strain of serotype 8 contained galactose also and heptose. The amino sugars galactosamine, glucosamine, glucosamine-6-phosphate, and 3-deoxy-D-manno-2-octylosonic ac-

id were found in a strain of serotype 8 (Conrad et al., 1996). SDS-PAGE analysis of the serotypes of *P. multocida* showed similarities but not identity. Capsulation did not affect the LPS profile (Rimler, 1990). Purified LPS of *P. multocida* was found to cause suppurative airsacculitis, pleuritis, and pneumonia in turkeys (Kunkle and Rimler, 1998). LPSs of *P. multocida* capsular type A were found to affect the humoral and cell-mediated immune response (Maslog et al., 1999). Similar to the analysis of OMP, high homogeneity of LPS profiles was found within *P. trehalosi* by SDS-PAGE electrophoresis. (Davies and Quirie, 1996).

Fatty acids The dominant fatty acids in *P. multocida* serotype 8 were 3-hydroxytetradecanoate and tetradecanoate (Conrad et al., 1996) and *P. multocida* could be distinguished from other Gram-negative bacteria by the presence of C_{14:0} 3OH (Dees et al., 1981). The fatty acids C_{14:0}, C_{16:1}, C_{16:0}, C_{14:0} 3OH, C_{18:2}, C_{18:1}, and C_{18:0} were found in human strains of *P. multocida* subsp. *multocida*, *P. multocida* subsp. *septica*, *P. canis*, *P. stomatis*, and *P. dagmatis*; however, a differentiation of these species was not possible on the basis of fatty acid analysis. (Holst et al., 1992). The fatty acid profiles of the species of the family *Pasteurellaceae* proved to be indistinguishable (Mutters et al., 1993). The growth medium was found to affect the fatty acid profiling of members of *Pasteurella* (Boot et al., 1999).

Fine structure Type 4 fimbriae and the corresponding genes of representative serovars of *P. multocida* have been characterized (Ruffolo et al., 1997; Doughty et al., 2000). Pili of rigid and curly types have been observed on both capsulated and noncapsulated strains of *P. multocida* (Rebers et al., 1988; Isaacson and Trigo, 1995).

Colonial and cultural characters On standard agar plates for fastidious Gram-negative bacteria like chocolate agar or Columbia blood agar, *Pasteurella* mostly appear as regular, smooth, convex, grayish, nontransparent, circular colonies. The colonies are generally small with a diameter of 0.5–2 mm after 24 h incubations at 37°C. β -hemolysis on bovine or ovine blood agar is not seen, but a greenish discoloration may occur. Occasionally, yellowish-pigmented colonies can be seen, a phenomenon that is commonly associated with strains of *P. dagmatis* and *P. canis*. Isolates of *P. multocida* from the respiratory tract of ruminants, pigs, and rabbits may form large, watery, mucoid colonies that may collapse after 48 h incubation. Growth in broth usually causes turbidity, but granular growth may occur.

Nutrition and growth conditions Growth requirements of *P. multocida* have been reviewed by Rimler and Rhoades (1989). Cysteine, nicotinamide, pantothenate, and thiamine were required for growth. Apart from these specific growth factors, the different species of *Pasteurella* seem to exhibit variation in nutritional needs and preferred growth conditions. Systematic investigations based on international reference strains, however, do not exist.

V-factor requirement, usually described as nicotinamide adenine dinucleotide (NAD), has been reported for *P. avium*, *P. volantium*, *Pasteurella* sp. A (Mutters et al., 1985b), and *P. multocida* subsp. *multocida* (Krause et al., 1987). Occurrence of V-factor-independent isolates of *P. avium*, *P. volantium*, and *Pasteurella* sp. A have subsequently been reported by Bisgaard (1993), Bragg et al. (1997), and Bisgaard et al. (1999). However, none of these isolates was shown to belong to the respective species based on genetic characterization. Sources of V-factor and demonstration of V-factor requirement is discussed elsewhere (see *Haemophilus*

chapter). The temperature range for growth is 22–44°C, with an optimum growth temperature of 35–37°C. Increased carbon dioxide may be required for surface growth of certain isolates. Metabolic pathways have been investigated for *P. multocida* and reviewed by Rimler and Rhoades (1989).

Genetics The genomic sequence was determined in the avian strain Pm 70 of *P. multocida* and found to be 2,257,487 bp long with 2014 predicted coding regions (May et al., 2001). Strains of *P. multocida* have been found to possess six (Liu et al., 1999; May et al., 2001) or five (Hunt et al., 1998) ribosomal operons. Similar, but not identical, genome structure, was determined by *I-CeuI* restriction analysis; however, a few bovine strains diverged (Liu et al., 1999). Few genes have been characterized and those exclusively in *P. multocida*. The catabolic pathways for asparagine, histidine, leucine, lysine, and phenylalanine were absent from the *P. multocida* genome (May et al., 2001).

Virulence genes Best understood is the formation and activity of the dermonecrotic toxin found in some representatives of capsular types A and D of *P. multocida* that cause atrophic rhinitis in pigs (Hunt et al., 2000). Other virulence factors are incompletely understood in *P. multocida*, the only species studied in more detail. Two gene regions showed homology to the filamentous hemagglutinin gene of *Bordetella pertussis*, coding for proteins important in host cell binding and immunity (Fuller et al., 2000; May et al., 2001). In addition, virulence genes were identified homologous to hemolysin, hemolysin-binding protein, secretion accessory protein, Ton-dependent transport, and adherence (Fuller et al., 2000). Genes of the RTX toxin gene family have not been found except in *P. aerogenes* (Kuhnert et al., 1997). The gene cluster coding for leukotoxin has been found in all serotypes of *P. trehalosi* and the sequence of *lktA* characterized in detail (Shewen and Wilkie, 1982; Burrows et al., 1993; Davies et al., 2001). Capsulation and piliation are potential virulence factors whose genes have been characterized (Boyce and Adler, 2000; Doughty et al., 2000).

Plasmids and transposons Plasmids of 1.3–100 kb have been found in *P. multocida* (Hunt et al., 2000). Plasmids have been found related to resistance to streptomycin, sulfonamides, tetracycline, penicillins, kanamycin, and chloramphenicol (Hunt et al., 2000). The possession of identical nonconjugative R plasmids of *P. multocida* did not follow the clonal population structure (Ikeda and Hirsh, 1990). The only transposable element identified in *P. multocida* has been Tn5706 (Hunt et al., 2000), but Tn10 was able to integrate into the chromosome of *P. multocida* and is important in generating mutants (Lee and Henk, 1996).

Population genetics The population structure of *P. multocida* and *P. trehalosi* have been concluded to be clonal as determined by MLEE (multilocus enzyme electrophoresis), with genetic diversity of between 0.289 and 0.474 (Davies et al. 1997; Blackall et al., 1998; May et al., 2001).

Antigenic structure *P. multocida* has been the subject of numerous antigenic and serologic studies. However, only a few serological typing systems have gained wide acceptance. Four types (I, II, III, and IV) were recognized by Roberts (1947), based on passive protection of mice by serum against live challenge organisms. The most commonly used capsule typing system used so far was developed by Carter (1955). This typing system is based on passive hemagglutination of erythrocytes sensitized by capsule antigen, and five capsule types (A, B, D, E, and F) are presently distinguished (Rimler and Rhoades, 1987, 1989). Presumptive

identification of capsular types A, D, and F by capsule depolymerizations with mucopolysaccharidases has been reported by Rimler (1994). Subsequent cloning and sequencing of the entire capsular biosynthetic loci of *P. multocida* strains X-73 (A:1) (Chung et al., 1998b) and M 1404 (B:2) (Boyce et al., 2000), and nucleotide sequence analysis of the biosynthetic region from each of the remaining three capsule types D, E, and F, identified capsule-specific regions and allowed development of a highly specific multiplex capsular PCR assay (Townsend et al., 2001).

A total of 11 different O or somatic types of *P. multocida*, based on the use of acid-treated cells and agglutinin-absorption procedures, were reported by Namioka (1978). A more commonly used system for somatic antigen typing based on gel diffusion precipitin tests was developed by Heddlestone et al. (1972). Sixteen serotypes (1–16) were recognized by Heddlestone et al. (1972). Serological typing of *P. multocida* has been reviewed and discussed by Rimler and Rhoades (1989). Comparative studies by Brogden and Packer (1979) indicated that a serotype determination by one method did not correlate with serotyping by other methods. The relationship between subspecies and serovars of *P. multocida* obtained by published serotyping systems remains to be elucidated. Average linkage cluster analysis of precipitate values obtained by crossed immunoelectrophoresis revealed a close antigenic relationship between species of the genus *Pasteurella* as defined by Mutters et al. (1985a, b). Avian and mammalian species clustered separately, just as *P. anatis* and *P. langaa* made up a separate cluster (Schmid et al., 1991), confirming subsequent 16S rDNA similarity studies. The immunogens of *Pasteurella* have been reviewed by Confer (1993).

Antibiotic sensitivity Minimum inhibitory concentrations (MICs) of selected antimicrobial agents have been reviewed by Rimler and Rhoades (1989). Most human isolates of *Pasteurella* are susceptible to penicillin (Holst et al., 1992), the antibiotic of choice for local wound infections. Other β -lactam antibiotics should be effective as well, although in the case of the carbapenems, meropenem was more active than imipenem (Jorgensen et al., 1991). Macrolides and, especially for urinary and respiratory infections, fluoroquinolones such as ciprofloxacin appear to be useful in human and animal infections (Gaillot et al., 1995; Hanan et al., 2000). Most strains are also susceptible to tetracyclines (Holst et al., 1992). Marbofloxacin, a fluoroquinolone exclusively for veterinary use could be useful in the treatment of diseases in dogs and cats (Spreng et al., 1995). Tilmicosin, a macrolide antibiotic for use in veterinary medicine, proved to be effective in the treatment of pasteurellosis in rabbits (McKay et al., 1996).

Ecology With the exception of certain strains of *P. multocida*, organisms classified as *Pasteurella* are usually regarded as opportunistic pathogens that may colonize and form part of the indigenous flora of the mucous membranes of the upper respiratory and lower genital tracts.

Although numerous investigations have been carried out to determine the prevalence of these organisms, lack of sensitive and specific tests for their isolation and identification have prevented insight into their real distribution (Bisgaard, 1993). Previous investigations have mainly focused on the respiratory tract. Detection of a high proportion of cloacal carriers in poultry of *P. multocida* (Muhairwa et al., 2000) emphasizes the importance of sampling mucosal membranes other than the respiratory tract.

P. multocida is distributed worldwide among terrestrial as well as aquatic species of mammals and birds, and any species of these

groups should be considered as a possible host. Although the host spectrum seems very large compared to other *Pasteurellaceae*, indications also exist that certain subtypes have developed, the disease potential of which seems to be limited to only a few species (see pathogenicity).

P. canis biovar 1, *P. stomatis*, *P. dagmatis*, and *Pasteurella* sp. B are mainly associated with the oral and nasal mucosa of dogs and cats (Bisgaard, 1993; Ganiere et al., 1993; Muhairwa et al., 2001). However, isolates of *P. dagmatis* from rodents and a scarlet macaw have also been reported (Bisgaard, 1993; Bisgaard et al., 1999). These isolates, however, remain to be characterized genetically.

P. gallinarum is normally associated with different pathological lesions in poultry, and its occurrence in healthy birds remains to be investigated (Bisgaard, 1993). A single isolate has been reported from a healthy duck (Muhairwa et al., 2001). Publications on isolation from animal species other than birds should be questioned (Bisgaard and Mutters, 1986a; Boot and Bisgaard, 1995; Frederiksen and Tønning, 2001). The natural habitats of the remaining species of *Pasteurella* belonging to the 16S rRNA cluster 18 of Olsen et al. (see Fig. BXII.γ.215 of the chapter describing the family *Pasteurellaceae*, this *Manual*) seem to be associated with birds only. Isolates of *P. avium* biovar 2 associated with pneumonia in cattle should be reinvestigated, since recent investigations have shown that nearly identical 16S rDNA sequences were observed for *P. multocida* and biovars 2 of *P. canis* and *P. avium* (Christensen et al., unpublished results). *P. multocida* and biovar 2 of *P. canis* and biovar 2 of *P. avium* also expressed similar OMP profiles (Abdullahi et al., 1990).

Pathogenicity Among species classified as *Pasteurella*, *P. multocida* has been recognized as an important veterinary pathogen for more than a century. It causes a wide range of diseases in animals. Capsule type A serotypes 1, 3, and 4 are recognized as the primary cause of fowl cholera in poultry and wild birds (Rhodes and Rimler, 1989). Disease may appear as an acute septicemia characterized by disseminated intravascular coagulation, petechial or ecchymotic hemorrhages, multifocal necroses, and fibrinous pneumonia. Chronic infections may involve a variety of local infections (Christensen and Bisgaard, 1997, 2000). *P. multocida* capsule types B and E are associated with hemorrhagic septicemia of cattle, water buffaloes, and occasionally other species, resulting in major economic losses, mainly in Southeast Asia (Carter and De Alwis, 1989; De Alwis, 1995). Respiratory diseases in cattle, including bronchopneumonia in feedlot cattle and enzootic pneumonia of calves less than 6 months old, are mainly associated with capsule type A (Frank, 1989). Outbreaks of septicemia in fallow deer have also been reported (Eriksen et al., 1999). Infections of major economic importance in pigs include atrophic rhinitis and bronchopneumonia (Chanter and Rutter, 1989; Gardner et al., 1994). These syndromes are caused by capsular types A and D. Severe cases of atrophic rhinitis are mainly associated with capsule type D. Pulmonary lesions may result in blood-borne dissemination to the kidneys in pigs (Buttenschön and Rosendal, 1990). In addition to major diseases in production animals, *P. multocida* is recovered from a wide range of sporadic infections in many other species, including laboratory animals (Manning et al., 1989), dogs and cats (Mohan et al., 1997), and other mammals (DiGiacomo et al., 1989).

Although *P. multocida* has challenged researchers for more than a century, mechanisms behind virulence and pathogenesis have remained unclear, as demonstrated in recent literature (Christensen and Bisgaard, 1997, 2000). Successful isolation of DNA fragments containing the toxin-encoding gene from tox-

genic *P. multocida* and cloning these into *E. coli* (Petersen and Foged, 1989; Kamps et al., 1990; Lax and Chanter, 1990) significantly improved our understanding of the pathogenesis of atrophic rhinitis. Sequencing of the entire genome of a common avian clone of *P. multocida* (May et al., 2001) provides a foundation for future research into virulence factors and mechanisms of pathogenesis and host specificity of this pathogen that may kill affected animals within 24 h.

P. multocida and other species of the 16S rRNA cluster 12 are considered as zoonotic pathogens (Bisgaard et al., 1994). Most human infections associated with *Pasteurella* species result from animal bites. The species usually observed in these infections are *P. multocida* subsp. *multocida* and subsp. *septica*, *P. canis*, *P. dagmatis*, and *P. stomatis* (Holst et al. 1992; Escande and Lion, 1993; Matsui et al., 1996). Like other opportunistic pathogens, *Pasteurella* species can be associated with different clinical syndromes, i.e., necrotizing fasciitis (Hamamoto et al., 1995), chronic lung abscess (Machiels et al., 1995), endocarditis (Genne et al., 1996), meningitis (Boocock and Bowley, 1995; Armstrong et al., 2000), pulmonary diseases such as pneumonia (Ory et al., 1998), peritonitis (Wallet et al., 2000), septicemia (Greif et al., 1986), periocular abscess and cellulitis (Hutcheson and Magbalon, 1999), and granulomatous hepatitis (Chateil et al., 1998). Usually human infections result from inoculation of animal secretions via bites or direct contact with animals carrying pasteurellae in their normal bacterial flora.

ENRICHMENT AND ISOLATION PROCEDURES

Species of *Pasteurella* are somewhat fastidious and V-factor-requiring species are traditionally isolated on enriched agar media supplemented with 5% serum or blood and cross-inoculated with a *Staphylococcus* or an *Acinetobacter* strain to provide the V-factor. However, in mixed cultures pasteurellae may be overgrown. Strains of *P. multocida* can be obtained from contaminated material by subcutaneous or intraperitoneal inoculation of mice (Muhairwa et al., 2001). After death, cultures are made from the spleen by ordinary methods. Several selective media have been developed for isolation of *P. multocida* (Rimler and Rhoades, 1989; Moore et al., 1994b; Lee et al., 2000a), and transport media have been investigated (Kawamoto et al., 1997). Comparative investigations, however, remain to be performed.

Short-term survival of pasteurellae for about a week is possible on complex solid media like chocolate agar stored at room temperature or preferably at 4°C in a plastic jar or plastic bag to avoid desiccation. In the case of V-factor-dependent pasteurellae, shorter intervals should be used. Cultures will remain viable for years when frozen at -70°C in liquid media such as protease peptone broth containing 20% glucose or glycerol. Recommended procedures for long-term survival also include lyophilization or storage in liquid nitrogen of cells in nutrient broth containing 20% glucose or glycerol. Both techniques preclude selection of mutants that occur during repeated subculturing.

DIFFERENTIATION OF THE GENUS *PASTEURELLA* FROM OTHER GENERA

Pasteurellae (Table BXII.γ.288) share the common features of the family *Pasteurellaceae* (see chapter *Pasteurellaceae*). They do not grow in Simmons citrate medium. Nitrate reduction is positive. No acid is produced from D-adonitol and L-sorbose. They do not contain arginine dihydrolase, but do produce alkaline phosphatase. In contrast to *Yersinia* they are nonmotile at 22°C, and do not grow in the presence of 4.5% NaCl. Differentiation

from *Kingella* and other fastidious Gram-negative rods is possible based on several features, including positive tests for oxidase and catalase and acid production from sucrose and other carbohydrates. Distinction from other genera of the family *Pasteurellaceae*, especially from *Actinobacillus*, has been difficult based on the original concepts of these genera. Despite reallocation of species and outlining of new genera as indicated by Olsen et al. (see Fig. BXII.γ.215 of the chapter describing the family *Pasteurellaceae*, this *Manual*), available data indicate that it will still be difficult to divide the family *Pasteurellaceae* into genera that are phenotypically and phylogenetically coherent.

The pattern of typical biochemical characteristics of *Pasteurellaceae*, combined with the common features of *Pasteurella* (see Table BXII.γ.288), describes true pasteurellae. Although the taxonomy of the genus is still under investigation, some phenotypic features can be defined that are useful in discriminating *Pasteurella* from both other named genera and unnamed and not formally recognized genus-like groups within the family *Pasteurellaceae*. True members of the genus never exhibit hemolysis on ordinary media. Oxidase reaction and usually catalase reaction are positive. Normally, gas is not produced from the fermentation of carbohydrates; occasionally, very small amounts of gas can be detected. The sole *Pasteurella* species exhibiting gas production is *P. dagmatis*. Differences in urease and ornithine decarboxylation can also be used for separation of the genera *Pasteurella* and *Actinobacillus*. With the exception of *P. dagmatis*, all pasteurellae are urease negative, whereas all actinobacilli are positive and, with the exception of *P. dagmatis* and *P. stomatis*, all *Pasteurella sensu stricto* (cluster 12 of Olsen et al.; see Fig. BXII.γ.215 of the chapter describing the family *Pasteurellaceae*, this *Manual*) are ornithine positive, whereas actinobacilli are negative. Differences in indole can also be used for separation of genuine pasteurellae (positive) and actinobacilli (negative), but single indole-negative isolates of *P. multocida* occur. Certain carbohydrates, like L-rhamnose and D-mannitol, are not cleaved. On the other hand, all pasteurellae produce acid from the six-carbon carbohydrates D-glucose, D-fructose, D-mannose, and D-galactose in contrast to many *Haemophilus* and *Actinobacillus* species. Differences in D-mannose and indole separate the genera *Mannheimia* and *Pasteurella*.

TAXONOMIC COMMENTS

In the mid-1980s, the taxonomy of the genus *Pasteurella* was studied by DNA-DNA hybridization (Mutters et al., 1985a, b). More recently, the genus was investigated in the context of the whole family by 16S rDNA sequencing (see chapter *Pasteurellaceae*). The results of the DNA hybridization studies led to the reclassification of the genus *Pasteurella sensu stricto*. According to this method, *Pasteurella* contains species interrelated at or above a DNA binding level of 55% based on the optical DNA hybridization method (cit. Mutters et al., 1985a). From the six *Pasteurella* species in the last edition of *Bergey's Manual of Systematic Bacteriology* only the type species of the genus, *Pasteurella multocida* and *P. gallinarum*, are recognized as true pasteurellae. *P. multocida* was subdivided into three subspecies—subsp. *multocida*, subsp. *septica*, and subsp. *gallicida*—according to DNA-binding data and differences in acid production from D-sorbitol and dulcitol; 1.4–2% 16S rRNA sequence variation was found among *P. multocida* subsp. *septica* and *P. multocida* subsp. *multocida* and subsp. *gallicida*, while subsp. *multocida* and subsp. *gallicida* were nearly identical (Boerlin et al., 2000; Petersen et al., 2001). Partial *atpD* DNA sequence comparison confirmed the 16S rRNA results in that the subspecies

TABLE BXII.γ.288. Phenotypic characters of *Pasteurella* species^a

Characteristic	Species classified with the 16S rRNA cluster 12 (genus <i>Pasteurella sensu stricto</i>)					Species classified with the 16S rRNA cluster 18					
	<i>P. multocida</i>	<i>P. canis</i>	<i>P. dagmatis</i>	<i>P. stomatis</i>	<i>Pasteurella</i> sp. B	<i>P. anatis</i>	<i>P. avium</i>	<i>P. gallinarum</i>	<i>P. langaa</i>	<i>P. volantium</i>	<i>Pasteurella</i> sp. A
Gram stain	—	—	—	—	—	—	—	—	—	—	—
Motility, 22°C and 37°C	—	—	—	—	—	—	—	—	— ^b	—	—
Catalase	+	+	+	+	+	+	+	+	— ^b	+	+
Oxidase	+	+	d	+	+	d	+	+	d	+	+
Glucose (Hugh and Leifson)	F	F	F	F	F	F	F	F	F	F	F
Symbiotic growth	—	—	—	—	—	—	+	—	—	+	+
Porphyrin test	+	+	+	+	+	+	+	+	+	+	+
β-Hemolysis (bovine blood)	—	—	—	—	—	—	—	—	—	—	—
Citrate (Simmons)	—	—	—	—	—	—	—	—	—	—	—
Mucate, acid	—	—	—	—	—	—	—	—	—	—	—
Malonate, base	—	—	—	—	—	—	—	—	—	—	—
H ₂ S/TSI	—	—	—	—	—	—	—	—	—	—	—
KCN, growth	—	—	—	—	—	—	—	—	—	—	—
Methyl-red, 37°C	—	—	—	—	—	— ^b	—	—	—	—	—
Voges–Proskauer, 37°C	—	—	—	—	—	—	—	—	—	—	—
Nitrate, reduction	+	+	+	+	+	+	+	+	+	+	+
Nitrate, gas	—	—	—	—	—	—	—	—	—	—	—
Urease	—	—	+	—	—	—	—	—	—	—	—
Alanine aminopeptidase	+	+	+	+	+	+	+	+	+	+	+
Arginine dehydrolase	—	—	—	—	—	—	—	—	—	—	—
Lysine decarboxylase	—	—	—	—	—	—	—	—	—	—	—
Ornithine decarboxylase	+	+	—	—	+	—	—	—	—	d	—
Phenylalanine deaminase	—	—	—	—	—	—	—	—	—	—	—
Indole	+	+	+	+	d	—	—	—	—	—	—
Phosphatase	+	+	+	+	+	+	+	+	+	+	+
Gelatinase	—	—	—	—	—	—	—	—	—	—	—
Tween 20	—	—	—	—	—	—	—	—	—	—	—
Tween 80	—	—	—	—	—	—	—	—	—	—	—
McConkey, growth	—	—	—	—	—	d	—	—	+	—	—
Pigment	—	—	—	—	—	—	—	—	—	—	—
Glycerol	d	—	d	—	—	(+)	—	d	—	—	d
<i>meso</i> -Erythrol	—	—	—	—	—	—	—	—	—	—	—
Adonitol	—	—	—	—	—	—	—	—	—	—	—
D(+)Arabitol	—	—	—	—	—	—	—	d	—	—	—
Xylitol	—	—	—	—	+	—	—	—	—	—	d
L(+)Arabinose	—	—	—	—	—	—	—	—	—	—	+/(+)
D(–)Arabinose	d	—	d	—	—	—	—	d	—	d	—
D(–)Ribose	+	+	+	+	+	+	+	+	+	(+)	+
D(+)Xylose	d	—	—	—	+	+	d	d	—	d	d
L(–)Xylose	—	—	—	—	—	—	—	—	—	—	—
Dulcitol	d	—	—	—	+	—	—	—	—	—	—
<i>meso</i> -Inositol	—	—	—	—	—	—	—	d	—	—	—
D(–)Mannitol	+	—	—	—	+	+	—	—	+	+	d
D(–)Sorbitol	d	—	—	—	—	—	—	—	—	d	—
D(–)Fructose	+	+	+	+	+	+	+	+	+	+	+
D(+)Fucose	—	—	—	—	—	—	—	—	—	—	—
L(–)Fucose	d	—	d	—	—	—	—	d	—	d	—
D(+)Galactose	+	+	+	+	+	+	+	+	+	+	+
D(+)Glucose, acid	+	+	+	+	+	+	+	+	+	+	+
D(+)Glucose, gas	—	—	— ^b	—	—	—	—	—	—	—	—
D(+)Mannose	+	+	+	+	+	+	+	+	+	+	+
L(+)Rhamnose	—	—	—	—	—	—	—	—	—	—	—
L(–)Sorbitose	—	—	—	—	—	—	—	—	—	—	—
Cellobiose	—	—	—	—	—	—	—	—	—	—	—
β-Glucosidase (NPG)	—	—	—	—	—	—	—	—	—	—	—
Lactose	—	—	d	—	—	d	—	d	+	d	—
ONPG	—	—	d	—	—	d	—	d	+	+	d
Maltose	—	—	+	—	+/(+)	—	—	+	—	+	d
D(+)Melibiose	—	—	—	—	—	—	—	—	—	—	—
Sucrose	+	+	+	+	+	+	+	+	+	+	+
Trehalose	d	d	d	+	+	+	+	+	—	+	+
D(+)Melizitose	—	—	—	—	—	—	—	—	—	—	—
Raffinose	—	—	d	— ^b	—	d	—	d	—	—	(+)
Dextrin	—	—	+/(+)	—	+/(+)	—	—	+	—	+	d
D(+)Glycogen	—	—	—	—	—	—	—	—	—	—	—
Inulin	—	—	—	—	—	—	—	—	—	—	—
Esculin	—	—	—	—	—	—	—	—	—	—	—
Amygdalin	—	—	—	—	—	—	—	—	—	—	—

(continued)

TABLE BXII.γ.288. (cont.)

Characteristic	Species classified with the 16S rRNA cluster 12 (genus <i>Pasteurella sensu stricto</i>)					Species classified with the 16S rRNA cluster 18					
	<i>P. multocida</i>	<i>P. canis</i>	<i>P. dagmatis</i>	<i>P. stomatis</i>	<i>Pasteurella</i> sp. B	<i>P. anatis</i>	<i>P. avium</i>	<i>P. gallinarum</i>	<i>P. langaa</i>	<i>P. volantium</i>	<i>Pasteurella</i> sp. A
Arbutin	—	—	—	—	—	—	—	—	—	—	—
Gentiobiose	—	—	—	—	—	—	—	—	—	—	—
Salicin	—	—	—	—	—	—	—	—	—	—	—
D(+)-Turanose	—	—	—	—	—	—	—	—	—	—	—
β-N-CH ₃ -Glucosamid	—	—	—	—	—	—	—	—	—	—	—
α-Fucosidase (ONPF)	—	—	—	—	—	—	—	—	—	—	—
α-Galactosidase	—	—	—	—	—	—	—	—	—	—	—
α-Glucosidase (PNPG)	d	d	+	+	+	+	+	+	—	+	+
β-Glucuronidase (PGUA)	—	—	—	—	—	—	—	—	—	—	—
α-Mannosidase	—	—	—	—	—	—	—	—	—	—	—
β-Xylosidase (ONPX)	—	—	—	—	—	—	—	—	—	—	—

^aSymbols: +, 90% or more of the strains positive within 1–2 d; (+), 90% or more of the strains positive within 3–14 d; —, less than 10% of strains positive within 14 d; d, 11–89% of the strains are positive; w, weak positive; F, fermentative; ONPG, *o*-nitrophenyl-β-D-galactopyranoside. Incubation temperature 37°C.

^bWeak positive reaction might occur.

of *P. multocida* differed, with *P. multocida* subsp. *septica* diverged the most from the others (Petersen et al., 2001).

The former biotype 6 or dog-type strains of *P. multocida* that contained D-mannitol and D-sorbitol negative strains of *P. multocida* were classified as the new species *P. canis*. Similar to *P. canis*, but ornithine negative, is *P. stomatis*, which proved to be closely related to *P. canis* and the former *Haemophilus avium*. This V-factor-dependent species of avian origin was classified as *P. avium*. Biotype Henriksen strains of *P. pneumotropica* were classified as *P. dagmatis*, *P. canis*, and *P. stomatis*. *P. dagmatis* is mostly isolated from bite wounds, mainly inflicted by dogs. In addition to the avian species, *P. gallinarum* and *P. avium*, three new species from avian hosts were described, the V-factor-dependent *P. volantium* and two V-factor-independent species, *P. langaa* and *P. anatis*. Two species remained unnamed, but were located in the genus, *Pasteurella* species A and *Pasteurella* species B. *Pasteurella* species A resembles a collection of V-factor-dependent strains formerly designated as *Haemophilus avium*-like. Based on DNA–DNA binding data, all these species were classified as *Pasteurella sensu stricto* (Mutters et al., 1985a, b). Based on 16S rRNA sequence data from selected strains, the genus was separated into two phylogenetic groups (Dewhirst et al., 1992, 1993).

The core group around the type species included the other species from mammalian hosts: *P. canis*, *P. stomatis*, *P. dagmatis*, and *Pasteurella* sp. B. The avian species *P. avium*, *P. gallinarum*, *P. volantium*, and *Pasteurella* sp. A form a separate cluster together with the type strain of *Haemophilus paragallinarum*, and reclassification of this group seems justified based on 16S rDNA phylogenetic analysis and the variations in *atpD* DNA sequences (Petersen et al., 2001).

P. langaa and *P. anatis* were not phylogenetically related to any of the present members of the genus, and should be excluded from the genus *sensu stricto*, just as reconsiderations are needed as to exclusion of the other avian taxa.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *PASTEURELLA*

Allocation of isolates to existing species of *Pasteurella* usually does not present a problem if, as suggested by Bisgaard (1993), extended characterization and reference strains/data are used. If sequencing facilities are available, partial or full sequencing of 16S rRNA genes should be included. Correct allocation to existing taxa and of newly discovered taxa, however, still remains

Both DNA hybridization and rRNA sequence data suggest the exclusion of species from the genus that were listed in the last edition of *Bergey's Manual of Systematic Bacteriology*. *P. ureae* was reclassified as *Actinobacillus ureae* (Mutters et al. 1986), while *P. pneumotropica* forms a large cluster with several new species requiring the rank of at least one new genus (Ryll et al., 1991). The *M. haemolytica* complex of ruminants contains distinct genetic and phenotypic groups. So far seven species have been outlined, five of which were named in the new genus *Mannheimia*. *P. aerogenes* (McAllister and Carter, 1974), *P. bettyae* (Sneath and Stevens, 1990), *P. caballi* (Schlater et al., 1989), *P. mairii* (Sneath and Stevens, 1990), *P. testudinis* (Snipes and Biberstein, 1982), and *P. trehalosi* (Sneath and Stevens, 1990) all represent misnamed pasteurellas.

Further taxa provisionally affiliated with *Pasteurella* based on phenotypic characters, such as the gas-producing SP-group, could not be integrated in the genus.

FURTHER READING

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- Dewhirst, F.E., B.J. Paster, I. Olsen and G.J. Fraser. 1992. Phylogeny of 54 representative strains of species in the family *Pasteurellaceae* as determined by comparison of 16S rRNA sequences. *J. Bacteriol.* 174: 2002–2013.
- Mutters, R., P. Ihm, S. Pohl, W. Fredericksen and W. Mannheim. 1985. Reclassification of the genus *Pasteurella* Trevisan 1887 on the basis of deoxyribonucleic acid homology, with proposals for the new species *Pasteurella dagmatis*, *Pasteurella canis*, *Pasteurella stomatis*, *Pasteurella anatis* and *Pasteurella langaa*. *Int. J. Syst. Bacteriol.* 35: 309–322.
- Mutters, R., K. Piechulla, K.H. Hinz and W. Mannheim. 1985. *Pasteurella avium* (Hinz and Kunjara 1977) comb. nov. and *Pasteurella volantium* sp. nov. *Int. J. Syst. Bacteriol.* 35: 5–9.

problematic, and cooperation with reference laboratories is recommended. The differential characteristics of species of *Pasteurella* are given in Table BXII.γ.289, while characters used for separation of subspecies of *P. multocida* are shown in Table BXII.γ.290.

TABLE BXII.γ.289. Characters used for separation of *Pasteurella* spp.^a

Characteristic	<i>P. multocida</i>	<i>P. canis</i>	<i>P. dagmatis</i>	<i>P. stomatis</i>	<i>Pasteurella</i> sp. B	<i>P. anatis</i>	<i>P. avium</i>	<i>P. gallinarum</i>	<i>P. langaa</i>	<i>P. volantium</i>	<i>Pasteurella</i> sp. A
Catalase	+	+	+	+	+	+	+	+	—	+	+
Symbiotic growth	—	—	—	—	—	—	+	—	—	+	+
Urease	—	—	+	—	—	—	—	—	—	—	—
Ornithine decarboxylase	+	+	—	—	+	—	—	—	—	d	—
Xylitol	—	—	—	—	+	—	—	—	—	—	d
L(+)-Arabinose	—	—	—	—	—	—	—	—	—	—	+/(+)
D(+)-Xylose	d	—	—	—	+	+	d	d	—	d	d
D(−)-Mannitol	+	—	—	—	+	+	—	—	+	+	d
Maltose	—	—	+	—	+/(+)	—	—	+	—	+	d
Dextrin	—	—	+/(+)	—	+/(+)	—	—	+	—	+	d
PNPG	d	d	+	+	+	+	+	+	—	+	+

^aSymbols: +, 90% or more of the strains positive within 1–2 d; (+), 90% or more of the strains positive within 3–14 d; —, less than 10% of the strains are positive within 14 d; d, 11–89% of the strains are positive.

List of species of the genus *Pasteurella*

1. ***Pasteurella multocida*** (Lehmann and Neumann 1899) Rosenbusch and Merchant 1939, 85^{AL} (*Bacterium multocidum multocidum* (sic) Lehmann and Neumann 1899, 196; *Pasteurella gallicida* (Burrill 1883) Buchanan 1925, 414.) *mul.to.ci' da*. L. adj. *multus* many; L. adj. suff. *-cidus* from; L. v. *caedere* to kill; M.L. fem. adj. *multocida* many killing, i.e., pathogenic for many species of animals.

Usually forms coccoid cells or short rods on solid media containing blood. Usually bipolar stained in Gram stain. Pleomorphic rods and short filaments can be seen in broth media and older cultures. Many strains produce capsules. Growth on solid media is best on blood containing agar like sheep blood or chocolate agar. Colonies reach a diameter of 1–2 mm with a light grayish color. Mucoid, smooth, and rough colonies are produced. Nonhemolytic. The indole-producing strains exhibit a distinct odor. The optimal growth temperature for avian strains might be as high as 42°C, compared to 36°C for mammalian strains. Growth can be observed in a wide mesophilic range from 25–42°C.

In addition to the features common to all members of the genus, the three subspecies share the following biochemical reactions. Acid usually produced from D-mannitol. Ornithine is decarboxylated. Usually no V-factor requirement, but V-factor-requiring strains may occur occasionally (Krause et al., 1987). Major differences in phenotypic characters have been reported for *P. multocida* (Heddleston, 1976; Biberstein et al., 1991; Bisgaard et al., 1991b; Fegan et al., 1995). The physiological characteristics of the species are presented in Tables BXII.γ.288, BXII.γ.289, and

BXII.γ.290. Several typing systems have been developed and applied for epidemiological studies of *P. multocida* (Nielsen and Rosdahl, 1990; Wilson et al., 1993; Blackall et al., 1999b, 2000; Fusing et al., 1999; Loubinoux et al., 1999; Blackall and Mifflin, 2000; Bowles et al., 2000; Gunawardana et al., 2000). Strains of the species are isolated from most mammals, including humans and birds, in which a wide range of diseases are reported due to *P. multocida*. The species belong to 16S rRNA cluster 12. The molecular weight of genomic DNA ranges from 1.5×10^9 to 1.9×10^9 .

The mol% G + C of the DNA is: 40.8–43.9 (T_m).

Type strain: ATCC 43137, NCTC 10322.

GenBank accession number (16S rRNA): AF294410, M35018.

- a. ***Pasteurella multocida* subsp. *multocida*** (Lehmann and Neumann 1899) Rosenbusch and Merchant 1939, 85^{AL} (*Bacterium multocidum multocidum* Lehmann and Neumann 1899, 196; *Pasteurella gallicida* (Burrill 1883) Buchanan 1925, 414.)

Characteristics are those of the species as given above. Differentiated from the other subspecies by its production of acid from D-sorbitol, but not from dulcitol. Other reactions useful in differentiating between the subspecies are shown in Table BXII.γ.290. The species belongs to 16S rRNA cluster 12. The molecular weight of genomic DNA ranges from 1.5 – 1.7×10^9 .

The mol% G + C of the DNA is: 40.8 to 43.9 (T_m).

Type strain: ATCC 43137, NCTC 10322.

GenBank accession number (16S rRNA): AF294410, M35018.

- b. ***Pasteurella multocida* subsp. *gallicida*** Muters, Ihm, Pohl, Frederiksen and Mannheim 1985a, 319^{VP} *gal.li.ci' da*. L. fem. n. *gallina* hen; L. adj. suff. *-cida* kill; M.L. adj. *gallicida* hen killing, referring to pathogenicity for poultry.

Characteristics are those of the species as given above. Differentiation from the other subspecies by its ability to form acid from D-sorbitol and dulcitol. Other reactions useful in differentiating between the subspecies are shown in Table BXII.γ.290. The species belongs to 16S rRNA cluster 12. The molecular weight of genomic DNA ranges from 1.5 – 1.8×10^9 .

TABLE BXII.γ.290. Characters used for identification of subspecies of *Pasteurella multocida*^a

Characteristic	<i>P. multocida</i>		
	subsp. <i>multocida</i>	subsp. <i>gallicida</i>	subsp. <i>septica</i>
Dulcitol	—	+	—
D(−)-Sorbitol	+	+	—
L(−)-Fucose	d	—	d
Trehalose	d	—	+
α-Glucosidase (PNPG)	d	—	+

^aSymbols: +, 90% or more of the strains positive; —, 90% or more of the strains negative; d, 11–89% of the strains positive.

The mol% G + C of the DNA is: 41.2–42.5 (T_m).

Type strain: ATCC 51689, NCTC 10204.

GenBank accession number (16S rRNA): AF224297, AF294412, AF326323.

- c. *Pasteurella multocida* subsp. *septica* Muters, Ihm, Pohl, Frederiksen and Mannheim 1985a, 319^{VP}
sep' ti.ca. M.L. fem. adj. *septica* poisoning, infecting.

Characteristics are those of the species as given above. Differentiation from the other subspecies by its negative reactions for acid production from D-sorbitol and dulcitol. Other reactions useful in differentiating between the subspecies are shown in Table BXII.γ.290. The species belongs to 16S rRNA cluster 12. The molecular weight of genomic DNA ranges from $1.5\text{--}1.6 \times 10^9$.

The mol% G + C of the DNA is: 41.5–43.5 (T_m).

Type strain: ATCC 51687, NCTC 11995, CIP A125.

GenBank accession number (16S rRNA): AF225205, AF294411, AF326325.

2. *Pasteurella anatis* Muters, Ihm, Pohl, Frederiksen and Mannheim 1985a, 320^{VP}
a.na' tis. L. fem. gen. n. *anas* duck, of a duck.

Cells are coccobacillary. Colonies are circular, smooth, and grayish-white with a diameter of 1.5–2 mm. Nonhemolytic. Strains have no V-factor requirement. They do not decarboxylate ornithine, do not produce indole, and do not hydrolyze urea. Acid is produced from trehalose, D-mannitol, and D-xylose. No acid is produced from L-arabinose, maltose, D-sorbitol, or dulcitol. Members of the species have been isolated from the intestinal and respiratory tracts of ducks. The species belongs to 16S rRNA cluster 18. The molecular weight of genomic DNA ranges from $1.8\text{--}1.9 \times 10^9$.

The mol% G + C of the DNA is: 39.9–42.3 (T_m).

Type strain: ATCC 43329, NCTC 11413.

GenBank accession number (16S rRNA): AF228001, M75054.

3. *Pasteurella avium* (Hinz and Kunjara 1977) Muters, Piechulla, Hinz, and Mannheim 1985b, 8^{VP} (*Haemophilus avium* Hinz and Kunjara 1977, 324.)
a'vi.um. L. fem. gen. pl. n. *avis* bird, of birds.

Coccoid to pleomorphic rods, occurring singly or in short chains. In liquid culture media, filamentous forms can be seen. Colonies on chocolate agar are usually smooth, convex, slightly yellowish or grayish-white. Nonhemolytic. The species exhibits the common biochemical characteristics of *Pasteurella sensu stricto*. Biotype 1 strains require V-factor, while biotype 2 strains, previously reported as the ornithine decarboxylase-negative type of Bisgaard Taxon 13, are V-factor independent. Acid is produced from glucose without gas formation and from D(+)galactose, D(+)mannose, D(–)fructose, and trehalose. No acid is produced from L(+)arabinose, L(+)rhamnose, raffinose, maltose, D(–)mannitol, D(–)sorbitol, or dulcitol. Indole is not produced, ornithine decarboxylase negative. Found in the hearts and infraorbital sinuses of chickens (biovar 1) and in the lungs of calves suffering from pneumonia (biovar 2). The species belongs to 16S rRNA cluster 18. The molecular weight of genomic DNA is 1.9×10^9 .

The mol% G + C of the DNA is: 43–45 (T_m).

Type strain: ATCC 29546, IPDH 2654.

GenBank accession number (16S rRNA): M75058.

Additional Remarks: The reference strain for biovar 2 is CCUG 16497 (Bisgaard's ornithine negative taxon 13).

4. *Pasteurella canis* Muters, Ihm, Pohl, Frederiksen and Mannheim 1985a, 320^{VP}
ca'nis. L. gen. n. *canis* of a dog.

Colonial morphology resembles that of *Pasteurella multocida*, although biotype 1 colonies of *P. canis* usually reach a diameter of only 1 mm. Nonhemolytic. Growth optimum is 36°C. Strains are V-factor independent. Ornithine decarboxylase positive. Acid is not produced from L-arabinose, raffinose, lactose, maltose, D-mannitol, D-sorbitol, or dulcitol. Urease negative. Biotype 1 strains exhibit positive reactions for indole; biotype 2 strains are indole negative. *P. canis* biotype 1 is recovered mainly from the oral cavities of dogs and is often isolated from dog-bite wounds in humans. Biotype 2 strains have been isolated from pneumonias from calves. The species belongs to 16S rRNA cluster 12. The molecular weight of genomic DNA ranges from $1.4\text{--}1.6 \times 10^9$.

The mol% G + C of the DNA is: 37.7–39.8 (T_m).

Type strain: ATCC 43326, NCTC 11621.

GenBank accession number (16S rRNA): M75049.

Additional Remarks: The reference strain for biovar 2 is CCUG 16498 (Bisgaard's ornithine positive taxon 13).

5. *Pasteurella dagmatis* Muters, Ihm, Pohl, Frederiksen and Mannheim 1985a, 319^{VP}
dag.ma'tis. Gr. fem. gen. n. *dagma* bite, from a bite.

Cells coccoid or short rods. Colonies on blood containing media are usually smooth and grayish white. Optimal growth temperature is 36°C under aerobic conditions. Nonhemolytic. Does not require V-factors. Produces small amounts of gas from D-glucose; produces acid from maltose. No acid is produced from D-xylose, L-arabinose, D-mannitol, D-sorbitol, and dulcitol. Tests for glycosides are negative. Positive reactions are obtained for indole and urease. Gelatin may be liquefied after more than 14 d of incubation. The biochemical characteristics are shown in Tables BXII.γ.288 and BXII.γ.289. *P. dagmatis* strains have been isolated from dogs and cats, as well as from human local infections resulting from animal bites or from septicemia and endocarditis (Frederiksen, 1989). The species belongs to 16S rRNA cluster 12. The molecular weight of genomic DNA ranges from $1.5\text{--}1.7 \times 10^9$.

The mol% G + C of the DNA is: 38.9–41.5 (T_m).

Type strain: ATCC 43325, NCTC 11617.

GenBank accession number (16S rRNA): M75051.

6. *Pasteurella gallinarum* Hall, Heddlestone, Legenhausen and Hughes 1955, 604^{AL}
gal.li.na' rum. L. fem. gen. pl. n. *gallina* hen, of hens.

Cells are coccoid to rod shaped; bipolar staining occurs. Colonies on blood containing solid media are circular, smooth, and convex with a grayish appearance and reach diameters of 1.5 mm. Nonhemolytic. Not V-factor dependent. Optimal growth temperature is 36°C. Produce acid from maltose and trehalose. No acid is produced from D-mannitol, D-sorbitol, dulcitol, and L-arabinose. Tests for ornithine decarboxylase, indole formation, and urease are negative. Production of acid from meso-inositol may occur. *P. gallinarum* is isolated from different lesions in fowl. The

species belongs to 16S rRNA cluster 18. The molecular weight of genomic DNA ranges from $1.5\text{--}1.6 \times 10^9$.

The mol% G + C of the DNA is: 41.2–44.8 (T_m).

Type strain: ATCC 13361.

GenBank accession number (16S rRNA): M75059.

7. ***Pasteurella langaa*** Muters, Ihm, Pohl, Frederiksen and Mannheim 1985a, 320^{VP}
lan' gaa. L. fem. n. *langaa* referring to the village of Langaa in Denmark.

Cells coccobacillary. Colonies similar to but smaller than those of *P. anatis*. Nonhemolytic. No V-factor requirement. Acid is produced from D-mannitol and lactose; tests for ornithine decarboxylation, indole formation, urease activity, acid from L-arabinose, D-xylose, raffinose, trehalose, maltose, and D-sorbitol are negative. Strains of the species are found in the respiratory tracts of apparently healthy chickens. The species belongs to 16S rRNA cluster 19. The molecular weight of genomic DNA ranges from $1.7\text{--}1.9 \times 10^9$.

The mol% G + C of the DNA is: 43.9–45.3 (T_m).

Type strain: ATCC 43328, NCTC 11411.

GenBank accession number (16S rRNA): M75053.

8. ***Pasteurella stomatis*** Muters, Ihm, Pohl, Frederiksen and Mannheim 1985a, 320^{VP}
sto.ma' tis. Gr. gen. n. *stoma* throat, of the throat.

Cells are coccoid to rod shaped with bipolar staining ends in the Gram stain. On blood containing solid media small colonies of about 1 mm in diameter with grayish or semitranslucent appearance are formed. Nonhemolytic.

Optimal growth temperature is 36°C. Strains fulfill the general criteria of the genus *Pasteurella sensu stricto*. Additionally they are V-factor independent, urease negative, and do not decarboxylate ornithine. No acid is produced from L-arabinose, D-xylose, raffinose, lactose, maltose, D-mannitol, D-sorbitol, or dulcitol. Indole positive. Occurs in the oral cavity or respiratory tract of dogs and cats. The species belongs to 16S rRNA cluster 12. The molecular weight of genomic DNA ranges from $1.5\text{--}1.6 \times 10^9$.

The mol% G + C of the DNA is: 40.4–43.5 (T_m).

Type strain: ATCC 43327, NCTC 11623.

GenBank accession number (16S rRNA): M75050.

9. ***Pasteurella volantium*** Muters, Piechulla, Hinz and Mannheim 1985b, 9^{VP}
vo.lan' ti.um. L. fem. gen. pl. n. *volantium* from fowl.

Cells coccoid. Occur singly or in short chains. Mesophilic, facultatively aerobic, and V-factor dependent. Non-hemolytic. On chocolate agar, colonies are smooth, convex, and may produce yellowish pigments. Phenotypic features are similar to *P. avium*, but acid production from maltose and D-mannitol is positive. Some strains decarboxylate ornithine, some strains produce acid from D-sorbitol. Strains are isolated from the wattles of domestic fowl, one strain has been isolated from human tongue (Kilian, 1976). The species belongs to 16S rRNA cluster 18. The molecular weight of genomic DNA ranges from $1.5\text{--}1.9 \times 10^9$.

The mol% G + C of the DNA is: 44–45 (T_m).

Type strain: ATCC 14385, NCTC 3438.

GenBank accession number (16S rRNA): M75070.

Other Organisms

1. *Pasteurella* species A (Muters, Piechulla, Hinz, and Mannheim 1985b)

Strains of the unnamed *Pasteurella* species A resemble a collection of heterogeneous strains formerly classified as *Haemophilus avium*. All strains have a V-factor requirement and produce acid from L-arabinose, but not from D-sorbitol; the ornithine decarboxylase test is negative. Strains are isolated from poultry and wild birds. The species belongs to 16S rRNA cluster 18. The molecular weight of genomic DNA ranges from $1.7\text{--}2.1 \times 10^9$.

The mol% G + C of the DNA is: 44–45.9 (T_m).

Deposited strain: IPDH 280, HIM 789-5.

GenBank accession number (16S rRNA): M75055.

2. *Pasteurella* species B (Muters, Piechulla, Hinz and Mannheim 1985b)

Cells are coccobacillary to rod shaped. Colonies are similar to *P. canis* or *P. stomatis*. Nonhemolytic. No V-factor dependency. Indole formation and ornithine decarboxylase tests are positive. Acid is produced from trehalose, maltose, D-xylose, and dulcitol. Acid is not produced from L-arabinose, D-mannitol, or D-sorbitol. Test for urease is negative. Strains are isolated from different animal hosts, but also from human dog-bite wounds or cat scratches. The species belongs to 16S rRNA cluster 12. The molecular weight of genomic DNA is 1.9×10^9 .

The mol% G + C of the DNA is: 38.9–40.0 (T_m).

Deposited strain: CCUG 19794, SSI P 683.

GenBank accession number (16S rRNA): M75052.

Species Incertae Sedis

1. ***Pasteurella aerogenes*** McAllister and Carter 1974, 920^{AL}
a.e.ro' gen.es. Gr. masc. n. *aer* air; L. v. *genere* to produce; M.L. adj. *aerogenes* gas-producing.

The species was excluded from the genus *Pasteurella* by DNA–DNA hybridization (Muters et al., 1985a) and 16S rRNA hybridization (De Ley et al., 1990). Original isolates obtained from pigs. The species belongs to 16S rRNA cluster 14. Other isolates associated with 16S rRNA cluster 5. The molecular weight of genomic DNA is $1.6\text{--}2.0 \times 10^9$.

The mol% G + C of the DNA is: 41.8 (T_m).

Type strain: ATCC 27883.

GenBank accession number (16S rRNA): M75048.

2. ***Pasteurella bettyae*** Sneath and Stevens 1990, 151^{VP}
be' tty.ae. M.L. gen. n. *bettyae* of Betty, to commemorate Elizabeth “Betty” O. King.

The species resembles the former CDC group HB-5 and was excluded genetically as well as by DNA–DNA hybridization data from the genus *sensu stricto* as by its phenotypic features, i.e., negative reactions for oxidase, no acid from D-galactose, fructose, D-mannose, or sucrose. Strains are obtained from human Bartholin gland abscesses and human finger infections. The species belongs to 16S rRNA cluster 9.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 23273, NCTC 10535.

GenBank accession number (16S rRNA): L06088.

3. *Pasteurella caballi* Schlater, Brenner, Steigerwalt, Moss, Lambert and Packer 1990, 320^{VP} (Effective publication: Schlater, Brenner, Steigerwalt, Moss, Lambert and Packer 1989, 2173.)

ca.ba'lli. Gr. n. gen. *cabilli* from a horse.

Strains of this taxon are different from true members of the genus by their aerogenic capacity and their negative catalase reaction. Some strains produce acid from *meso*-inositol and from L-rhamnose. Isolates are obtained from respiratory and genital tract infections in horses and also from humans who have had contact with horses. Cases of wound infections (Bisgaard et al., 1991a) and a wound infection after horse bite (Escande et al. 1997) have been reported. The species belongs to 16S rRNA cluster 9.

The mol% G + C of the DNA is: 41–42 (T_m).

Type strain: ATCC 49197.

GenBank accession number (16S rRNA): AF224291.

4. *Pasteurella granulomatis* Ribeiro, Carter, Frederiksen and Riet-Correa 1990, 105^{VP} (Effective publication: Ribeiro, Carter, Frederiksen and Riet-Correa 1989, 1402.)

gran.nu.lo'ma.tis. L. dim. n. *granulum* grain; Gr. suff. *oma* a swelling or tumor; M.L. gen. n. *granulomatis* of a granuloma.

The nonhemolytic species was transferred to the genus *Mannheimia* as *Mannheimia granulomatis* Angen, Mutters, Caugant, Olsen and Bisgaard 1999a (see chapter on *Mannheimia*).

The mol% G + C of the DNA is: 39.2 (T_m).

Type strain: 26, ATCC 49244.

5. *Pasteurella lymphangitidis* Sneath and Stevens 1990, 151^{VP} *lymph.ang.iti'dis*. M.L. gen. n. *lymphangitidis* pertaining to lymphangitis, inflammation of the lymph nodes.

DNA–DNA hybridization data exclude the former BL organisms isolated from lymphangitis in *Bos indicus* in Southern India from the genus *Pasteurella* (Mutters et al., 1985a). The biochemical characteristics of a negative oxidase reaction, lack of nitrate reduction, and no acid formation from D-galactose and D-mannose phenotypically exclude this group from the genus.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 49635, NCTC 10547.

6. *Pasteurella mairii* corrig. Sneath and Stevens 1990, 151^{VP} *mai'ri.i*. M.L. gen. n. *mairii* of Mair, after Nicholas S. Mair, who isolated the organism.

DNA hybridization data showed only low degrees of relatedness with *Pasteurella sensu stricto* as well as with other selected *Pasteurella*-named species (Mutters et al., 1985a). Strains are isolated from abortion in pigs and from septicemia in piglets. The species belongs to 16S rRNA cluster 14.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 49633, NCTC 10699.

GenBank accession number (16S rRNA): AF024532.

7. *Pasteurella pneumotropica* Jawetz 1950, 179^{AL}

pneu.mo.tro'pi.ca Gr. n. *pneumon* lung; Latinized Gr. adj. *tropicus* tropic, circle, M.L. fem. adj. *pneumotropica* having an affinity for the lungs.

Based on the molecular data, the two biovars Jawetz and Heyl resemble two distinct species outside the genus *Pasteurella*. The former Henriksen biovar has been reclassified as *Pasteurella dagmatis*. Overall, the *P. pneumotropica* complex remains heterogeneous, containing several unclassified new species (Ryll et al., 1991). The species belongs to 16S rRNA cluster 15. The molecular weight of genomic DNA is $1.5\text{--}1.6 \times 10^9$.

The mol% G + C of the DNA is: 40.3–42.8 (T_m).

Type strain: ATCC 35149, NCTC 8141.

GenBank accession number (16S rRNA): M75083.

8. *Pasteurella testudinis* Snipes and Biberstein 1982, 209^{VP} *tes.tu'di.nis*. L. n. *testudo* tortoise; L. gen. n. *testudinis* of the tortoise.

The species has been recovered from healthy Californian desert tortoises (*Gopherus agassizii*) and from tortoises with upper respiratory tract disease (Snipes and Biberstein, 1982; Snipes et al., 1995). The species belongs to 16S rRNA cluster 21. The molecular weight of genomic DNA is 1.6×10^9 .

The mol% G + C of the DNA is: 46.8–47.4 (T_m).

Type strain: ATCC 33688, UCD 90-23-79n.

GenBank accession number (16S rRNA): L06090.

9. *Pasteurella trehalosi* Sneath and Stevens 1990, 152^{VP} *tre.hal.osi*. M.L. gen. n. *trehalosum* pertaining to the carbohydrate trehalose.

The species includes the biovar T of the former *Pasteurella haemolytica*. Genotypic investigations have shown that the trehalose-positive biovar does not belong to the genus *Pasteurella* (Mutters et al., 1985a; De Ley et al., 1990; Dewhirst et al., 1993). Hemolysis on sheep blood agar and lack of acid formation from D(+) galactose also excludes the species from *Pasteurella*. Although these strains show no close affiliation to the genus *Pasteurella*, they were recently named *Pasteurella trehalosi* (Sneath and Stevens, 1990). The species belongs to 16S rRNA cluster 10. (The type strain has not been sequenced.) The molecular weight of genomic DNA is 1.8×10^9 .

The mol% G + C of the DNA is: 42.6 (T_m).

Type strain: S110, ATCC 29703, NCTC 10370.

Genus II. *Actinobacillus* Brumpt 1910, 849^{AL}

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Ac.ti.no.ba.cil'lus. Gr. n. *actis* a ray; L. dim. masc. n. *bacillus* a small staff or rod; M.L. masc. n. *Actinobacillus* ray bacillus or rod.

Cells, measuring $0.4 \pm 0.1 \times 1.0 \pm 0.4 \mu\text{m}$, are spherical, oval, or rod-shaped (Phillips, 1984). **Most often bacillary but sometimes interspersed with coccid elements that may lie at the pole of a larger form, producing the characteristic “Morse-code”**

form. Cell forms up to $6 \mu\text{m}$ in length may appear when grown on media containing glucose or maltose. Cells are single or arranged in pairs or, more rarely, in chains. Endospores are not formed. Gram negative, but staining is irregular. Not acid fast.

India ink may demonstrate small amounts of extracellular slime in wet preparations. **Nonmotile.** Organisms are aerobic, microaerobic, facultatively anaerobic, or chemoorganotrophic, having both respiratory and fermentative types of metabolism. After growth for 24 h on blood agar, translucent colonies, usually 1–2 mm in diameter appear. Surface colonies have low viability and may die in 2–7 days. Growth may be very sticky upon primary cultivation, making it difficult to remove colonies completely from the agar surface. Optimum growth temperature is 37°C. Temperature range for growth is 25–42°C.

Glucose is fermented, with production of acid but not gas. Inulin is not fermented, and inositol is fermented by *A. rossii* and infrequently by *A. porcinus* (Sneath and Stevens, 1985; Møller and Kilian, 1990; Møller et al., 1996a). Only *A. indolicus* is indole positive. **Nitrate reduction is positive** or variable (variable in *A. porcinus* [Møller et al., 1996a] and in *A. seminis* [van Tonder, 1979]). Actinobacilli are parasitic or commensal in humans, sheep, cattle, horses, pigs, other mammals, and birds. They are often commensals in the alimentary and genital tract and respiratory mucosa but can also occur as pathogens and cause a variety of lesions.

Genus *Actinobacillus* belongs to the family *Pasteurellaceae*. Virtually complete 16S rRNA sequencing has shown that this family is located in the *Gammaproteobacteria* (Dewhirst et al., 1992). Families closely related to the *Pasteurellaceae* are the *Enterobacteriaceae*, *Vibrionaceae*, and *Aeromonadaceae*. More distant relatives within the *Gammaproteobacteria* are the *Pseudomonas fluorescens* complex, the *Moraxellaceae*, and the *Cardiobacteriaceae*. DNA–rRNA hybridization found that the members of the *Pasteurellaceae* are most closely related to members of the *Enterobacteriaceae*, the *Vibrionaceae*, the *Aeromonadaceae*, and the genus *Alteromonas* (De Ley et al., 1990).

When 16S rRNA sequences were determined for over 70 strains of species in the family *Pasteurellaceae*, the strains were found to fall in seven clusters (Dewhirst et al., 1992, 1993). The first cluster, which is split into three subclusters, includes *Haemophilus sensu stricto*. This cluster contains the type strains of *H. influenzae*, *H. aegyptius*, *H. aphrophilus*, *H. haemolyticus*, *H. paraphrophilus*, *H. segnis*, and *A. actinomycetemcomitans*. It also contains *H. aphrophilus* NCTC 7901 and *A. actinomycetemcomitans* FDC Y4, ATCC 29522, ATCC 29523, and ATCC 29524. Cluster 2, which splits into two subclusters, contains *P. mairii* CCUG 27189, Bisgaard taxon 6 CCUG 15568, and the type strains of *A. seminis* and *P. aerogenes*. Cluster 3, including the genus *Pasteurella sensu stricto*, contains the type strains of *P. multocida*, *P. anatis*, *P. avium*, *P. canis*, *P. dagmatis*, *P. gallinarum*, *P. langaa*, *P. stomatis*, *P. volantium*, *P. trehalosi*, *H. haemoglobinophilus*, *H. parasuis*, *H. paracuniculus*, *H. paragallinarum*, “*A. salpingitidis*”, and *A. capsulatus*. It also comprises Bisgaard taxa 2, 3, 7, and 13, *Pasteurella* species A CCUG 18782, *Pasteurella* species B CCUG 19974, *Haemophilus* taxon C CAPM 5111, *H. parasuis* type 5 Nagasaki, and *P. volantium* NCTC 4101. Four subclusters exist in Cluster 3. The species belonging to *Pasteurella sensu stricto* fall in three of the four disconnected clusters. Cluster 4 contains the genus *Actinobacillus sensu stricto*, which includes the type strains of *A. lignieresii*, *A. equuli*, *A. pleuropneumoniae*, *A. suis*, *A. ureae*, *A. hominis*, *H. paraaerolyticus*, *H. paraphrohaemolyticus*, *H. ducreyi*, and *M. haemolytica*. This cluster also comprises *Actinobacillus* species strain CCUG 19799 (Bisgaard Taxon 11), *A. suis* ATCC 15557, *Haemophilus* “minor group” strain 202, *P. bettyae*, and Bisgaard taxa 5, 8, and 9 (*A. arthritidis* and genomospecies 2). Cluster 4 falls in two subclusters. Each species in these subclusters branch slightly deeper than do those in the cluster above, giving no logical point at which to end the genus

Actinobacillus sensu stricto. The type strains of *P. pneumotropica* and *A. muris* form a fifth group. The sixth group is constituted by the type strain of *H. parainfluenzae* and by *H. parainfluenzae* ATCC 29242. Two strains of Bisgaard Taxon 14 and the type strain of *P. testudinis* form a seventh cluster.

The mol% G + C of the DNA is: 35.5–46.9.

Type species: *Actinobacillus lignieresii* Brumfit 1910, 849.

FURTHER DESCRIPTIVE INFORMATION

Cell morphology Actinobacilli show marked pleomorphism. Bacillary forms predominate, but coccobacilli and longer filaments can also be seen. The latter can break up into short bacillary forms or granules, which may give an appearance of streptobacilli or streptococcal chains.

Capsules are produced by most species. Extracellular slime can be seen especially in *A. equuli* and *A. suis*.

Cell constituents

FATTY ACIDS Cellular fatty acid patterns of established and newly described taxa in the family *Pasteurellaceae*, as determined by capillary gas chromatography, have been found to be homogenous with minor variations (Mutters et al., 1993). In this case, discrimination of species or groups within the family could not be achieved. Separation from the *Neisseriaceae* and *Moraxella* was possible. All *Pasteurellaceae* strains investigated show proportions of >15% for unbranched tetradecanoic acid C_{14:0}, hexadecanoic acids C_{16:1}, and C_{16:1 cis 9}, and moderate amounts of 3-hydroxy-dodecanoic acid C_{12:0 3OH}, tetradecanoic acid C_{14:5}, pentadecanoic acid C_{15:0}, octadecanoic acid C_{18:0}, eikosoic acids C_{20:0} and C_{20:4 cis 5,8,11}. In contrast to this, however, Boot et al. (1999) have found that principal component analysis of fatty acid data obtained under standardized growth conditions may discriminate among *Pasteurellaceae* species.

By using a column with a nonpolar stationary phase, it is possible to analyze free fatty acids from bacterial cells directly in the gas chromatograph without derivatization (Brondz and Olsen, 1983; Brondz et al., 1983). While *A. actinomycetemcomitans* strains can be divided into three groups based on their free fatty acid content, *H. aphrophilus* is homogeneous. However, the distribution of bound cellular fatty acids from whole cells is not so specific as to allow discrimination between *A. actinomycetemcomitans* and *H. aphrophilus* (Calhoon et al., 1981; Jantzen et al., 1981; Brondz and Olsen, 1984b).

The lipid content of lipopolysaccharide (LPS) from *A. actinomycetemcomitans* constitutes 35.4%; in *H. aphrophilus* it is only 18.4% (Brondz and Olsen, 1989).

3-Hydroxytetradecanoic acid and tetradecanoic acid, which are the sole LPS acids, make up 21.1% and 14.3%, respectively, of fatty acids in *A. actinomycetemcomitans*, and 10.9% and 7.5% in *H. aphrophilus*. Variation in the lipid content of LPS acids may explain in part the difference in periodontopathogenic potential between *A. actinomycetemcomitans* and *H. aphrophilus*.

When fatty acids of isolated outer membrane vesicles in species of *Actinobacillus*, *Haemophilus*, and *Pasteurella* were analyzed by gas chromatography and the data treated with multivariate analysis, it was found that *A. actinomycetemcomitans* can be assigned to one group (class 1) and *H. influenzae* to another (class 2) (Olsen et al., 1995). *H. aphrophilus*, *H. paraphrophilus*, and *P. multocida* are distinct from these classes, with the exception of the type strain of *H. aphrophilus*, which falls in class 1 (borderline). The abundance of 3-OH fatty acids indicates that the outer membrane vesicles are rich in LPS. Multivariate analysis of fatty acid

data from such vesicles can distinguish two groups of *A. actinomycetemcomitans* (Brondz et al., 1995). One group contains strains ATCC 33384^T, ATCC 29522, SUNY 366, HK 435, and Q 1247; the other group comprises ATCC 29524, ATCC 29523, FDC 2112, FDC 511, and SUNY 489.

PHOSPHOLIPIDS The distribution of phospholipids, as determined by thin layer chromatography, is uniform within the family *Pasteurellaceae* (Mutters et al., 1993). Phosphatidylethanolamine has been observed in major amounts, phosphatidylglycerine in smaller amounts, and lysophosphatidylethanolamine in only petty amounts.

LIPOQUINES The lipoquine content can be used to discriminate between groups within the family *Pasteurellaceae*, but it does not necessarily reflect the degree of genomic relatedness (Mutters et al., 1993). *Actinobacillus* exhibits high amounts of menaquinone-7 (MK-7) (30%) and demethylmenaquinone-7 (DMK-7) (54%). Representative strains of *Haemophilus* contains mostly DMK-6, with smaller amounts (~10% each) of MK-8 and DMK-7. A group consisting of *A. actinomycetemcomitans*, *H. aphrophilus*, *H. segnis*, and *P. bettyae* has a profile similar to that of *Haemophilus*, but does not contain MK-8. Another group comprising *M. haemolytica* and Bisgaard Taxon 3 has a lipoprotein pattern between those of *Actinobacillus* and *Haemophilus*, whereas the patterns of "*Histophilus*", "*A. salpingitidis*", and *A. rossii* are completely different.

Of 16 strains of members of the *Pasteurellaceae* studied by high-performance liquid chromatography, 12 have been found to produce menaquinones, mostly as minor components, in addition to demethylmenaquinones (Kroppenstedt and Mannheim, 1989). These organisms represent the genus *Actinobacillus*, the genus *Pasteurella sensu stricto*, the *Mannheimia haemolytica* complex, *H. paragallinarum*, *H. parasuis*, *H. aphrophilus*, *P. aerogenes*, and Bisgaard Taxon 14. *P. pneumotropica* biotype Jawetz, members of the genus *Haemophilus sensu stricto*, and *A. actinomycetemcomitans* do not contain menaquinones. The view that the family *Pasteurellaceae* can be differentiated from the *Enterobacteriaceae* and other fermenting Gram-negative bacteria by a lack of menaquinones therefore cannot be maintained.

CARBOHYDRATES There is a common pattern of cellular sugars among all members of the family *Pasteurellaceae*, and characteristic carbohydrate profiles discriminate between groups, often to the species level (Mutters et al., 1993). The sugar patterns distinguish, for example, *P. stomatis* from Bisgaard's Taxon 16, mannose-positive *A. hominis* strains from *A. equuli*, and *A. actinomycetemcomitans* from *H. aphrophilus*, *H. segnis*, and intermediate strains. Sugar profiles, established by gas chromatography of whole cells after methanolysis and derivatization with trifluoroacetic acid anhydride have demonstrated that *H. aphrophilus* differs from *A. actinomycetemcomitans*, *A. ureae*, *H. paraphrophilus*, *H. influenzae* type b, *M. haemolytica*, and *P. multocida* by its lack of D-glycero-D-mannoheptose (Brondz and Olsen, 1984d, 1985a). Whole defatted cells, which are intermediates between whole cells and isolated cell membranes, give the same sugar profiles as whole cells, reaffirming that D-glycero-D-mannoheptose is present in *A. actinomycetemcomitans* and absent in *H. aphrophilus* (Brondz and Olsen, 1984a). Further, multivariate analysis of quantitative chemical and enzymatic characterization data has demonstrated that D-glycero-D-mannoheptose is a true chemotaxonomic marker for these organisms (Brondz et al., 1990). LPS is the primary source of D-glycero-D-mannoheptose in *A. actino-*

mycetemcomitans and *H. paraphrophilus* (Brondz and Olsen, 1984c, 1985b). Whereas *A. actinomycetemcomitans* LPS contains 7.8% D-glycero-D-mannoheptose and 11.3% L-glycero-D-mannoheptose, *H. aphrophilus* LPS contains 17.4% L-glycero-D-mannoheptose (Brondz and Olsen, 1989).

PROTEINS One-dimensional SDS-PAGE of whole cell proteins has been used to characterize 39 strains representing all biovars established within the taxa 2 and 3 complex of Bisgaard and 2 strains belonging to the avian *M. haemolytica*-"*A. salpingitidis*" complex (Bisgaard et al., 1993). There is no correlation between results obtained from protein profiling and those previously established by DNA-DNA hybridization. Comparison of phena defined by protein profiling with species/groups established by DNA-DNA hybridization, chemotyping, and biotyping has shown that the best correlation exists between DNA-DNA hybridization and biotyping.

Both one- and two-dimensional electrophoresis of cellular proteins separate *A. actinomycetemcomitans* and *H. aphrophilus* (Calhoun et al., 1981; Jellum et al., 1984). The high resolution of the 2-D electrophoresis distinguishes strains within species, in contrast to 1-D electrophoresis. The 2-D whole-cell protein profiles of *A. actinomycetemcomitans*, *A. ureae*, *H. paraphrophilus*, *H. aphrophilus*, *H. influenzae*, *P. multocida*, and *M. haemolytica* have also been compared (Olsen et al., 1987). The protein patterns of the genera *Actinobacillus*, *Haemophilus*, and *Pasteurella* are quite distinct. *A. actinomycetemcomitans* can easily be distinguished from the other species tested, including *A. ureae*, *H. aphrophilus*, *H. paraphrophilus*, and *M. haemolytica*. The protein patterns of *P. multocida* and *M. haemolytica* also differ. However, it is difficult to distinguish strains of *H. paraphrophilus* and *H. aphrophilus*.

When outer membrane proteins of *A. actinomycetemcomitans* and *H. aphrophilus* strains were compared by means of SDS-PAGE and immunoblots with rabbit antiserum, the most prominent feature was found to be the absence of a heat-modifiable protein in *H. aphrophilus* (Bolstad et al., 1990). Another study has demonstrated a remarkable degree of homology between the outer membrane proteins of *A. actinomycetemcomitans* and *H. aphrophilus* (DiRienzo and Spieler, 1983). In addition, the principal heat-modifiable outer membrane protein of *A. actinomycetemcomitans* shows N-terminal sequence homology and immunologic cross reactivity with the OmpA proteins of other Gram-negative bacteria (Wilson, 1991).

POLYAMINES Analysis of polyamines is a promising method for discriminating members of the family *Pasteurellaceae* (Busse et al., 1997). Members of the genus *Actinobacillus sensu stricto* contain 1,3-diaminopropane as the predominant compound. In addition, in most of the species of the genus *Haemophilus*, 1,3-diaminopropane is the major substance. In contrast, however, "*H. intermedius* subsp. *gazogenes*" and *H. parainfluenzae* contain high levels of 1,3-diaminopropane, cadaverine, and putrescine. *H. parainfluenzae* is phylogenetically only distantly related to the type species of the genus *Haemophilus* (Dewhirst et al., 1993). The phylogenetic diversity of *Pasteurella* is also to some extent reflected by different polyamine patterns.

ENZYMES Multivariate analysis of data (API-ZYM micro-method) from 19 enzyme activities in the outer membrane vesicles/fragments from clinically important species of the *Actinobacillus*-*Haemophilus*-*Pasteurella* group has enabled distinction among *A. actinomycetemcomitans*, *H. aphrophilus*, and *H. paraphrophilus* (Myhrvold et al., 1992). *A. actinomycetemcomitans* is divided

into two strain groups (classes). *A. lignieresii* falls outside or on the borderline of the *A. actinomycetemcomitans* class. *A. ureae*, *H. influenzae*, *H. parainfluenzae*, *M. haemolytica*, and *P. multocida* appear outside the *A. actinomycetemcomitans*, *H. aphrophilus*, and *H. paraphrophilus* classes. The API-ZYM method also distinguishes whole cells of *A. actinomycetemcomitans*, *A. lignieresii*, and *H. aphrophilus* (Slots, 1981).

The genetic diversity of 52 field isolates, mainly from Australia, and 15 reference strains of *A. pleuropneumoniae* has been examined with multilocus enzyme electrophoresis (MEE) (Hampson et al., 1993), and 33 electrophoretic types (ETs), with a mean genetic diversity per locus of 0.312, were found. Australian strains of serovars 1, 2, 5, and 7 belong to the same clonal lines as strains of these serovars from other countries. Distinct clones of serovars 3, 7, 11, and 12 are also detected. The type strains of serovars 1, 9, and 11 are placed in the same ET, and strains of *A. pleuropneumoniae* biovar 2 are closely related to biovar 1 strains.

MEE has also been used for clonal analysis of the *A. pleuropneumoniae* population in a geographically restricted Danish area (Møller et al., 1992). Sixty-six percent of 250 isolates from lungs of pigs with pleuropneumonia and from tonsils of apparently healthy pigs at slaughter belong to three ETs, out of a total of 37 ETs detected. While five biotype 2 isolates constitute a separate ET closely related to biotype 1 isolates, the type strain of the species constitutes its own ET, with a genetic distance of 0.30 from its nearest neighbor.

Genetic relationships among isolates assigned to *A. actinomycetemcomitans*, *H. aphrophilus*, and *H. paraphrophilus* can be examined based on electrophoretically demonstrable allelic variation in 14 structural genes coding for metabolic enzymes (Caugant et al., 1990). Among the 51 isolates that have been analyzed, 25 ETs are detected, with a genetic diversity of 0.753 per locus. Eleven ETs are seen with cluster analysis; these represent the genotypes of all 17 isolates assigned to *A. actinomycetemcomitans*. The remaining 14 ETs represent the genotypes of 34 isolates of *H. aphrophilus* and *H. paraphrophilus*. All strains of *H. aphrophilus* are closely related, except for ATCC 13252. Strains assigned to *H. paraphrophilus* include distantly related lineages, some of which are similar to *H. aphrophilus*. There is no significant overall genetic similarity between *A. actinomycetemcomitans* and the *Haemophilus* species examined.

Bacteriolysis The ability of ethylenediaminetetraacetic acid (EDTA) and hen egg white lysozyme to induce bacteriolysis has been tested in major clinical species of the *Actinobacillus-Haemophilus-Pasteurella* group (Olsen and Brondz, 1985). *A. actinomycetemcomitans* lyses more easily than does *H. aphrophilus*. The former can be divided into two strain groups according to lysis patterns, while *H. aphrophilus* is homogenous. In one group of *A. actinomycetemcomitans* (group I), EDTA has a considerable lytic effect, which is not increased by supplementation with lysozyme. In the other group of *A. actinomycetemcomitans* (group II), the lytic effect of EDTA is much less, but lysozyme has a considerable supplementary lytic effect. Maximal lysis of *A. actinomycetemcomitans* (ATCC 29522) occurs at pH 8.0 in the presence of EDTA and at pH 7.6 with EDTA-lysozyme. In *H. aphrophilus* (ATCC 33389^T) maximal lysis occurs at pH 9.0 with EDTA and at pH 9.2 with EDTA-lysozyme. When *H. influenzae*, *H. paraphrophilus*, *P. multocida*, *M. haemolytica*, and *A. ureae* were similarly tested, the group I strains of *A. actinomycetemcomitans* were found to be the most rapidly lysed by EDTA. *H. paraphrophilus* is the least sensitive of the Gram-negative organisms examined, but it is less resistant

than *Micrococcus luteus*. The latter organism is the most sensitive to lysozyme, followed by *A. ureae* and the group II strains of *A. actinomycetemcomitans*. The group I strains of *A. actinomycetemcomitans*, *H. paraphrophilus*, and *M. haemolytica* are the least sensitive.

Multivariate analysis of 41 quantitative variables, including data from lysis kinetics after exposure to both EDTA and EDTA-lysozyme, cell sugar and fatty acid contents, methylene blue reduction, and API ZYM enzymatic assessment of whole cells and outer membrane vesicles/fragments, divides *A. actinomycetemcomitans* strains into two groups (Brondz and Olsen, 1993). One group contains ATCC 33384^T, ATCC 29522, FDC 2112, and FDC 2043; the other group comprises ATCC 29524, ATCC 29523, FDC 2097, FDC 511, and FDC Y4. Both groups are distinct from members of the genera *Haemophilus* and *Pasteurella*.

Cultural and biochemical characteristics The sticky properties of actinobacilli cultures when grown on nutrient agar or blood are marked with some species, but may be lost on subcultivation. The same stickiness may be revealed when cultures are grown in liquid media, especially those containing glucose. The increased viscosity is especially seen with *A. equuli* and to a lesser extent with *A. suis*. Stickiness can be demonstrated by drawing a strand of sticky material between the surface of the broth and the inoculating loop.

A few species exhibit hemolytic activity when cultured on blood agar. Evidence has been given for hemolytic *A. lignieresii* in horses (Samitz and Biberstein, 1991). A narrow zone of hemolysis may occasionally surround *A. minor* colonies when they are tested for the Christie Atkins and Munch-Petersen (CAMP) phenomenon. Usually, *A. pleuropneumoniae* shows complete hemolysis on ovine and bovine blood agar, but occasionally, biovar 1 isolates may demonstrate incomplete or no hemolysis. However, the CAMP-reaction is always positive (Møller and Kilian, 1990). A zone of complete hemolysis always surrounds *A. suis* when it is cultured on ovine or bovine blood agar. In contrast, only partial hemolysis is observed when *A. suis* is grown on horse blood agar. Hemolytic *A. suis*-like organisms have been isolated from the oral cavity of horses (Bisgaard et al., 1984; Samitz and Biberstein, 1991).

Pigmented colonies are not produced, except in the case of *A. suis*, which reveals a creamy yellow color when its broth-grown cultures are centrifuged and the pellet resuspended in saline (Kim et al., 1976).

Usually, it takes 24–48 h for fermentation of carbohydrates and alcohols to occur, but some species or strains may need up to 7 days. Acid, but no gas is formed. Some substrates are fermented rather slowly. *A. succinogenes* accumulates high concentrations of succinic acid (Guettler et al., 1999).

A small number of *A. equuli* strains are gelatinase positive when tested by the gelatine plate method of Frazier (1926) (Fredriksen, 1973; Vallée et al., 1974).

Genome size/pulsed-field gel electrophoresis The genome size of the type strain (serotype 1) of *A. pleuropneumoniae*, as determined by pulsed-field gel electrophoresis (PFGE) of *Asd* and *ApaI* digested chromosomal DNA, was 2404 ± 40 kb (Chevallier et al., 1998). The chromosome sizes for the reference strains of the other serotypes ranged between 2.3 and 2.4 Mb. Different PFGE patterns demonstrate heterogeneity of the chromosomal structure among different field strains of serotype 1, 5a, and 5b, while strains of serotype 9 show homogenous *ApaI* patterns. PFGE suggests that strains of *A. pleuropneumoniae* se-

rototype 2 obtained from Europe and North America are clonally related.

When *A. equuli* strains from 174 horses were examined by PFGE and restriction enzyme electrophoresis, a high degree of strain variability was found within each horse population (10 farms) and some variability over time was found between strains isolated from the same horse (Sternberg, 1998).

Valcarcel et al. (1997), using PFGE, found that the genome of *A. actinomycetemcomitans* contains a single chromosome of 2.3 Mb, estimated from the sum of restriction fragments generated with rare cutting endonucleases. Large plasmids with sizes ranging from 35–300 kb have been detected, and extrachromosomal elements constitute over 20% of the total genome in some strains.

16S rRNA data/phylogeny Phylogeny of the family *Pasteurellaceae* has been determined by comparison of 16S rRNA sequences (Dewhirst et al., 1992, 1993). The family has been divided into seven clusters, of which the fourth cluster contains the type strains of *A. lignieresii*, *A. equuli*, *A. pleuropneumoniae*, *A. suis*, *A. ureae*, *H. parahaemolyticus*, *H. parainfluenzae*, *H. paraphrohaemolyticus*, *H. ducreyi*, and *M. haemolytica*. This cluster also includes *Actinobacillus* sp. strain CCUG 19799 (Bisgaard Taxon 11), *A. suis* ATCC 15557, *H. ducreyi* ATCC 27722 and HD 3500, *Haemophilus* "minor group" strain 202, and *H. parainfluenzae* ATCC 29242. *A. hominis* is closely related to *Actinobacillus sensu stricto* species in Cluster 4A, while Bisgaard taxa 5, 8, 9 (*A. arthritidis* and genomospecies 2), and *P. bettyae* fall in Cluster 4B. The type strain of *A. actinomycetemcomitans* falls in the first cluster and also contains *A. actinomycetemcomitans* FDC Y4, ATCC 29522, ATCC 29523, and ATCC 29524, and *H. aphrophilus* NCTC 7901. The type strain of *A. seminis* is included in the second cluster, and the type strains of *A. capsulatus* and "*A. salpingitidis*" in the third cluster. *A. muris* is related to *P. pneumotropica*, and both are found in Cluster 5. The branching is extremely complex. For further information regarding these clusters, see genus definition at the head of this chapter.

The restriction pattern profiles of *Asd*-, *Apa*I-, and *Nhe*I-digested chromosomes of different serotypes of *A. pleuropneumoniae* demonstrate a high degree of polymorphism (Chevallier et al., 1998). Analysis of the macrorestriction pattern polymorphism has provided a determination of phylogenetic relationships between the serotype reference strains. These relationships reflect to some extent groups of serotypes known to cross-react serologically.

Fussing et al. (1998b) has characterized and determined intraspecies relatedness of *A. pleuropneumoniae*, as well as its relationship to *A. lignieresii* and *A. suis*, by sequence analysis of the ribosomal operon. Amplification and sequence analysis of the 16S–23S rDNA ribosomal intergenic sequence (RIS) from the three species gives two RISs differing by about 100 bp. The larger RISs differ among the three species. A species-specific area has been found for *in situ* probe detection of *A. pleuropneumoniae*.

DNA–DNA hybridization DNA–DNA hybridization has been used extensively by the Mannheim group to determine the genetic relationships within and among genera of the family *Pasteurellaceae* (e.g., Mannheim and Pohl, 1980; Mannheim et al., 1980; Mannheim, 1983). DNA–DNA hybridization demonstrates the relative closeness of *Actinobacillus* and *Pasteurella* (Mannheim, 1983). Recently a micro-well format DNA hybridization method has been developed and validated on 23 strains belonging to *Actinobacillus* species, avian [*Pasteurella*] *haemolytica*-like bacteria

and *Mannheimia* species (Christensen et al., 2000). Based on DNA homology studies, Escande et al. (1984) have suggested that the genus *Actinobacillus* be limited to *A. lignieresii*, *A. suis*, *A. capsulatus*, an unnamed equine *Actinobacillus* species (strain CCM 5500), and the generically misnamed *Pasteurella ureae*. *A. actinomycetemcomitans*, "*A. salpingitidis*", and *A. seminis* have been suggested to not belong in the genus.

The degree of genetic relatedness of *A. pleuropneumoniae* to selected members of the family *Pasteurellaceae* has been determined by free solution DNA–DNA hybridization (Borr et al., 1991). Representative strains of all 12 serotypes of *A. pleuropneumoniae* form a homogenous group exhibiting 74–90% sequence homology with *A. pleuropneumoniae* serotype 1. All serotypes show a high degree of genetic relatedness to *A. lignieresii*. Little homology has been demonstrated between *A. pleuropneumoniae* strains and selected *Haemophilus* and *Pasteurella* species.

A. pleuropneumoniae serotype reference strains and 204 field strains representing all 12 serotypes and biovars 1 and 2 have been analyzed for the presence of a number of toxin genes by DNA–DNA hybridization with specific gene probes (Beck et al., 1994). The patterns of *apx* genes and those of the expressed Apx toxins in biovar 1 field strains are the same as the patterns in the corresponding serotype reference strain. Nearly all strains expressed their *apx* genes and produced the same Apx toxins as their serotype reference strain.

The DNA homology between serovars (1–13) of *M. haemolytica* and other species (*P. multocida* and *actinobacilli*) has been found to be low (Bingham et al., 1990), suggesting essentially no genetic relationship of *M. haemolytica* with the ATCC reference strains of *Pasteurella* or *Actinobacillus*.

"*H. somnus*" DNA shows 43%, 46%, and 58% homology with the DNA from *A. lignieresii*, *H. influenzae*, and *H. parainfluenzae*, respectively (Gonzales and Bingham, 1983).

Based on DNA relatedness Coykendall et al. (1983) have suggested that *A. actinomycetemcomitans* is distinct from *H. aphrophilus* and *H. paraphrophilus*. Furthermore, DNA–DNA hybridization has been used by Tønnum et al. (1990) to identify clinical isolates as *A. actinomycetemcomitans* and *H. aphrophilus*. The results obtained with hybridization are supported by interstrain-to-intra-strain genetic transformation ratios. Distinction between *A. actinomycetemcomitans* and *H. aphrophilus* is generally clear-cut with respect to both hybridization and genetic transformation. Distinction between these two species has also been obtained with DNA–DNA hybridization using total genomic DNA probes (Tønnum et al., 1989).

DNA–rRNA hybridization DNA–rRNA hybridization studies demonstrate at least seven rRNA branches in the family *Pasteurellaceae* (De Ley et al., 1990). The *A. actinomycetemcomitans*, *H. aphrophilus*, *H. ducreyi*, and "*Histophilus ovis*" rRNA branches are separate and quite remote from the three authentic genera of the family. DNA–rRNA hybridization with labeled probes is therefore an excellent method to decide whether an organism belongs to the family *Pasteurellaceae* or not.

When chromosomal DNA from *A. actinomycetemcomitans* is digested with the restriction endonucleases *Cla*I, *Bam*HI, *Bgl*II, or *Hind*III and hybridized to the *rrnB* ribosomal RNA operon of the *Escherichia coli* chromosome, isolates from the same individual and, belonging to the same serotype are found to be genetically identical, but those belonging to different serotypes are non-identical (Saarela et al., 1993). Isolates of the same or different serotypes are genetically nonidentical in different individuals.

Typing systems

BIOCHEMICAL TYPING The possibility of using biochemical differences among strains of a given serotype as epidemiological markers has been examined in *A. pleuropneumoniae*. Sirois and Higgins (1991) applied 38 biochemical and physiological tests to distinguish 67 strains belonging to serotype 1 and 5. Three fermentation tests have allowed the classification of serotype 1 strains into six phenotypic groups and serotype 5 strains into four of these groups. Groups II and III contain exclusively serotype 1 strains, whereas the majority of strains in groups I and IV belongs to serotype 1 and 5, respectively. The latter contains almost all the serotype 5 strains studied.

SEROTYPING Phillips (1967) has conducted serological and antigenic structure typing of *A. lignieresii*. Serological characterization of *A. pleuropneumoniae* in Australian pigs from 1993–1996 demonstrates that the dominant serovars are 12, 1, and 7 (Blackall et al., 1999a). In pigs from Quebec, serotype 1 is predominant, while serotype 5 is second in prevalence (Mittal et al., 1992). Both subtypes of serotype 5 (5a and 5b) are present in Quebec. Serotypes 3, 6, 7, 8, 10, and 12 have been isolated in small numbers, whereas serotypes 4, 9, and 11 are not present. Quantitation of the type- and group-specific antigens by the coagglutination test distinguishes serotype 1 and 9 strains from those of serotype 11 (Mittal et al., 1993).

Long-chain lipopolysaccharides have been used for the detection of antibodies in animals infected with *A. pleuropneumoniae* serotypes 4 and 7 (Gottschalk et al., 1997).

Using a monoclonal antibody-based polystyrene agglutination test, Dubreuil et al. (1996) have identified serotype 5a and 5b strains of *A. pleuropneumoniae*. Mittal et al. (1988b) have found serotype 6 strains of *A. pleuropneumoniae* to be antigenically closer to serotype 8 than to serotypes 3 or 5. A combination of serological tests, such as coagglutination followed by 2-mercaptoethanol tube agglutination, and immunodiffusion tests distinguish serotype 6 strains from those of other cross-reacting serotypes. Serotype-3 strains of *A. pleuropneumoniae* possess epitopes shared with almost all other serotypes (Mittal et al., 1988a).

Serotype-specific rabbit antisera against *A. actinomycetemcomitans* have been used to serotype 177 isolates of this organism from 136 periodontally healthy or diseased subjects by indirect immunofluorescence and/or immunodiffusion assays (Asikainen et al., 1991). Serotype b is dominant in subjects with periodontal disease, and serotype c is the most common serotype in healthy subjects. Most subjects are infected with only one serotype, and *A. actinomycetemcomitans* infections are relatively serotype-stable. Nonserotypeable *A. actinomycetemcomitans* isolates may originate from serotypeable isolates, especially those of serotype c (Paju et al., 1998).

Sensitized latex particles (SLP) for detection of capsular polysaccharides have been applied to the typing and diagnosis of *A. pleuropneumoniae* (Inzana, 1995). The SLP are agglutinated by all strains of the homologous serotype, but not by heterologous serotypes or strains of *A. suis*, *P. multocida*, or *H. parasuis*.

RESTRICTION MAPPING Minor differences in the cleavage patterns of 17 serotype 7 strains of *A. pleuropneumoniae*, detectable by restriction endonuclease analysis (REA), enable grouping of these strains into seven profiles (Wards et al., 1998).

DNA fingerprinting has also been used to differentiate 12 reference strains, each representing one serotype, and 11 field strains of *A. pleuropneumoniae* belonging to serotype 9, which is

the most frequent *Actinobacillus* species in the Czech Republic (Rychlik et al., 1994). Nine of the reference strains can be distinguished from each other and from serotypes 1, 9, and 11. A close relationship exists between the latter three. Among the 12 serotype 9 strains, four DNA types can be identified. REA is also powerful for typing and discerning genetic heterogeneity and homogeneity among *A. actinomycetemcomitans* strains (Han et al., 1991).

Genomic DNA fingerprinting by restriction-fragment-end-labeling classifies 31 strains of *A. actinomycetemcomitans* into 29 distinct types based on polymorphism (van Steenberg et al., 1995). Serotype-specific fragments can be assigned for the three serotypes investigated.

Restriction fragment-length polymorphism (RFLP) has been used to study the epidemiology and pathogenesis of *A. actinomycetemcomitans* (Di Rienzo and Slots, 1990). RFLP type B of *A. actinomycetemcomitans*, corresponding to reference strain JP2, seems to be particularly virulent, as indicated by the presence of this type in three subjects who converted from a healthy periodontal state to localized juvenile periodontitis.

RIBOTYPING Ribotyping is more efficient than biochemical fingerprinting in detecting differences between isolates of *A. equuli* from healthy and diseased horses (Sternberg and Brandstrom, 1999). There is a poor correlation between the methods. In *A. pleuropneumoniae*, ribotyping has been evaluated as an epidemiological tool and applied to study infections caused by this organism in Danish pig herds (Fussing et al., 1998a). This assessment was based on 13 reference strains and 106 epidemiologically unrelated field strains representing the nine serotypes of biotype 1 and serotype 14 of biotype 2. The enzymes *CfoI* and *HindIII* were used to provide ribotype patterns. Ribotyping of the reference strains resulted in 10 *CfoI* types and 11 *HindIII* types. From the Danish herd strains, 17 different *CfoI* ribotypes and 24 *HindIII* ribotypes were established. Combining *HindIII*- and *CfoI*-ribotyping divides the herd strains into 26 different types.

The genomic relationships among 112 *A. pleuropneumoniae* serotype 2 strains collected throughout Europe and North America have been examined by Fussing (1998). *HindIII* ribotyping of the strains produced five ribotypes with high similarity. Sequence analysis of the ribosomal intergenic region of strains representing each ribotype and each country demonstrates no differences. PFGE patterns of strains representing all countries show high similarity, suggesting that strains of *A. pleuropneumoniae* serotype 2 are clonal.

Six different typing methods have been compared for *A. actinomycetemcomitans* (van Steenberg et al., 1994). Ribotyping provides unequivocal distinction among different strains, as compared to restriction endonuclease results. Serotyping, biotyping, and SDS-PAGE patterns of outer membrane proteins are reproducible, but have a low discriminatory potential. Ribotyping shows intrafamilial similarity among *A. actinomycetemcomitans* isolates (Alaluusua et al., 1993) and supports separation of *A. actinomycetemcomitans* from *H. aphrophilus* and *H. paraphrophilus* (Sedláček et al., 1993).

PCR-ribotyping, repetitive extragenic palindromic element (REP)-based PCR and enterobacterial repetitive intergenic consensus (ERIC)-based PCR have been used for strain differentiation in epidemiological studies of *A. seminis* (Appuhamy et al., 1998). PCR-ribotyping gives the simplest pattern and is useful in differentiating *A. seminis* strains and distinguishing this species from related species.

REA OF PCR-AMPLIFIED 16S RRNA GENES *A. actinomycetemcomitans*, *H. aphrophilus*, and *H. paraphrophilus* can be rapidly distinguished by REA of PCR-amplified 16S rRNA genes (Riggio and Lennon, 1997).

MULTIPLEX PCR A multiplex PCR assay has been developed for detection of *A. pleuropneumoniae* and identification of serotype 5 isolates (Lo et al., 1998). DNA sequences specific to the conserved export and serotype-specific biosynthesis regions of the capsular polysaccharide of *A. pleuropneumoniae* serotype 5 are applied as primers to amplify 0.7 and 1.1 kb DNA fragments, respectively. The multiplex PCR assay rapidly detects *A. pleuropneumoniae* and distinguishes serotype 5 strains from other serotypes.

AP-PCR Hennessy et al. (1993) have recommended the use of M13 and T3–T7 oligodeoxynucleotide primers for diagnostic and epidemiological identification of *A. pleuropneumoniae* by arbitrarily primed (AP) PCR techniques. In *A. actinomycetemcomitans*, the isolates of a given AP-PCR genotype usually belong to the same serotype (Asikainen et al., 1995).

TOXIN TYPING A PCR method that has been developed for toxin typing of *A. pleuropneumoniae* is useful in differentiating strains for diagnostics and epidemiology and in detecting serotypes with atypical toxin patterns (Frey et al., 1995, 1996).

Antigenic structure Actinobacilli have a complex antigenic structure. Heat-stable somatic antigens and heat-labile surface antigens have been demonstrated in *A. lignieresii*, *A. equuli*, *A. pleuropneumoniae*, and *A. suis* (Phillips, 1967; Mráz, 1969a; Kim, 1976; Nielsen, 1990). Species may be divided into different antigenic types. Six antigenic types, 1–6, have been detected in *A. lignieresii*, with 2 subtypes, 1a and 4c, based on heat-stable antigens (Phillips, 1967). Most cattle strains in the UK belong to type 1 and most sheep strains to types 2, 3, and 4. Diverging distributions may be found in other geographic areas. In Japan, type 3 is found in cattle and the predominant type is 5 (Nakazawa et al., 1979). In *A. lignieresii* heat-labile antigens associated with extracellular slime may cause cross reaction in agglutinating tests if unheated suspensions of the different antigenic types are used. In *A. equuli*, both heat-labile and heat-stable antigens have been demonstrated, and a number of antigenic groups have been detected. Mráz (1968) has demonstrated the presence of antigenic groups in this organism, and Kim (1976) has distinguished 28 groups based on heat-stable O antigens. There is no evidence of host specificity among groups.

In *A. pleuropneumoniae*, 14 serovars have been recognized based on soluble surface antigens composed of capsular polysaccharides and lipopolysaccharides (in addition serovars 1 and 5 can be subdivided into 1a, 1b and 5a, 5b, respectively) (Jolie et al., 1994; Nielsen et al., 1997b).

A. rossii shows serological cross-reactions with *A. lignieresii*, *A. equuli*, *A. suis*, *A. seminis*, and *M. haemolytica* (Ross et al., 1972).

No detailed study has been made of the antigens of *A. suis*, but there is a marked uniformity in the antigenic structure of 16 strains taken from Britain, Denmark, Germany, and the United States (Kim et al., 1976).

There are antigenic cross relationships between *A. suis*, *A. lignieresii*, and *A. equuli* (Haupt, 1934; Valleé et al., 1963, 1974; Wetmore et al., 1963; Bouley, 1966; Kim, 1976). Furthermore, actinobacilli and pasteurellae share antigens (Mráz, 1969a, 1977; Ross et al., 1972).

Pulverer and Ko (1972) have found 6 heat-stable agglutinating antigens in *A. actinomycetemcomitans* and have established 24 ag-

glutinating patterns among 100 strains. King and Tatum (1962) have detected 3 precipitating antigens distributed singly among different strains. This has been confirmed by Zambon et al. (1983), who named the 3 serotypes a, b, and c. Later, serotypes d and e were added (Saarela et al., 1992; Gmür et al., 1993). Zambon et al. (1983) have found that *A. actinomycetemcomitans* cross-reacts with *H. aphrophilus*.

Tube agglutination with somatic antigens reveals 5 serotypes in "*A. salpingitidis*" that differ from other actinobacilli and closely related pasteurellae (Mráz et al., 1976). Presumed existence of a thermostable envelope antigen has been found by gel precipitation, according to Ouchterlony (1968). Its low specificity demonstrates the resemblance of all species within the genus, as well as in the group of *Actinobacillus*–*Pasteurella* (Mráz et al., 1976).

Pathogenicity and sources Actinobacilli have low pathogenicity in experimental animals. Efforts to produce specific lesions with *A. lignieresii* in laboratory animals have usually failed. Chick and mice embryos may be more susceptible than guinea pigs and rabbits (Mráz, 1969a, b). Mice that are injected intraperitoneally and chick embryos affected by injection into the allantoic cavity usually die (Holmes, 1998). *A. equuli* is not pathogenic in mice, and *A. equuli* and *A. suis* are not detrimental to guinea pigs and rabbits. *A. pleuropneumoniae* causes infection in mice by the intraperitoneal and intranasal route, and *A. suis* infects mice by the intraperitoneal route. *A. ureae* is slightly pathogenic for guinea pigs, mice, and rabbits (Henriksen and Jysum, 1961).

Actinobacilli were first found associated with cattle, but have since been isolated from other animals and humans (Peel et al., 1991). In domestic animals, they occur both as pathogens and commensals. As commensals, actinobacilli are present in the alimentary, respiratory, and genital tracts of normal animals. *A. lignieresii* has a commensal role in the mouth and rumen of healthy cattle and sheep (Phillips, 1961, 1964; Bisgaard et al., 1986). Invasion and establishment within tissue usually promote the transition from commensalism to pathogenicity. This implies that actinobacilli often operate as opportunists. A number of human infections are the result of animal bites. Actinobacilli usually cause sporadic disease, but sometimes a group of animals is affected by a common trigger factor.

A. lignieresii produces actinobacillosis in cattle and sheep. Several predisposing factors exist, and there is considerable variation in pathogenicity between strains, some of which do not cause disease at all (Magnusson, 1928). Lignières and Spitz (1902) isolated an organism from actinomycotic lesions of soft tissue in cattle that they named "l'actinobacille" due to the nature of the lesion, described as actinobacillosis. In cattle, chronic granulomatous lesions are found most frequently on the tongue ("wooden tongue") and in other soft tissues of the head and upper alimentary tract. Subcutaneous injection into cattle may produce an abscess similar to that occurring naturally. Granules consisting of rods surrounded by clubs are present in the pus. Lymph nodes in the region are invariably affected. Christiansen (1917) was the first to describe the organism in sheep as "*Bacterium purificiens*". Tunnicliff (1941) concluded that it was similar to the organism isolated from cattle. Skin, lungs, testes, and mammary glands are the sites most commonly affected in sheep.

In addition to infections in cattle and sheep, *A. lignieresii* has been isolated from an epidural abscess (Chladek and Ruth, 1976), an intramandibular phlegmon (Zaharija et al., 1979), and an enlarged tongue (Baum et al., 1984) of horses. *A. lignieresii* infections in humans bitten by horses and sheep have been re-

ported (Dibb et al., 1981; Clark et al., 1984; Peel et al., 1991; Benaoudia et al., 1994). *A. arthritidis* has been isolated from the cavum oris of apparently healthy horses, as well as from joints of horses with arthritis, and from horses and foals with septicemia (Christensen et al., 2002a).

A. suis is an opportunistic pathogen of swine of all ages and has been isolated from healthy pigs (Cutlip et al., 1972; van Ostaaijen et al., 1997). Generally, *A. suis* is associated with septicemia and death in suckling and newly weaned piglets. However, septicemia in mature swine has also been reported (Miniats et al., 1989; Yaeger, 1996). Localized infections, such as endocarditis, polyarthritis, subcutaneous abscesses, and pneumonia, have also been observed (Zimmermann, 1964; Mair et al., 1974). Human infection has been reported as the result of a pig bite (Escande et al., 1996). It is likely that equine strains of organisms, described as *A. suis* or hemolytic variants of *A. equuli* may constitute a separate group of organisms, provisionally designated taxon 11 (Bisgaard et al., 1984).

A. equuli causes septicemia, often in association with nephritis, arthritis, pneumonia, pleuritis, or enteritis in young foals, a condition referred to as "sleepy foal disease" or "joint ill." Adult horses may suffer from arthritis, pneumonia, endocarditis, pericarditis, nephritis, meningitis, and metritis. Abortion may occur in pregnant animals. Golland et al. (1994) have described 15 cases of peritonitis in horses. In piglets that are a few days old, *A. equuli* may cause septicemia (Pedersen, 1977), while older pigs may suffer from arthritis, endocarditis, osteomyelitis, nephritis, and pneumonia. *A. equuli* occurs as a commensal in the equine intestinal tract and mouth. It has not been isolated from normal swine. An *A. equuli*-like bacterium has been isolated from an infected horse bite wound in a human being (Peel et al., 1991).

A. hominis was originally isolated from patients with chronic lung disease, but it is also an etiologic agent of septicemia in hepatic failure (Wüst et al., 1991) and pleural empyema (Friis-Møller and Kharazmi, 1988). It has not been isolated from animals.

A. ureae has been isolated from the human respiratory tract, sputum, and cerebrospinal fluid (Henriksen and Jysum, 1961; Jones, 1962; Verhaegen et al., 1988). In 11 cases, it has been the primary pathogen of meningitis (Kingsland and Guss, 1995). It has also been isolated from bone marrow infection (Avlami et al., 1997), chronic bronchitis, pneumonia, sepsis, and peritonitis (Kingsland and Guss, 1995). Strains vary in virulence and have no animal host (Kolyvas et al., 1978).

A. pleuropneumoniae is the etiologic agent of a highly contagious and often fatal acute or chronic pleuropneumonia in swine. As opposed to other actinobacilli, *A. pleuropneumoniae* is a primary pathogen. Pigs that survive the infection often become subclinical carriers. *A. pleuropneumoniae* has been isolated from 42% of tonsils from 303 chronically infected slaughterhouse swine (Møller and Kilian, 1990). It is rarely recovered from the nasal cavity and the lungs. Although isolates of all serovars may cause disease, significant differences in their pathogenic potential have been reported (Rosendal et al., 1985; Jacobsen et al., 1996). Several virulence factors have been described, including capsules, lipopolysaccharides, outer membrane proteins, transferrin binding proteins, and Apx toxins (Haesebrouck et al., 1997). The capsule consists of derivatized repeating oligosaccharides. The DNA region involved in the export of capsular polysaccharide by *A. pleuropneumoniae* serovar 5a has been characterized (Ward and Inzana, 1997). Among the different bio- and serovars of *A. pleuropneumoniae*, three exotoxins have been

described: ApxI and ApxII, which are hemolytic, and ApxIII, which is non-hemolytic (Frey et al., 1993; Jansen, 1994). They are toxic to porcine alveolar macrophages and neutrophils and belong to the pore-forming RTX toxins, which are widely spread among pathogenic Gram-negative bacteria, including the *Pasteurellaceae* family (Frey et al., 1994). *A. lignieresii*, *A. equuli*, and *A. suis* also possess RTX cytotoxin genes, and *A. suis* produces proteins similar to ApxI and ApxII (Burrows and Lo, 1992; Kamp et al., 1994).

A. scotiae has been isolated from porpoises (*Phocoena phocoena*) stranded on the Scottish coastline (Foster et al., 1998). One strain (M2000/95/1^T) was recovered from the brain, lung, spleen, liver, mesenteric lymph node, blood, and small intestine of one animal that died of septicemia.

A. delphinicola has been isolated from harbor porpoises, a Sowerby's beaked whale, and a striped dolphin (Foster et al., 1996). The pathological significance of this organism for the sea mammal species is not clear.

A. indolicus, *A. minor*, and *A. porcicus* are isolated from the upper respiratory tracts of pigs, where they constitute part of the residing mucosal flora (Møller et al., 1996a). They are not pathogenic to pigs, as indicated by inoculation experiments (Rosendal et al., 1985; Brandreth and Smith, 1986).

A. rossii has been isolated from the vaginas of postparturient sows and aborted piglets (Ross et al., 1972; Sneath and Stevens, 1990).

A. capsulatus has been described in rabbits, where it acts as an agent of arthritis (Arseculeratne, 1962; Phillips, 1984), in hamsters (Krause et al., 1989), and in the lungs, livers, and/or spleen tissue of snowshoe hares (Zarnke and Schlatter, 1988). Isolation of *A. capsulatus* in pure culture from animals with fatal infections indicates that it may act as a primary pathogen. Inoculation experiments have shown that domestic rabbits injected intravenously with *A. capsulatus* develop lung and liver lesions similar to those observed in naturally infected hares (Arseculeratne, 1962).

A. muris has been isolated from the pharyngeal flora of healthy white laboratory mice and from uteri and vaginas of apparently healthy mice (Bisgaard, 1986).

A. succinogenes has been cultured from the bovine rumen (Guettler et al., 1999). It produces succinate-pathway enzymes in much higher amounts than do other succinic-producing bacteria (van der Werf et al., 1997). Its general properties are suggestive of a microorganism adapted to a symbiotic role in the rumen.

A. seminis has been associated with epididymitis and orchitis in rams (Baynes and Simmons, 1960, 1968; van Tonder, 1973, 1979; Erasmus et al., 1982; Sponenberg et al., 1983; De Wet and Erasmus, 1984), ovine contagious epididymitis (Ajai et al., 1980; Burgess, 1982), mastitis in ewes (Alsenosy and Dennis, 1985), polyarthritis and posthitis in sheep (Watt et al., 1970) and abortion in sheep (Foster et al., 1999).

A. actinomycetemcomitans is indigenous to humans. The pathogenicity of *A. actinomycetemcomitans* in humans has been doubted, since it is detected with actinomycetes in more than 30% of actinomycotic lesions (Heinrich and Pulverer, 1959b). However, recent evidence implicates *A. actinomycetemcomitans* in the etiology of destructive periodontal diseases in man (e.g., Zambon, 1985; Zambon et al., 1988; Haffajee and Socransky, 1994). It has also been isolated from abscesses (abdominal [Garner, 1979], brain [Martin et al., 1967; Garner, 1979], facial [Page and King, 1966], hand [Mauff et al., 1983], mediastinal [Garland and Prichard, 1983], and thyroid [Burgher et al., 1973]), as well as from actinomycosis (Colebrook, 1920), endarteritis (Symbas et

al., 1967), endocarditis (Mitchell and Gillespie, 1964), meningitis (Page and King, 1966), pneumonia (Meyers et al., 1971), septicemia (Page and King, 1966), urinary tract infection (Townsend and Gillenwater, 1969), and vertebral osteomyelitis (Muhle et al., 1979).

"*A. salpingitidis*" is isolated from the respiratory tract and oviduct of clinically healthy domestic fowl, from salpingo-peritonitis in hens, and from organs of dead chicks and young turkeys (Mráz et al., 1976). It is a commensal in the respiratory tract and oviduct of domestic fowl. Independent infections arise on the basis of previous weakening; the others are of mixed or secondary character (e.g., mycoplasmosis, organ tuberculosis, leukocytosis, parasitic invasion). "*A. salpingitidis*" is pathogenic for 5-day-old chicks and eventually for white mice. Guinea pigs and rabbits are not susceptible.

ENRICHMENT AND ISOLATION PROCEDURES

Enriched media (blood agar or serum agar) used for the isolation of pathogens from animal tissues can also be used for the primary isolation of actinobacilli from tissues. For the V-factor-dependent actinobacilli (*A. indolicus*, *A. minor*, *A. pleuropneumoniae* biovar 1, and *A. porcinus*), enriched media should include NAD, or a nurse-strain should be used as described in the *Haemophilus* chapter. For primary isolation of *A. pleuropneumoniae* from pleuropneumonic lungs, enriched¹ bovine blood agar plates cross-streaked with a β -hemolytic *Staphylococcus aureus* strain will support the growth and enhance the hemolytic pattern. Media suitable for serotyping and genetic characterization of *A. pleuropneumoniae* are chocolate agar (see Genus *Haemophilus*, this edition), PPLO agar (Nicolet, 1971), and brain heart infusion broth supplemented with 10 mg/l NAD (Møller et al., 1996a).

Selective media may prevent overgrowth of actinobacilli when they are present in mixed cultures. Such media have been used to recover *A. lignieresii* from the mouths of normal cattle (Till and Palmer, 1960) and from the rumens of normal cattle and sheep (Phillips, 1961, 1964), *A. indolicus*, *A. minor*, and *A. porcinus* from the upper respiratory tract (Møller and Kilian, 1990), and *A. actinomycetemcomitans* from the periodontal pocket (Slots, 1982b). Selective and indicator media for isolation of *A. pleuropneumoniae* from the pharyngeal tonsils of chronically infected swine have been described (Møller et al., 1993; Sidibé et al., 1993; Jacobsen and Nielsen, 1995).

Gram and Ahrens (1998) have developed a PCR method for specific detection of *A. pleuropneumoniae*. When used to detect *A. pleuropneumoniae* from tonsils of carrier pigs, the PCR method, combined with cultivation, shows a high sensitivity as compared to cultivation alone. The PCR amplifies the outer membrane lipoprotein (*omlA*) gene. A multiplex PCR for specific detection of *A. pleuropneumoniae* serovar 5 has been developed by Lo et al. (1998). DNA sequences specific to the conserved export and serotype-specific biosynthesis regions of the capsular polysaccharide of *A. pleuropneumoniae* serovar 5 are used as primers. Frey et al. (1995) have developed a PCR method that allows determination of the activator, structural, and secretion genes of the three toxins ApxI, ApxII, and ApxIII in *A. pleuropneumoniae*. The profile of the individual genes in the three Apx operons is con-

stant for a given serotype. Despite the high similarity between 16S rDNA from *A. pleuropneumoniae* and *A. lignieresii*—with differences in only three positions—a species-specific oligonucleotide for *in situ* detection of *A. pleuropneumoniae* has been constructed (Fussing et al., 1998b).

MAINTENANCE PROCEDURES

Most strains die on blood agar or serum agar within 4 d (Phillips, 1984). Cultures grown in Robertson's cooked medium may be stored for up to 4 weeks, and when lyophilized in sterile rabbit serum, 20% peptone solution, or skim milk, for up to 20 years. When heavy suspensions of cultures are grown on nutrient agar, blood agar, chocolate agar, or PPLO agar and washed off in the solutions described for lyophilization, they may be stored at -70°C for up to 2 years without loss of viability.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

V-factor requirement V-factor requirement is demonstrated on agar medium lacking the V factor by cross-inoculation of the plate with a nurse strain (e.g., *Staphylococcus* or *Pseudomonas*) or application of a V-factor-impregnated paper disk (Genus *Haemophilus*, this edition).

Fermentation of carbohydrates Fermentation of carbohydrates for the V-factor-dependent species can be examined in phenol red broth base (Difco) supplemented with 1% of the respective carbohydrates and 10 $\mu\text{g/ml}$ NAD (Genus *Haemophilus*, this edition).

Micromethods for examination of V-factor-dependent actinobacilli The micromethods do not require growth and are recommended for the detection of indole production, urease, ornithine and lysine decarboxylases, arginine dihydrolase, and glycosidase activities (Genus *Haemophilus*, this edition).

DIFFERENTIATION OF THE GENUS *ACTINOBACILLUS* FROM OTHER GENERA

Actinobacilli are distinguished from enterobacteria by their smaller genome, (e.g., *A. pleuropneumoniae*, with 2404 ± 40 kb) (Chevallier et al., 1998), obligatory parasitism, and absence of motility. They differ from the vibrios by their smaller genome, absence of flagella, and other phenotypic properties.

Complex genomic relationships have been shown by DNA–DNA hybridization to exist between members of the genera of the family *Pasteurellaceae* (Pohl, 1979). The phenotypic distinction of *Actinobacillus* and *Pasteurella* is difficult to determine with the existing composition of the genera (Sneath and Stevens, 1985). DNA–DNA hybridization studies show that the traditional genus names in *Pasteurellaceae* can be kept only if the genera are restricted to entities clustering at or above 50% DNA binding level with the type species of the genera (Mutters et al., 1985a). This is a practical choice, since there is no consensus as to which level genera should be defined within different families. rRNA–DNA hybridization identifies seven rRNA branches, in addition to numerous unclustered taxa within the family *Pasteurellaceae* (De Ley et al., 1990). 16S rRNA sequencing of 70 strains representing different taxa within *Pasteurellaceae* has been performed by Dewhirst et al. (1992, 1993). The rRNA branches (*P. multocida*, *A. lignieresii*, *H. influenzae*, *H. aphrophilus*, *A. actinomycetemcomitans*, and *H. ducreyi*) correspond to the 16S rRNA subclusters 3B, 4A, 1C, 1B, 1A, and 2B, respectively (Dewhirst et al., 1993). These findings indicate that the genera defined by DNA–DNA hybrid-

1. Meat blood agar: 500 g of chopped, lean meat cooked for 20 min in 1 liter of water, supplemented with 10 g/l Bacto peptone (Difco, Detroit, USA), 5 g/l NaCl, and 10 g/l bacteriological agar no. 1 (Oxoid, Unipath Ltd., Basingstoke, England), and cooked for 20 min. pH adjusted to 7.7–7.6 by NaOH. Autoclaved at 120°C for 20 min, and 5% calf's blood added at a temperature of 45°C .

ization do not represent phylogenetic groups and that further reorganization within the family is necessary (Angen, 1997).

The close similarity between actinobacilli and the genera *Haemophilus* and *Pasteurella* necessitates comparison of biochemical characters at the species level. Boot and Bisgaard (1995) have reclassified 30 *Pasteurellaceae* strains isolated from rodents. Strains previously reported as *Actinobacillus* spp. have been reclassified as *P. pneumotropica* biotype Jawetz, *P. dagmatis*, or taxon 22. These findings underline the importance of extended characterization of isolates and comparison with reference strains to avoid misclassification within the family *Pasteurellaceae*. The main features of differentiation between these genera and others are given in the *Pasteurellaceae* chapter, Table BXII.γ.287.

TAXONOMIC COMMENTS

In *Bergey's Manual of Determinative Bacteriology*, 9th Ed. (Holt et al., 1994), the genus *Actinobacillus* included *A. actinomycetemcomitans*, *A. capsulatus*, *A. equuli*, *A. hominis*, *A. lignieresii*, *A. muris*, *A. pleuropneumoniae*, *A. rossii*, *A. seminis*, *A. suis*, and *A. ureae*. The new species *A. minor*, *A. porcineus*, and *A. indolicus*, created by Møller et al. (1996a), and *A. delphinicola*, established by Foster et al. (1996), were later added to the genus. Most recently, *A. scotiae* and *A. succinogenes* have been included in the genus *Actinobacillus* (Foster et al., 1998; Guettler et al., 1999) and Bisgaard Taxon 9 has been classified as *A. arthritis* and genomospecies 2 (Christensen et al., 2002a).

Within the family *Pasteurellaceae*, genomic relationships have indicated that several groups/complexes stand for new genera: the *H. aphrophilus* group, the "*A. salpingitidis*" group, the "*Histophilus ovis*" / "*H. somnus*" group, and the former *P. haemolytica* and *P. pneumotropica* complexes. The rRNA branches containing *H. aphrophilus*, *A. actinomycetemcomitans*, *H. ducreyi*, and *H. ovis* represent candidates for new genera (De Ley et al., 1990). Based on DNA hybridization data, Escande et al. (1984) and Mutters et al. (1989) have proposed that *Actinobacillus* be limited to *A. lignieresii*, *A. equuli*, *A. capsulatus*, *A. suis*, *A. ureae*, *A. hominis*, *A. pleuropneumoniae*, *A. arthritis*, and Bisgaard's taxa 5 and 11. 16S rRNA sequencing (Dewhirst et al., 1992) has demonstrated true actinobacilli in cluster 4A, which contains *A. lignieresii*, *A. pleuropneumoniae*, Bisgaard's Taxon 11, *A. ureae*, *A. equuli*, *A. suis*, and *H. parahaemolyticus*. *A. hominis* has subsequently been added (Dewhirst et al., 1993). Sequencing of additional strains includes *A. arthritis*, Bisgaard's taxa 5, 8, and *P. bettyae* in cluster 4 (Dewhirst et al., 1993). Of these, Bisgaard's taxa 8 and 9 are closely related to *Actinobacillus sensu stricto*. *A. capsulatus* does not cluster with the true actinobacilli (Dewhirst et al., 1992, 1993). This organism, along with *A. seminis*, *P. mairii*, *P. aerogenes*, and *H. paracuniculus*, may represent a new genus within *Pasteurellaceae* (Dewhirst et al., 1993).

Dewhirst et al. (1992) have pointed out that in cluster 4, each species branches slightly deeper than the one above, giving no logical point at which to end the genus *Actinobacillus sensu stricto*. In addition, Møller et al. (1996a) have emphasized the problem of defining natural genera in the family *Pasteurellaceae*, because NAD-dependent species are distributed throughout the family. Although new genera may be warranted to accommodate species such as *A. actinomycetemcomitans*, *H. aphrophilus*, *H. parasuis*, *H. segnis* and others, the natural borders for such genera may not be easy to define. Reorganization without a logical basis may cause unnecessary confusion and frustration among clinical microbiologists (Møller et al. (1996a)).

It is clear from phylogenetic studies that *A. actinomycetemco-*

mitans is not a true actinobacillus, because *Actinobacillus sensu stricto* falls in cluster 4A and *A. actinomycetemcomitans* in cluster 1A (Dewhirst et al., 1992). Furthermore, the representatives of serotypes a, b, and c of *A. actinomycetemcomitans* diverge in the established phylogenetic tree at a level greater than that seen between many species. This may warrant the establishment of sub-species from these serotypes. However, not all serotypes have yet been sequenced.

According to rRNA-DNA relatedness results (De Ley et al., 1990) and 16S rRNA sequencing data (Dewhirst et al., 1992, 1993), *A. actinomycetemcomitans*, "*A. salpingitidis*", *A. muris*, and *A. seminis* should be excluded from *Actinobacillus sensu stricto*. *A. actinomycetemcomitans* and *H. aphrophilus* are related to the genus *Haemophilus sensu stricto* (Dewhirst et al., 1992). Because *A. actinomycetemcomitans* is phenotypically different from *Haemophilus* and has a long evolutionary branch, it may be placed in its own genus (Dewhirst et al., 1992). Potts et al. (1985) have suggested that *A. actinomycetemcomitans* be transferred to the genus *Haemophilus*. However, the International Committee on Taxonomy Subcommittee on *Pasteurellaceae* has rejected this proposal because *A. actinomycetemcomitans*, although similar to *H. aphrophilus*, is not convincingly related to *H. influenzae*, the type species of the genus *Haemophilus*. Previously, Kilian (1976) had pointed out that *A. actinomycetemcomitans* and *H. aphrophilus* do not require hemin and NAD for growth and therefore, at that time, did not meet the criteria for inclusion into the genus *Haemophilus*.

Most recently, the Dewhirst group has established a phylogenetic tree for the family *Pasteurellaceae* based on 114 nonredundant 16S rRNA sequences (Family *Pasteurellaceae*, Fig. BXII.γ.215). The tree contains 21 phylogenetic clusters. One of these is the *Actinobacillus sensu stricto*, which contains *A. lignieresii*, *A. suis*, *A. equuli*, *A. hominis*, *A. ureae*, and *A. pleuropneumoniae*. *H. parahaemolyticus* was also included. Some unnamed species, such as Bisgaard Taxa 8 and 9 (*A. arthritis* and genomospecies 2), have also joined this group, as have taxa represented by strains Kunstyr 570 (MCCM 00149), Bisgaard F64 (CCUG 28015), and SSI P585 (CCUG 19799). The genus *Actinobacillus* should probably include these organisms, as proposed by Escande et al. (1984) and Mutters et al. (1989).

The remaining clusters, except for *Haemophilus sensu stricto*, *Pasteurella sensu stricto*, *Mannheimia*, and *Lonepinella*, contain generically misclassified species of *Actinobacillus*, *Haemophilus*, and *Pasteurella* and unnamed species of *Pasteurellaceae* from a variety of culture collections. Thus, *A. minor* falls with *Haemophilus* "minor group" 202 and *A. pleuropneumoniae* with *H. parahaemolyticus*, each in two subclusters of the Porcine Cluster. *A. rossii* and *A. porcineus* comprises the Rossii Cluster, together with strains considered to be part of the *P. aerogenes* complex, but do not appear with the type strain of *P. aerogenes*, which is in the Seminis Cluster. In addition, the more deeply branching and unstable Bisgaard Taxon 7 may belong to the Rossii Cluster. *A. indolicus* falls in the Parasuis Cluster, together with *H. parasuis* and a taxon represented by *H. parasuis* strain Nagasaki. *A. capsulatus* is contained in the Capsulatus Cluster, together with *H. paracuniculus* and a third taxon represented by *A. capsulatus* strain SSI P796. Previously, DNA-DNA hybridization had placed *A. capsulatus* in the *Actinobacillus sensu stricto* (Escande et al., 1984; Mutters et al., 1989). However, the sequences for five isolates (SSI P243^T, P244, P572, P662, and P796) fall in the Capsulatus Cluster. This is in contrast to a horse isolate (SSI P585 [CCUG 19799]), which falls in the middle of the *Actinobacillus sensu stricto* Cluster, as well as a hamster "*A. capsulatus*" strain (MCCM 149) and a duck isolate

(Bisgaard F64 [CCUG 28015]), which form the deepest branch of the *Actinobacillus sensu stricto* Cluster. These findings suggest that *A. capsulatus*-like strains from species other than rabbit might represent unique species. *A. actinomycetemcomitans* falls in the Aphrophilus Cluster, together with *H. segnis*, *H. aphrophilus*, *H. paraphrophilus*, and an *H. segnis*-like taxon represented by strain MCCM 00337. This is consistent with rRNA–DNA studies by De Ley et al. (1990), which suggested that the Aphrophilus Cluster be separated from the *Haemophilus sensu stricto*. *A. seminis* is contained in the Seminis Cluster with *P. mairii* and *P. aerogenes*. The Rodent Cluster comprises 9 species, with *A. muris*, *P. pneumotropica*, and “*H. influenzae-murium*” as named species. This cluster also contains Bisgaard Taxon 17 (rat), strain Kunstyr 507 (MCCM

02120; hamster), Mannheim Michael A (CCUG 28028; guinea pig), Kunstyr 246 (CCUG 28030; hamster), Mannheim A/5a (MCCM 00235; rat), and Forsyth A3 (mouse). “*A. salpingitidis*” falls in the Avian Cluster, together with *P. volantium*, *P. gallinarum*, *P. avium*, *P. anatis*, *H. paragallinarum*, Bisgaard Taxon 2 and 3, *Haemophilus* taxon C, *Pasteurella* species A, and strains representing species such as Wuthe DS 1801 (MCCM 01557), “*Actinobacillus salpingitidis*” CCUG 28018, and *Haemophilus* sp. CCUG 19808. *A. delphinicola* and *A. scotiae* fall in the Ducreyi Cluster, together with *H. ducreyi*, *P. langaa*, and *Phococnobia uteri*, while *A. succinogenes* appears in the Somnus Cluster with “*H. somnus*” and Bisgaard Taxon 6 and 10.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *ACTINOBACILLUS*

Differential characteristics of species of *Actinobacillus* are given in Tables BXII.γ.291 and BXII.γ.292.

In the following list of *Actinobacillus* species, *A. lignieresii*, *A. suis*, *A. equuli*, *A. hominis*, *A. ureae*, and *A. pleuropneumoniae* are

considered species belonging to *Actinobacillus sensu stricto*. The other species in the list are pending placement in other genera of the family *Pasteurellaceae*.

List of species of the genus *Actinobacillus*

1. *Actinobacillus lignieresii* Brumpt 1910, 849^{AL}

lig.ni.e.re' si.i. M.L. gen. n. *lignieresii* of Lignières, named for J. Lignières, one of the bacteriologists who first isolated this organism.

Cells are small and rod-shaped, often showing shorter coccobacillary forms. In media with fermentable carbohydrates such as glucose and maltose, long bacillary or filamentous forms can be seen. Shorter bacillary or filamentous forms are more common on media containing blood or serum. Small granules can be detected scattered among the cells and often lying at the pole of the bacillary or filamentous form. This gives the characteristic “Morse-code” form (Phillips, 1960).

The rods, which measure $1.15\text{--}1.25 \times 0.4 \mu\text{m}$, are non-motile, nonsporing, and non-acid-fast. Slime, but not a capsule, can be demonstrated in wet India ink preparations. Cells stain easily, particularly with carbol fuchsin. They may show bipolar staining.

Viscous colonies, 1–2 mm in diameter, are formed on primary isolation. The sticky nature of the colonies is lost with repeated subcultivation. The colony size may increase up to 4 mm upon prolonged cultivation. *A. lignieresii* grows on MacConkey agar. There is no hemolysis on sheep blood agar. The growth is poor and not visible for some days in a gelatine stab. There is no liquefaction. Broth cultures are turbid with little deposit. Growth is improved by the addition of serum.

The best fermentation with carbohydrates is obtained in a peptone–Lemco base with 1% carbohydrate and bromothymol blue as an indicator (Holmes, 1998). Tests should not be prolonged beyond 14 d. The organism produces acid without gas promptly from maltose, mannitol, and sucrose, but with a delayed reaction in lactose (5–7 d). Cellobiose, melibiose, rhamnose, salicin, and trehalose are not fermented. Acid is also produced in litmus milk, but no clot is formed. The oxidase reaction is generally positive, while the catalase reaction varies between strains. Esculin and sodium hippurate are not hydrolysed. Produces actinobacillosis in cattle and sheep and has been isolated from an epidural abscess, an intramandibular phlegmon, and an

enlarged tongue of a horse. *A. lignieresii* infections in humans bitten by horses or sheep have occurred.

Recently Christensen et al. (2002a) have shown that equine isolates are phenotypically indistinguishable from *A. lignieresii* and that they belong to a separate species tentatively designated *genospecies* 1.

The optimum temperature for growth is 37°C. Only slight growth occurs at 20°C and no growth at 44°C.

The mol% G + C of the DNA is: 41.8–42.6 (T_m) (Boháček and Mráz, 1967).

Type strain: ATCC 49236, NCTC 4189.

GenBank accession number (16S rRNA): M75068.

2. *Actinobacillus actinomycetemcomitans* (Klinger 1912) Topley and Wilson 1936, 279^{AL} (*Bacterium actinomycetemcomitans* Klinger 1912, 198.)

ac.ti.no.my.ce.tem.co' mi.tans. Gr. n. *actis* a ray; Gr. n. *myces* a fungus; M.L. n. *actinomyces* ray fungus; L. part. adj. *comitans* accompanying; M.L. part. adj. *actinomycetemcomitans* accompanying an actinomycete.

Cells are nonmotile cocci, measuring $0.7 \pm 0.1 \times 1.0 \pm 0.4 \mu\text{m}$, or are rod-shaped (Phillips, 1984; Zambon, 1985). They may occur singly, in pairs, or in small clumps. The rod-shaped cells are seen more frequently in agar cultures than in broth or gelatine cultures. While cells may be distinct rod forms in culture, many more coccal forms are seen in actinomycotic lesions.

Colonies on agar are small, <0.5 mm in diameter after 24 h, but enlarge to 2–3 mm after 24 h (Holmes, 1998). They adhere to the agar medium, are difficult to break up, and are described as starlike (Colebrook, 1920; Blix et al., 1990) or like “crossed cigars” (Heinrich and Pulverer, 1959a). Colonies may sometimes have a rough surface and pitting after several days of cultivation. They become mucoid and nonadherent after repeated subculture (Slots, 1982a).

Cells have a microcapsule. Primary broth colonies form tiny, discrete colonies that cling to the sides of glass tubes, while subsequent cultures become less adherent and exhibit uniform turbidity (Zambon, 1985). The organism grows poorly in air but well in 5% CO₂ (capnophilic) or under

TABLE BXII-γ.291. Differential characteristics of *Actinobacillus* species^a

Characteristics	1. <i>A. lignieresii</i>	2. <i>A. actinomycetemcomitans</i>	3. <i>Actinobacillus arthritidis</i>	4. <i>A. capsulatus</i>	5. <i>A. delphiniicola</i>	6. <i>A. equuli</i>	7. <i>A. hominis</i>	8. <i>A. indolicus</i>	9. <i>A. minor</i>	10. <i>A. muros</i>	11. <i>A. pleuropneumoniae</i>	12. <i>A. portinus</i>	13. <i>A. rossii</i>	14. <i>A. scoliae</i>	15. <i>A. seminis</i>	16. <i>A. succinogenes</i>	17. <i>A. suis</i>	18. <i>A. ureae</i>	19. " <i>A. salpingtoides</i> "
<i>Host:</i>																			
Bovine rumen	+																		
Cattle																			
Cetaceans					+														
Domestic fowl																			
Horse			+																+
Lamb																			
Humans		+																	
Mouse																			
Rabbit				+						+									
Sheep	+																		
Swine						+													
V-factor dependency	-	-	-	-	-	-	-	+	+	-	d	+	+	-	-	+	+	+	+
Catalase	d	+	+	+	-	d	-	+	d	+	d	-	+	-	d	+	+	+	+
Oxidase	+	+	+	+	+	d	+	d	d	+	d	-	+	+	d	+	+	+	+
Hemolysis on SBA	-	-	-	- ^c	-	-	-	d	- ^c	-	+	-	-	c	-	-	+	+	-
Urease	-	-	-	+	-	-	+	+	+	-	+	-	+	+	-	-	+	+	+
Indole	-	-	-	-	-	-	+	+	+	-	+	-	+	+	-	-	+	+	-
ONPG ^e	d	- ^f	+	+	-	d	+	+	+	-	+	+	d	+	-	+	-	+	+
<i>Acid production from:</i>																			
D-Mannitol	+	d	+	+	-	+	+	-	-	+	+	d	+	-	d ^f	+	-	+	+
Raffinose	d	-	+	+	-	+	+	+	+	+	d	d	d	-	-	+	+	- ^g	+
Salicin	-	-	-	+	-	-	d	-	-	+	-	-	-	-	-	+	+	+	-
D-Sorbitol	d	-	+	+	-	d	-	-	-	-	-	d	+	-	-	+	-	+	-d
Sucrose	+	-	+	+	-	+	+	+	+	+	+	d	-	-	-	+	+	+	+
Trehalose	-	-	-	+	-	+	+	d	d	+	-	d	-	-	-	-	+	+	+
D-Xylose	+	d	+	+	-	+	+	d	d	-	+	d	+	-	-	+	+	- ^g	+
Mol% G + C content ^h	41.8-42.6	42.7	nd	42.4	nd	40.0-41.8	40.9	35.5	38.2	46.9	42.2-43.2	41.4	41.9	40.5	43.7	45	40.5	41.2-43.7	39.6-42.9 (mean 41.5)

^aFor symbols see standard definitions; nd, not determined. *A. lignieresii*, *A. equuli*, *A. hominis*, *A. pleuropneumoniae*, *A. suis*, and *A. ureae* belong to *Actinobacillus sensu stricto*. Remaining species will eventually be transferred to a number of genera in the family *Pasteurellaceae*.

^bDifferent reactions are given by different authors.

^cSlight hemolysis on SBA may be apparent if incubated longer than 48 h.

^dHemolysis on SBA may be very slight or absent.

^eONPG reaction, hydrolysis of orthonitrophenyl-β-D-galactopyranoside.

^fLate reaction.

^gMost strains positive.

^hFor methods of mol% G + C estimation, see text.

TABLE BXII-γ.292. Additional characteristics of *Actinobacillus* species.^a

Characteristic	<i>A. lignieresii</i>	<i>A. actinomycetemcomitans</i>	<i>A. arthritidis</i>	<i>A. capsulatus</i>	<i>A. delphiniicola</i>	<i>A. equuli</i>	<i>A. hominis</i>	<i>A. indolicus</i>	<i>A. minor</i>	<i>A. muris</i>	<i>A. pleuropneumoniae</i>	<i>A. porcicus</i>	<i>A. rossii</i>	<i>A. scottiae</i>	<i>A. seminis</i>	<i>A. succinogenes</i>	<i>A. suis</i>	<i>A. ureae</i>	<i>"A. salpingitidis"</i>
Growth on MacConkey agar	+	+	d	+	+	+	+	+	+	+	+	d	+	+	+	+	+	+	+
Phosphatase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Gelatinase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Hydrogen sulfide produced	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Methyl red test	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Voges-Proskauer test	d	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Lysine decarboxylase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Ornithine decarboxylase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Arginine dihydrolase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
<i>Fermentation (acid, no gas) of:</i>																			
Adonitol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
L-Arabinose	d	+	d	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Cellobiose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Dextrin	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Dulcitol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Esculin	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
D-Fructose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
D-Galactose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
D-Glucose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Glycerol	d	+	d	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Inositol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Inulin	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Lactose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Maltose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
D-Mannose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Melibiose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
L-Rhamnose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Starch	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

^aSymbols: see standard definitions.

anaerobic conditions (Holm, 1954; Slots, 1982a). Optimum growth at 37°C (Heinrich and Pulverer, 1959a).

Slots (1982a) has studied 135 biochemical characters in 6 reference strains of *A. actinomycetemcomitans* and 130 strains freshly isolated from the oral cavity. They all decompose hydrogen peroxide, are oxidase negative and benzidine positive, reduce nitrate, produce strong alkaline and acid phosphatases, and ferment fructose, glucose, and mannose. Variable fermentation results have been obtained with dextrin, maltose, mannitol, and xylose. Carbohydrate fermentation has been used to subgroup *A. actinomycetemcomitans* into 3–10 biochemical groups based on the variable fermentation of key sugars, including dextrin, galactose, maltose, mannitol, and xylose (King and Tatum, 1962; Pulverer and Ko, 1970; Slots et al., 1980). *A. actinomycetemcomitans* is indole negative and produces catalase.

A number of chemotaxonomic techniques have been applied to establish the relationship between *A. actinomycetemcomitans* and closely related species (Olsen, 1993; Olsen et al., 1999). Indigenous to humans where recent evidence implicates *A. actinomycetemcomitans* in the etiology of destructive periodontal diseases. It has also been isolated from abscesses and from cases of actinomycosis, endarteritis, endocarditis, meningitis, pneumonia, septicemia, urinary tract infections, and vertebral osteomyelitis.

The mol% G + C of the DNA is: 42.7 (T_m).

Type strain: ATCC 33384, NCTC 9710.

GenBank accession number (16S rRNA): M75039.

3. **Actinobacillus arthritidis** Christensen, Bisgaard, Angen and Olsen 2002a, 1244^{VP}
arthritidis. N.L. fem. gen. n. *arthritidis* of arthritis; from Gr. n. *arthron* joint; L. suff. *-idis* used in names of inflammations.

Cells are pleomorphic without distinct morphology. On bovine blood agar the colonies are nonhemolytic, circular, raised, and regular with an entire margin. The surface of the colonies is generally smooth, shiny, grayish and non-transparent. Mucoid and watery cultures also exist. A diameter of 2–3 mm is often observed after 24 h aerobic incubation at 37°C. The species is positive in the following tests: catalase, oxidase, fermentative reaction in Hugh–Leifson medium with (+)-D-glucose, porphyrin test, nitrate reduction urease, alanine aminopeptidase, phosphatase, ONPG, α -galactosidase, and production of acid from (–)-D-ribose, (+)-D-xylose, (–)-D-mannitol, (–)-D-sorbitol, (–)-D-fructose, (+)-D-galactose, (+)-D-glucose, (+)-D-mannose, lactose, sucrose, and raffinose. Negative results are obtained in Gram staining. Motility at 22 and 37°C, symbiotic growth, β -hemolysis, Simmons citrate, mucate-acid, malonate-base, H₂S/TSI, growth in the presence of KCN, methyl red and Voges–Proskauer at 37°C, production of gas from nitrate, arginine dihydrolase, lysine decarboxylase, ornithine decarboxylase, phenylalanine deaminase, indole production, gelatinase, hydrolysis of Tweens 20 and 80, pigment production, α -glucosidase (PNPG), β -glucosidase (NPG), α -fucosidase, (ONPF), α -glucuronidase (PGUA), α -mannosidase, and β -xylosidase. Negative for production of acid from *m*-erythritol, adonitol, (+)-D-arabitol, xylitol, (–)-L-xylose, dulcitol, *m*-inositol, (+)-D-fucose, (+)-L-rhamnose, (–)-L-sorbose, cellobiose, trehalose, (+)-D-melezitose, (+)-D-glycogen, inulin, esculin, amygd-

alin, arbutin, gentiobiose, salicin, (+)-D-turanose, and β -N-CH₃ glucosamide. No gas produced from (+)-D-glucose.

Strains of *A. arthritidis* have been isolated from the oral cavity of apparently healthy horses, as well as from diseased foals and horses (mainly those with septicemia and arthritis). Isolates phenotypically indistinguishable from *A. arthritidis*, except for being sorbitol negative are tentatively designated genomospecies 2 (Christensen et al., 2002a).

The mol% G + C of the DNA is: unknown.

Type strain: Wetmore 1706, ATCC 13376, CCUG 24862.

GenBank accession number (16S rRNA): AF247712.

4. **Actinobacillus capsulatus** Arseculeratne 1962, 38^{AL}
cap.su.la'tus. L. n. *capsula* a small chest, capsule; M.L. masc. adj. *capsulatus* encapsulated.

Cells are rod-shaped. Filamentous bacilli that fragment into minute, coccoid bodies are seen in old cultures (Phillips, 1984). Moniliform bodies are formed in 5-d-old cultures on Loeffler's serum. Capsules are present. Primary cultures will not grow on nutrient agar or in nutrient broth. However, subcultures grow as pinpoint colonies or as a faint turbidity. Colonies on sheep blood agar are very sticky. "Flower head" colonies are produced on rabbit blood agar (Phillips, 1984). Growth is sparse on Loeffler's serum, but profuse on Dorset egg, where colonies survive for 10 d. In serum broth, small discrete mural colonies occur.

Optimum temperature is 37°C. No growth at 22°C. Cells are killed after 10 min at 60°C. Differential biochemical characters are given in Tables BXII.7.291 and BXII.7.292.

A. capsulatus has been isolated from rabbits, hamsters, and snowshoe hares.

The mol% G + C of the DNA is: 42.4 (T_m) (Mannheim et al., 1980).

Type strain: Frederiksen P243, ATCC 51571, NCTC 11408.

GenBank accession number (16S rRNA): M75062.

5. **Actinobacillus delphinicola** Foster, Ross, Malnick, Willems and Garcia 1996, 652^{VP}
del.phi.ni'co.la. L. n. *delphinus* dolphin; L. n. *cola* dweller; M.L. n. *delphinicola* dolphin dweller.

Cells are nonmotile, facultatively anaerobic, pleomorphic rods (Foster et al., 1996). Added CO₂ is required for growth.

Colonies on CSBA (Columbia agar supplemented with 5% citrated sheep blood) incubated at 37°C in an atmosphere containing 10% added CO₂ are circular, entire, low, convex, smooth, gray, and 0.75–1 mm in diameter after 24 h. They are nonhemolytic or weakly hemolytic. Blood and serum enhance growth. There is no growth on MacConkey agar. Growth occurs at 42°C, but not at 22°C. Acid is produced from glucose and mannose, but not from adonitol, dulcitol, galactose, inositol, inulin, lactose, maltose, melibiose, raffinose, rhamnose, salicin, sorbitol, sucrose, trehalose, and xylose.

Catalase negative and oxidase positive. Nitrate is not reduced. Voges–Proskauer test is positive. Urease, H₂S, and indole tests are negative. Alkaline phosphatase, esterase–lipase, leucine arylamidase, and acid phosphatase are produced in large amounts. Naphthol-AS-BI-phosphohydrolase is generated, usually in large amounts, and esterase is produced in lesser amounts, as assessed by the API ZYM system.

Some strains may hydrolyze arginine and decarboxylate ornithine and/or lysine. Isolated from harbor porpoises, a Sowerby's beaked whale, and a striped dolphin.

The mol% G + C of the DNA is: not determined.

Type strain: M906/93, NCTC 12870.

GenBank accession number (16S rRNA): X89377.

Additional Remarks: The 16S rRNA sequence is not from the type strain.

6. ***Actinobacillus equuli*** (van Straaten 1918) Haupt 1934, 513^{AL} (*Bacillus equuli* van Straaten 1918, 75.)
e. quu'li. L. n. *equulus* foal; L. gen. n. *equuli* of a foal.

Cells are rod-shaped, but vary markedly depending upon the growth medium. Longer, filamentous forms, similar to those of *A. lignieresii*, are seen when the medium contains glucose or maltose. On nutrient or blood agar, *A. equuli* colonies are so viscous that a string of material is formed between the medium and inoculating loop when an attempt is made to lift the colony from the medium (Holmes, 1998). Stickiness is not lost upon repeated subculturing. When first isolated from clinical material, colonies are usually rough, but they may become smooth on repeated subculture. The viscous character is also present in liquid culture, and it is not lost on repeated subculture.

Growth occurs between 20°C and 39°C. Some strains will grow at 44°C. Glycerol and mannose are usually fermented slowly. Fermentation may be delayed with dextrin, fructose, maltose, melibiose, raffinose, sucrose, trehalose, and xylose. Most strains fail to ferment arabinose, cellobiose, salicin, or sorbitol, but some strains ferment them either promptly or slowly.

Causes "sleepy foal disease". Adult horses may also be infected and abortions may occur in pregnant animals. *A. equuli* may also infect pigs.

The mol% G + C of the DNA is: 40.0–41.8 (T_m) (Boháček and Mráz, 1967).

Type strain: ATCC 19392, NCTC 8529.

GenBank accession number (16S rRNA): M75072.

7. ***Actinobacillus hominis*** Friis-Møller 1985, 375^{VP} (Effective publication: Friis-Møller 1981, 156.)
ho' mi.nis. L. masc. n. *homo* man; L. gen. n. *hominis* of man.

Cells are small, nonmotile, polymorphic rods with small capsules (Friis-Møller, 1981). The capsule is best developed on chocolate agar. On nutrient agar, cultures grow with capsulated and noncapsulated variants. No growth is observed on MacConkey agar and 7.5% salt agar.

On 5% blood agar, colonies are 1.0 mm, grayish-white, and soft mucoid, without hemolysis after 24 h at 35°C. After 24 h, the colonies are flatter, with concentric rings. Colonies are up to 2 mm on nutrient agar, and on chocolate agar are 1.5–2.5 mm and very mucoid. There is no CO₂ or V-factor-requirement. Growth is very poor at 22°C, and no growth is seen at 4°C. In semisolid medium, colonies are facultatively anaerobic.

Strains are CAMP negative and possess very strong alkaline and acid phosphatase and esterase (C₄) activity, as shown by API ZYM tests. Nearly all strains have valine arylamidase and some α -fucosidase activity. All isolates have α -galactosidase and some α -glucosidase activity (Friis-Møller, 1981). Other biochemical characters are listed in Tables BXII.γ.291 and BXII.γ.292. Originally isolated from

patients with chronic lung disease, but also isolated from cases of septicemia in hepatic failure, and pleural empyema. Not isolated from animals. Originally isolated from patients with chronic lung disease, but also isolated from cases of septicemia in hepatic failure, and pleural empyema. Not isolated from animals.

The mol% G + C of the DNA is: 40.9 (Bd).

Type strain: P 578, CCUG 19800, ATCC 49457, P578 NCTC 11529.

GenBank accession number (16S rRNA): L06076.

8. ***Actinobacillus indolicus*** Møller, Füssing, Grimont, Paster, Dewhirst and Kilian 1996a, 956^{VP}
in.do'li.cus. M.L. adj. *indolicus* pertaining to indole, which is formed by the organism.

Cells are small rods with lengths varying from 0.8–2.4 μ m (Møller et al., 1996a). Colonies are grayish and opaque on chocolate agar and 2 mm in diameter after 48 h of incubation. Cultures have a characteristic pungent smell. Isolates show satellite growth on blood agar plates when cross-inoculated with an appropriate feeder strain.

Galactose, glucose, mannose, maltose, and sucrose are fermented. Arabinose, inulin, esculin, salicin, mannitol, sorbitol, and inositol are not fermented. Lysine and ornithine are not decarboxylated. Cells are V-factor-dependent, X-factor-independent, nonmotile, nonhemolytic, urease negative, and catalase and indole positive.

Isolated from the upper respiratory tracts of pigs.

The mol% G + C of the DNA is: 35.5 (HPLC).

Type strain: 46KC2, CIP 105316, CCUG 39029.

GenBank accession number (16S rRNA): U65584.

9. ***Actinobacillus minor*** Møller, Füssing, Grimont, Paster, Dewhirst and Kilian 1996a, 955^{VP}
mi' nor. L. comp. adj. *minor* less, smaller, referring to the name of this taxon used previously (Møller and Kilian, 1990).

Cells appear as small rods, varying in length from 1.6 to 2.4 μ m. They are V-factor-dependent but X-factor-independent. Furthermore, they are nonmotile, nonhemolytic, urease positive, and indole negative. Colonies on chocolate agar are smooth, grayish, and approximately 0.8 mm in diameter after 48 h of incubation. Isolates show satellite growth on blood agar plates when cross-inoculated with a nurse strain. Glucose, mannose, maltose, sucrose, and raffinose are fermented, but arabinose, inulin, esculin, mannitol, sorbitol, and inositol are not. Lysine and ornithine are not decarboxylated. *A. minor* has been isolated from the upper respiratory tracts of pigs.

The mol% G + C of the DNA is: 38.2 (HPLC).

Type strain: NM305, CIP 105314, CCUG 38923.

GenBank accession number (16S rRNA): U65582.

10. ***Actinobacillus muris*** Bisgaard 1988, 220^{VP} (Effective publication: Bisgaard 1986, 1)
mu' ris. L. n. *mus* the mouse; L. gen. n. *muris* of the mouse.

The name *Actinobacillus muris* was used by Wilson and Miles (1964) as a name for the organism called *Streptobacillus moniliformis*. It was clearly a junior synonym and therefore illegitimate (Bisgaard, 1986). Bisgaard (1986) proposed the name *A. muris* for the organism listed here, since it was not on the Approved List of Bacterial Names.

Cells are nonmotile rods. The type strain is identical to

P. ureae strain Ackerman 80-443D (Ackerman and Fox, 1981). Surface cultures on bovine blood agar are circular, low convex, and regular with an entire margin (Bisgaard, 1986). Colonies have a smooth, shiny, opaque, and grayish surface and, at 37°C, reach a diameter of 1.5–2.0 mm after 24 h and 3.5–4.0 mm after 48 h. No hemolysis. Growth is butyrous and easily emulsified and cultures have no characteristic odor. Colonies are easily removed from the agar surface.

Ribose, mannitol, fructose, glucose, mannose, cellobiose, maltose, melibiose, sucrose, trehalose, melezitose, raffinose, and salicin are fermented. Catalase, oxidase, nitrate, and urease tests are positive.

The genome molecular mass is 1.4×10^9 Da (Bisgaard, 1986). *A. muris* has been isolated from the pharynx, uteri, and vaginas of mice.

The mol% G + C of the DNA is: 46.9 (T_m) (Piechulla et al., 1985a).

Type strain: Ackerman 80-443D, ATCC 49577, CCUG 16938, NCTC 12432.

GenBank accession number (16S rRNA): AF024526.

11. **Actinobacillus pleuropneumoniae** (Shope 1964) Pohl, Bertschinger, Frederiksen and Mannheim 1983, 513^{VP} (*Haemophilus pleuropneumoniae* Shope 1964, 362.) *pleu.ro.pneu.mo'niae*. Gr. n. *pleura* lung, sac; Gr. n. *pneumon* the lungs; M.L. fem. gen. n. *pleuropneumoniae* of pleuropneumonia.

Matthews and Pattison (1961) were the first to describe this organism, but they called it *H. parainfluenzae* and did not propose a new species (Kilian and Biberstein, 1984). Pohl et al. (1983) transferred the organism from the genus *Haemophilus* to *Actinobacillus* on the basis of phenotypic and DNA relatedness. It contains two provisional biotypes. Biovar 1 consists of V-factor-dependent strains hitherto classified as *H. pleuropneumoniae* (Matthews and Pattison) Shope strains, as described by Kilian et al. (1978), with strain Shope 4074 (CCM 5869 ATCC 27088) as the type strain. Biovar 2 consists of V-factor-independent strains and is represented by strain Bertschinger 2008/76 Frederiksen P597 = HIM 677-314.

According to the emended description by Pohl et al. (1983), cells are nonmotile, nonsporeforming, small, and coccoid to rod-shaped, occurring singly, in pairs, or in short chains. Cells are not acid-fast and contain demethylmenaquinone and ubiquinone.

Colonies on chocolate agar are smooth and grayish white and reach 3 mm in diameter after 48 h in a candle jar at 37°C. On bovine or sheep blood agar, the majority of isolates produce a narrow zone of complete hemolysis. The hemolysin acts synergistically with *Staphylococcus aureus* β -toxin, resulting in a positive CAMP reaction. Capsulated strains produce iridescent colonies on clear agar media (e.g., Levinthal's agar and PPLO agar). Biovar 1 isolates show satellite growth on blood agar plates when cross-inoculated with a nurse strain.

Aerobic, microaerophilic in primary isolation, facultatively anaerobic. No additional CO₂ is required for growth in subculture. Mesophilic. Chemoorganotrophic, with pronounced growth response to fermentable carbohydrates in peptone media. Minimal nutritional requirements are not known.

Biovar 1 is the dominant cause of porcine pleuropneumonia around the world. A total of 14 serovars have been recognized (and biovars 1 and 5 subdivided into 1a, 1b and 5a, 5b, respectively [Nielsen, 1990; Nielsen et al., 1997b]). X factors are not required. Attention is drawn to the cultural resemblance of *A. pleuropneumoniae* biovar 2 to *A. suis* and *Mannheimia varigena* (Bisgaard's Taxon 15) (Angen et al., 1999a; Genus *Pasteurella*, this edition).

Some biochemical reactions are given in Tables BXII.γ.291 and BXII.γ.292. A urease-negative variant of *A. pleuropneumoniae* has been described (Blanchard et al., 1993).

The mol% G + C of the DNA is: 43.2 or 42.2 (T_m) (Kilian et al., 1978); 42.2 (T_m) (strain Bertschinger 2008/76).

Type strain: 4074, ATCC 27088, NCTC 12370.

GenBank accession number (16S rRNA): M75074.

Additional Remarks: GenBank accession number for 16S rRNA of MCCM 00189 is AF224283.

12. **Actinobacillus porcinus** Møller, Fussing, Grimont, Paster, Dewhirst and Kilian 1996a, 956^{VP} *por.ci' nus*. M.L. adj. *porcinus* pertaining to pigs, hogs.

Cells are small rods (length 0.2–2.4 μ m) (Møller et al., 1996a). Colonies on chocolate agar are smooth and translucent to grayish. Some isolates grow poorly on chocolate agar, and their colonies are only 0.2 mm in diameter. Others grow readily on chocolate agar and form colonies that are approximately 0.6 mm in diameter after 48 h of incubation. Isolates show satellite growth on blood agar plates when cross-inoculated with a nurse strain. The fermentation of most carbohydrates is variable. No fermentation of inulin, esculin, and salicin. *A. porcinus* is V-factor-dependent, X-factor-independent, nonmotile, nonhemolytic, and urease, catalase, and indole negative. Lysine and ornithine are not decarboxylated. Isolated from the upper respiratory tracts of pigs.

The mol% G + C of the DNA is: 41.4 (HPLC).

Type strain: NM319, CIP 105315, CCUG 38924.

GenBank accession number (16S rRNA): U65583.

13. **Actinobacillus rossii** Sneath and Stevens 1990, 151^{VP} *ros.si' i*. M.L. gen. n. *rossii* of Ross, named after R.F. Ross, who, with his colleagues, isolated the organism (Ross et al., 1972).

Phenon 17 of Sneath and Stevens (1985); group "Ross" of Kilian and Frederiksen (1981b).

Cells are small, nonmotile bacilli, seldom coccobacilli that are often >2 mm long (Sneath and Stevens, 1990). Endospores are not formed. Growth is aerobic and facultatively anaerobic. Mesophilic. Surface colonies grown aerobically on sheep blood agar are round, grayish, semitransparent, and about 2 mm in diameter after 48 h at 37°C. Occasionally, weak hemolysis without marked greening of the erythrocytes is seen. *A. rossii* produces acid from inositol and sorbitol, but not from sucrose. The organism also reduces Janus green (Sneath and Stevens, 1985). Isolated from the vaginas of postparturient sows and aborted piglets.

The mol% G + C of the DNA is: 41.9 (T_m) (Mannheim et al., 1980).

Type strain: ATCC 27072, NCTC 10801.

14. **Actinobacillus scotiae** Foster, Ross, Patterson, Hutson and Collins 1998, 933^{VP}

sco.ti.ae. L. n. *scotiae* classical Latin name of Scotland; L. gen. n. *scotiae* of Scotland, where the isolates were collected from porpoises.

Cells are pleomorphic, nonmotile, Gram-negative rods. They are facultatively anaerobic. Added CO₂ is required for growth. Colonies on Columbia agar (Difco) supplemented with 5% citrated sheep blood that are incubated at 37°C in an atmosphere of 10% added CO₂ are 0.5 mm in diameter after 24 h. They are weakly hemolytic. Blood or serum enhances growth. No requirement of X or V factors. No growth on MacConkey agar. Growth occurs at 25°C, but not at 42°C. Catalase negative and oxidase positive. Nitrate is reduced. Acid is produced from glucose, lactose, and maltose, but not from adonitol, arabinose, dulcitol, inositol, mannose, melibiose, raffinose, rhamnose, salicin, sorbitol, sucrose, trehalose, or xylose. Some strains produce acid from galactose. Urease and Voges-Proskauer positive, but indole negative. Some strains generate ornithine decarboxylase. Lysine decarboxylase and arginine dihydrolase are not produced. The type strain is positive for ornithine decarboxylase and negative for acid production from galactose. *A. scotiae* has been isolated from porpoises.

The mol% G + C of the DNA is: 40.5 (*T_m*).

Type strain: M2000/95/1, ATCC 27072, NCTC 12922.

GenBank accession number (16S rRNA): Y09653.

15. ***Actinobacillus seminis*** (ex Baynes and Simmons 1960) Sneath and Stevens 1990, 151^{VP}
sem.i'nis. L. n. *semen* seed; L. gen. n. *seminis* of semen.

Phenon 14 of Sneath and Stevens (1985); "*Actinobacillus seminis*" of Kilian and Frederiksen (1981b) and Phillips (1984).

Cells are small, nonmotile bacilli and coccobacilli that are often >2 µm long (Sneath and Stevens, 1990). Gram reaction is negative. Endospores are not formed. Growth is aerobic and facultatively anaerobic, as well as mesophilic. On first isolation, 5% CO₂ is often required. On sheep blood agar, surface colonies grown aerobically are round, grayish, semitransparent, and about 1 mm in diameter after 48 h at 37°C. Colonies do not produce hemolysis, but sometimes produce greening of the erythrocytes. *A. seminis* has been associated with epididymitis and orchitis in rams; ovine contagious epididymitis; mastitis in ewes; and polyarthrititis, posthitis, and abortion in sheep.

A. seminis has been associated with epididymitis and orchitis in rams; ovine contagious epididymitis; mastitis in ewes; and polyarthrititis, posthitis, and abortion in sheep.

The organism produces slow and weak fermentation of carbohydrates and no acid from mannose. It also gives a negative phosphatase reaction (Sneath and Stevens, 1985).

The mol% G + C of the DNA is: 43.7 (*T_m*) (Mannheim et al., 1980).

Type strain: ATCC 15768, NCTC 10851.

GenBank accession number (16S rRNA): M75047.

16. ***Actinobacillus succinogenes*** Guettler, Rumler and Jain 1999, 214^{VP}
suc.ci.no'ge.nes. M.L. n. *acidum succinicum* succinic acid; Gr. v. *gennaio* to produce; M.L. adj. *succinogenes* succinic-acid-producing.

Cells are nonmotile, pleomorphic, Gram-negative rods (0.8 × 1 µm). Filamentous cells are occasionally seen.

"Morse code" forms and chains are common in actively growing broth cultures. Cells within an extracellular matrix are common in aerobically grown cultures. Chemoorganotrophic. Yeast extract and CO₂ stimulate growth. Colonies on TSB agar are circular, entire, gray, translucent, and 1–1.5 mm in diameter after 24 h incubation at 37°C with CO₂. Growth occurs at 37–39°C, but not at 20°C or 45°C. Catalase and oxidase positive. Alkaline phosphatase positive. Acid, but not gas, is produced from glucose and fructose within 24 h. Acid is also produced from arabinose, ribose, xylose, galactose, mannose, mannitol, sorbitol, amygdalin, arbutin, esculin, salicin, cellobiose, maltose, lactose, sucrose, raffinose, β-gentiobiose, arabinol, gluconate, and 5-ketogluconate. Acid is not produced from dulcitol, inositol, inulin, glycerol, erythritol, arabinose, xylose, adonitol, methyl β-xyloside, sorbose, rhamnose, methyl α-D-mannoside, methyl α-D-glucoside, N-acetylglucosamine, melibiose, trehalose, melezitose, amidon, glycogen, xylitol, turanose, xylose, tagatose, fucose, arabinol, and 2-ketogluconate. β-Galactosidase, arginine-β-arylamidase, and leucine arylamidase are produced in large amounts. Nitrates are reduced to nitrites. Indole and urease are not produced. Rapid and extensive growth is observed when utilizing arabinose, cellobiose, fructose, galactose, glucose, lactose, maltose, mannitol, mannose, sucrose, xylose, or salicin in media with yeast extract and corn steep liquor. Marked production of succinic acid in presence of CO₂ and high substrate concentration. *A. succinogenes* has been cultured from the bovine rumen.

The mol% G + C of the DNA is: 45 (HPLC) (Mesbah et al., 1989).

Type strain: 130Z, ATCC 55618.

GenBank accession number (16S rRNA): AF024525.

17. ***Actinobacillus suis*** van Dorssen and Jaartveld 1962, 456^{AL}
su'is. L. n. *sus* the pig, swine; L. gen. n. *suis* of the pig.

Cells are usually rod-shaped. Considerable variability may be seen, with long rods and filaments of variable length among cells grown on media both with and without glucose. Colonies on nutrient agar or blood agar are sticky and adhere to the medium. Growth is less viscous than in *A. equuli*. Stickiness increases with prolonged incubation of up to 72 h. In older colonies, a transparent border zone is developed, giving the appearance of a fried egg. The marked adherence to the medium may be lost upon repeated subculturing. Viscous growth also occurs in nutrient broth, but is not so marked as in *A. equuli*. After centrifugation, broth-grown sedimented cells are creamy yellow. Nutrient agar and nutrient broth cultures die within 15 d at 4°C.

A. suis usually ferments arabinose, dextrin, and galactose promptly, but a few strains may show slow fermentation with these substrates. Acid is also formed from cellobiose, lactose, maltose, melibiose, salicin, and sucrose, but not from mannitol and rhamnose. Glycerol and mannose are usually slowly fermented. Most strains of *A. suis* (and *A. equuli*), but not *A. lignieresii*, ferment melibiose and trehalose (Holmes, 1998). All strains of *A. suis*, but no strains of *A. equuli* and *A. lignieresii*, hydrolyze esculin. *A. suis* is an opportunistic pathogen of swine. Human infections have been reported as the result of pig bites.

The mol% G + C of the DNA is: 40.5 (*T_m*) (Mráz, 1968).

Type strain: ATCC 33415, CCM 5586.

GenBank accession number (16S rRNA): AF015299.

18. *Actinobacillus ureae* (Jones 1962) Muters, Pohl and Mannheim 1986, 343^{VP} (*Pasteurella ureae* Jones 1962, 150.)
u.re' ae. Gr. n. *urum* urine; M.L. gen. n. *ureae* of urine.

Cells are small, rod-shaped, and pleomorphic depending on the growth medium. Occasionally, bipolar staining is seen. Growth is best on media containing blood or serum. It occurs from 25–40°C, with optimal growth at 37°C. Differential biochemical characters are given in Tables BXII.γ.291 and BXII.γ.292. *A. ureae* occurs infrequently in the noses of healthy humans and occasionally causes ozaena and other infections in the respiratory tract.

The mol% G + C of the DNA is: 41.2–43.7 (T_m) (Pohl, 1979; Muters et al., 1984).

Type strain: Henriksen 3520/59, ATCC 25976, NCTC 10219.

GenBank accession number (16S rRNA): M75075.

19. “*Actinobacillus salpingitidis*” (Kohlert 1968) Mráz, Vladík and Boháček 1976.

Cells are coccobacteria of $0.5 \pm 0.1 \times 1.5 \pm 0.5 \mu\text{m}$, with a tendency for bipolar staining (Mráz et al. 1976). Longer polymorphic rods and curved filaments occur. In the microscope, cells appear individually, in pairs, and in short chains. Long chains can be found in broth cultures. Nonmotile. Gram negative.

Colonies on nutrient agar are 0.1 mm in diameter. Some strains show a mixture of colony sizes (0.1–0.5 mm). They glisten in transmitted light. Older colonies of larger types reach 1 mm and acquire a grayish appearance. Colonies on nutrient agar with sheep's blood are 0.2–1 mm in diameter, exhibiting a 0.5–1.5 mm zone of expressive or total hemolysis. After 2 d of incubation, they measure 0.5–2 mm. The larger zones turn grayish and have an umbonate profile. Colonies on nutrient agar with lamb's blood, according to Smith (1962), have a simple zone of expressive or total hemolysis. MacConkey agar with crystal violet (BioQuest)

yields visible growth during 48 h of incubation. Endo agar provides no growth, except in sporadic cases. Nutrient broth yields homogeneous or fine-granulated turbidity with possible larger flakes. From the third day of incubation, a whitish-gray sediment is formed. With slight shaking the content of the test tube may be easily redispersed. Growth in litmus milk gives no changes or sporadic delayed discoloration.

Metabolism is fermentative. During 2 days of incubation, acid, but no gas, is formed from glucose, xylose, levulose, mannose, galactose, saccharose, trehalose, and mannitol; within 14 d acid is also formed from lactose (3–12 d), raffinose (2–10 d), and glycerol (2–7 d). Results are variable and, for a great part, delayed for arabinose, maltose, inulin, starch, dextrin, sorbitol, and inositol, while rhamnose, cellobiose, melibiose, adonitol, dulcitol, salicin, and esculin remain negative.

Nitrites are produced from nitrates. Urea is not hydrolysed. Indole is not produced. Hydrogen production is variable. The methyl red test is negative. Acetoin is not produced. Ammonia production is variable. Catalase is positive. Oxidase is usually positive, and cytochrome oxidase is negative. Arginine dihydrolase and lysine and ornithine decarboxylases are negative. β -Galactosidase is positive. Facultatively anaerobic. Growth at 30–40°C. Temperature optimum is 37°C. Isolated from the respiratory tract and oviduct of healthy domestic fowl, from salpingo-peritonitis in hens, and from the organs of dead chicks and young turkeys.

The mol% G + C of the DNA is: 39.6–42.9; mean value 41.5 (T_m).

Deposited strain:

GenBank accession number (16S rRNA): L06077.

Additional Remarks: Suggested neotype strain: CCM 5974 (strain 556/71 of Mráz et al., 1976) = CCUG 23139.

Genus III. *Haemophilus* Winslow, Broadhurst, Buchanan, Krumwiede, Rogers and Smith 1917, 561^{AL}

MOGENS KILIAN

Hae.mo'phi.us. Gr. n. *haima* blood; Gr. n. *philos* lover; M.L. masc. *Haemophilus* blood-lover.

Minute to medium-sized **coccobacilli or rods**, generally less than 1 μm in width and variable in length, sometimes forming threads or filaments and showing **marked pleomorphism**. Gram negative. **Nonmotile. Aerobic or facultatively anaerobic. Require pre-formed growth factors present in blood**, particularly X factor (protoporphyrin IX or protoheme) and/or V factor (nicotinamide adenine dinucleotide [NAD] or NAD phosphate [NADP]). Even after specific growth factors have been provided, growth is best on complex media. Optimum temperature, 35–37°C. **Nitrates are reduced to or beyond nitrites**. Oxidase and catalase reactions vary among strains. **Chemorganotrophic. All species can attack carbohydrates fermentatively**, yielding acetic, lactic, and succinic acids as end products in glucose broth. Occur as **obligate parasites on the mucous membranes of humans and a variety of animal species**. Several 16S rRNA sequence signatures for the family *Pasteurellaceae* have been demonstrated, but none of these is specific for the genus *Haemophilus* as presently defined (Dewhirst et al., 1992).

The mol% G + C of the DNA is: 37–44.

Type species: ***Haemophilus influenzae*** (Lehmann and Neumann 1896) Winslow, Broadhurst, Buchanan, Krumwiede, Rogers, and Smith 1917, 561 (*Bacterium influenzae* Lehmann and Neumann 1896, 187.)

FURTHER DESCRIPTIVE INFORMATION

Phylogenetic position The genus *Haemophilus* is located in the *Gamma*proteobacteria and is a member of the family *Pasteurellaceae* (Sneath and Johnson, 1973; Pohl, 1979, 1981a; Mannheim et al., 1980; Mannheim 1981; Chuba et al., 1988; De Ley et al., 1990; Dewhirst et al., 1992, 1993).

Cell morphology Cells of *Haemophilus* species tend to occur as individual short rods or coccobacilli. Filament formation is environmentally influenced and develops as cultures age and are under less than optimum conditions of growth (Fig. BXII.γ.216). Capsules are present in some species, notably *H. influenzae*, *H.*

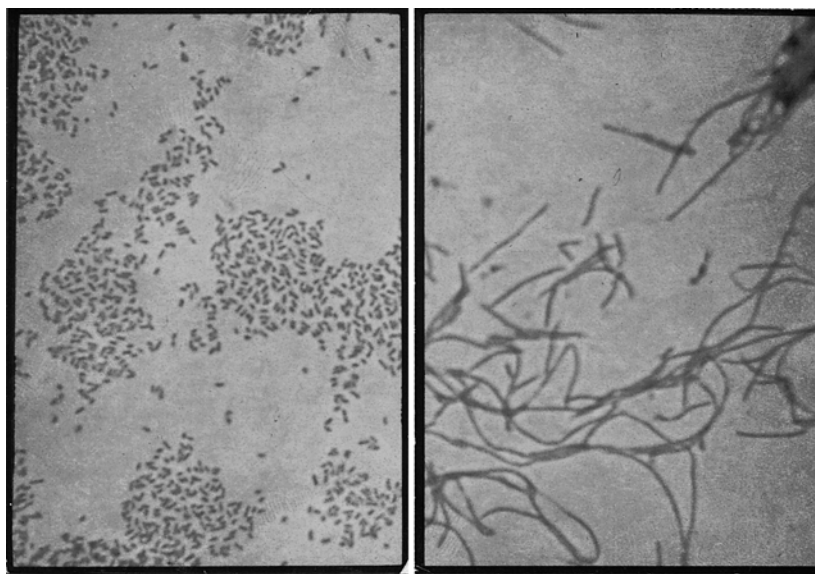


FIGURE BXII.γ.216. Cell morphology of a strain of *H. influenzae* (left) and a strain of *H. parainfluenzae* (right), illustrating the extent of pleomorphism that may be observed in bacteria belonging to this genus.

paragallinarum, and *H. parasuis* and are of particular interest in *H. influenzae* and *H. paragallinarum*, where they play a part in pathogenesis, determination of type specificity, and induction of anti-infective immunity. Occasional capsulate strains of *H. parainfluenzae* have been described (Sims, 1970).

Cytoplasmic membrane composition The membrane phospholipids of *Haemophilus* spp. are similar to those of other Gram-negative bacteria, with phosphatidylethanolamine being the major type. Smaller quantities of phosphatidylglycerine have also been detected. It has been suggested that trace amounts of lysophosphatidylethanolamine may be a characteristic marker of *Pasteurellaceae* in general (Mutters et al., 1993).

Cell wall composition The cell walls resemble those of other Gram-negative bacteria in structure, composition, and endotoxic activity. Unlike the lipopolysaccharide (LPS) of *Enterobacteriaceae*, *H. influenzae* LPS lacks long polysaccharide O chains and, thus, may be described more accurately as a lipooligosaccharide (LOS). It consists of a membrane-anchoring lipid A, an inner core of a singly phosphorylated 2-keto-3-deoxyoctulosonic acid (KDO) linked to three heptose molecules, and an outer core oligosaccharide consisting of a heteropolymer of glucose and galactose and a sialylated terminal lactosamine (Zamze and Moxon, 1987; Phillips et al. 1993; Roche and Moxon, 1995; Risberg et al., 1997). There is marked intra- and inter-strain heterogeneity in the composition and linkage of these saccharide units in the outer core of *H. influenzae* LPS and hence in the epitopes expressed (van Alphen et al., 1990; Roche and Moxon, 1995). The lipid A backbone is a glucosamine disaccharide substituted by two phosphate groups, as in enterobacterial lipid A, but with a simpler fatty acid composition of only tetradecanoic acid and its 3-hydroxylated derivative (Maskell et al., 1992). KDO is present in lesser amounts than in the LPS of *Enterobacteriaceae*: 0.4–1.5% of LPS in *H. influenzae*, compared to 5–8% in *Salmonella* spp. (Flesher and Insel, 1978).

Numerous genes are involved in the LPS biosynthesis by *H. influenzae*. These include the *lic* genes organized in three chromosomal loci (*lic1*, *lic2*, and *lic3*) (Weiser et al., 1990; Maskell et

al., 1992; Hood et al., 1996). The first gene (*lic1A*, *lic2A*, *lic3A*) of each of these multigene loci contains variable numbers of tandem repeats of the tetramer CAAT which, by facilitating slipped-strand mispairing, are responsible for the variable expression of LPS epitopes (Weiser et al., 1989; Maskell et al., 1992; High et al., 1996). An additional source of variation is the on-off switching in the expression of phosphorylcholine (ChoP). Choline is taken up from the environment by *H. influenzae* and incorporated into its LPS as ChoP under control of the *licA* gene in the *lic1* locus. The translational switch is based on variation in the number of CAAT repeats in the *licA* gene (Kolberg et al., 1997; Weiser et al., 1997). The frequency of spontaneous, reversible gain and loss (phase variation) of LPS oligosaccharide epitopes expressed by genes in the three chromosomal loci *lic1*, *lic2*, and *lic3* is 10^{-2} – 10^{-3} /generation (Weiser et al., 1990).

As in other members of the family *Pasteurellaceae*, the number of cell wall fatty acids found in *Haemophilus* species is comparatively low. The general pattern is characterized by relatively large amounts of *n*-tetradecanoate ($C_{14:0}$), 3-hydroxy-tetradecanoate ($C_{14:0\ 3OH}$), hexadecanoate ($C_{16:1}$), and *n*-hexadecanoate ($C_{16:0}$). The three C_{18} fatty acids, octadecadienoate ($C_{18:2}$), octadecanoate ($C_{18:1}$), and *n*-octadecanoate ($C_{18:0}$) are also present, but in low concentrations. Small species-to-species variations in the relative amounts of these fatty acids have been demonstrated, but discrimination based on such differences cannot be achieved (Jantzen et al., 1981; Brondz and Olsen, 1984b; Brondz and Olsen, 1989; Mutters et al., 1993).

Analyses of polyamine patterns of whole cells of representative strains of species within the family *Pasteurellaceae* reveal unexpected diversity compared to that observed for other Gram-negative rods. At least seven different polyamine patterns are present among members of the family. All current species of *Haemophilus*, with the exception of *H. felis*, show a predominance of 1,3-diaminopropane (Table BXII.γ.293) (Busse et al., 1997). However, the extent of intra-species diversity is not yet known.

Outer membrane proteins (OMPs) have been the subject of significant interest, particularly in *H. influenzae*, as a means of typing and as potential vaccine candidates. There have been 25–

TABLE BXII.γ.293. Polyamine patterns of *Haemophilus* species and selected related taxa^a

Polyamine content (percentage of total polyamines) ^b	<i>Haemophilus influenzae</i>		<i>Haemophilus aegyptius</i>		<i>Haemophilus aphrophilus</i>		<i>Haemophilus ducreyi</i>		<i>Haemophilus felis</i>		<i>Haemophilus paracuniculus</i>		<i>Haemophilus paragaillinarum</i>		<i>Haemophilus parahaemolyticus</i>		<i>Haemophilus parainfluenzae</i>		<i>Haemophilus segnis</i>		<i>Actinobacillus actinomycetemcomitans</i>		<i>"Histophilus ovis"</i>	
	Type	MCCM 00678	Type	MCCM 00678	Type	MCCM 02064	Type	MCCM 02064	Type	MCCM 02064	Type	MCCM 02064	Type	MCCM 02064	Type	MCCM 02064	Type	MCCM 02064	Type	MCCM 02064	Type	MCCM 00330	MCCM 02133	
DAP	96.2	91.4	62.9	0	98.2	79.1	2.5	33.4	71.6	84.9	73.7	70.6	86.7	40.2	65.4	99.7	97.6	88.5	78.7					
PUT	1.3	8.2	22.3	0	0.4	0.1	70.4	1.9	0.2	0.3	2.6	3.3	0	21.3	32.4	0	0	5.9	1.1					
CAD	0	0.5	14.7	0	0.2	0.5	5.6	0	0.2	0.6	0.5	tr	0	39.5	0.4	0	0	1.3	0.6					
TYR	0.9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
NSPD	0	0	0	0	0	tr	17.7	40.0	tr	tr	0	0	0	0	0	0	0	0	0					
SPD	1.4	tr	0	0	0.9	11.5	3.7	20.5	9.3	9.1	20.5	24.7	5.5	0	1.7	0.3	1.2	4.3	10.9					
HSPD	0	0	0	0	0	0	0	0	0.8	tr	0	0	0	0	0	0	0	0	0					
SPM	0.1	tr	0	0	0.2	8.9	tr	4.2	17.9	5.1	2.7	1.5	7.8	0	0	0	1.2	tr	8.8					

Compiled from Busse et al. (1997)

^aCompiled from Busse et al. (1997).^bAbbreviations: DAP, 1,3-diaminopropane; PUT, putrescine; CAD, cadaverine; TYR, tyramine; NSPD, *syn*-homosermidine; SPD, spermidine; HSPD, *syn*-homospermidine; SPM, spermidine; tr, trace (less than 0.05 mmol/g dry weight).

35 OMPs detected in *H. influenzae* strains, with only a few accounting for 80% of the total OMPs (Loeb and Smith, 1982). The nomenclature of the major proteins has been reviewed by van Alphen (1993). The major OMPs are P1 (MW 43-50 kDa), P2 (a doublet of MW 43-50 kDa with porin activity), P4 (a lipoprotein of MW 30 kDa), P5 (2 conformers of the same protein), and P6 (a lipoprotein of MW 16.6 kDa). The protein originally designated P3 consists of both the non-heat-modified form of P1 and the heat-modified form of P5. The size polymorphism of proteins P1, P2, and P5 forms the basis for OMP typing of *H. influenzae*. While P2 is conserved in *H. influenzae* serovar b strains, it is extremely diverse and shows antigenic drift in noncapsulated strains (Duim et al., 1994; Murphy, 1994). P4 and P6 are reported to be conserved in *H. influenzae* (Granoff and Munson, 1986; Nelson et al., 1988; Barenkamp, 1992; van Alphen, 1993; Gilsdorf, 1998). OMP profile diversity has also been detected among isolates of *H. paragallinarum* (Blackall et al., 1990b).

Stable wall-deficient variants (L-forms) have been produced experimentally from *H. influenzae* (Roberts et al., 1974).

Fine structure The cell wall of *Haemophilus* species is typical of Gram-negative bacteria, having an ultrastructure composed of multiple wavy outer membranes and a poorly defined plasma membrane, with an intervening electron-transparent space (Sherwin and Wilkins, 1973; Kilian and Theilade, 1975; Doern and Buckmire, 1976; Kilian and Theilade, 1978; Holt et al., 1980). The entire cell wall, including the cell membrane, averages 20 nm in thickness. Vesicular structures ("blebs") have been demonstrated on the outer wall of several *Haemophilus* species (Holt et al., 1980). These are morphologically identical to lipopolysaccharide vesicles and are released into the surroundings. Methods for the isolation of the inner and outer membranes from *H. influenzae* serovar b cells have been devised by Loeb et al. (1981).

Fresh clinical isolates of *H. influenzae* usually express peritrichous pili with hemagglutinating properties. However, after serial subcultivations most isolates, either capsular or noncapsular, do not express such pili. Exceptions are *H. aegyptius* and the original Brazilian purpuric fever (BPF)-associated clone of *H. influenzae*, which are more stably piliated (Gilsdorf et al. 1997; Read et al., 1998). The hemagglutinating pili are 4.7–18 nm in diameter and 209–453 nm in length, with a hollow core. They are composed of a pilin protein subunit of M_r 21,000. Pilus expression is governed by a 5-gene operon (*hifA*–*hifE*) and oscillates between phase off and phase on at a rate of about 10^{-4} /generation. This phase variation is likely due to slip-strand mispairing in a poly(TA) dinucleotide tract that separates the divergent, bidirectional *hifA* and *hifB*–*E* promoters, altering the spacing upstream of the transcriptional start sites (van Ham et al., 1993). Some strains, including the previously mentioned conjunctival *Haemophilus* taxa (*H. aegyptius* and BPF-clone), possess duplicated pilus gene clusters, which may explain the higher stability of pilus expression (Read et al., 1998). In addition to the hemagglutinating pili, other types of adherence organelles that do not mediate hemagglutination have been described for *H. influenzae* (Jacques and Paradis, 1998). Some strains of *H. influenzae* biotype IV, which constitute a distinct evolutionary lineage (see below) and may be isolated from neonatal infections or the female genital tract, possess peritrichous pili that do not hemagglutinate, but mediate binding to HeLa cells (Rosenau et al., 1993). Fimbrial structures that are thinner than pili have been demonstrated in otitis media isolates of noncapsular *H. influenzae* (Bak-aletz et al., 1988). In addition, different morphological types of

hemagglutinating and non-hemagglutinating pili were described by Brinton et al. (1989), but their relationship to those mentioned above has not been clarified. Fine tangled pili composed of a unique major protein subunit of M_r 24 kDa have been described in strains of *H. ducreyi* (Brentjens et al., 1996). Surface-associated thin hairlike structures of an unidentified nature have been demonstrated in *H. aphrophilus*, *H. paraphrophilus* (Holt et al., 1980), occasional strains of *H. parainfluenzae* (Kahn and Gromkova, 1981), and *H. parasuis* (Munch et al., 1992).

Colonial and cultural characters Surface colonies of *Haemophilus* species on sufficiently rich media are usually nonpigmented or slightly yellowish, flat, and convex, and they attain a diameter of 0.5–2.0 mm within 48 h at 37°C. *H. influenzae* undergoes spontaneous phase variation in colony opacity. Weiser et al. (1995) have identified a gene locus that contributes to opacity variation and is associated with the ability to colonize the nasopharynx of infant rats. Most species produce smooth colonies, but some variation is seen, particularly in *H. parainfluenzae* and *H. aphrophilus*. Some species show β hemolysis on blood agar (see Table BXII.γ.294). Growth in broth media usually shows even turbidity, but strains of *H. aphrophilus*, *H. paraphrophilus*, and *H. parainfluenzae* show granular growth with heavy deposits, due to aggregation of cells.

Nutrition and growth conditions Apart from the specific growth factor requirements, the various species of *Haemophilus* exhibit some variation in their nutritional needs and preferred growth conditions. The most universally satisfactory propagative media are chocolate agar and Levinthal media (agar and broth). The former has the virtue of relative ease of preparation, the latter that of transparency, which facilitates the recognition of colonial phases and dissociation phenomena. Colonial iridescence, a property highly correlated with encapsulation, is most readily recognized on Levinthal's agar.

The most critical ingredients of any medium for *Haemophilus* spp., whether for propagation or for characterization, are the growth factors X and/or V ("vitamin-like"). All members of the genus require one or both of these growth factors (few exceptional isolates, see below), and the exact requirement is one of the key characteristics of the individual species (see Table BXII.γ.294). X factor is usually protoporphyrin IX but, under conditions where iron cannot be obtained from another source, the iron-containing protoheme is required (Lwoff and Lwoff, 1937; White and Granick, 1963; Pidcock et al., 1988). Blood or blood derivatives, including hemin, are the traditional and adequate sources of X factor. The customary 5% blood used in blood and chocolate agar is ample. When crystalline hemin is used, required amounts vary between 0.1 and 10 µg/ml for *H. influenzae* (Gilder and Granick, 1947; Brumfitt, 1959; Biberstein and Spencer, 1962; Evans et al., 1974), while the optimum is 200 µg/ml for *H. ducreyi* (Hammond et al., 1978a). The X factor is heat stable and remains active in autoclaved media. The requirement for X factor is substantially reduced, but not eliminated, during anaerobic growth (Gilder and Granick, 1947). The X factor requirement is due to lack of some or all of the enzymes involved in the biosynthesis of protoporphyrin from δ -aminolevulinic acid (Biberstein et al., 1963; White and Granick, 1963) (Fig. BXII.γ.217). Isolates of *H. aphrophilus* often become independent of X factor upon subcultivation (King and Tatum, 1962; Sutter and Finegold, 1970; Kilian, 1976). The biochemical basis for this is not known, but it may suggest that the apparent X

TABLE BXII.γ.294. Differential characteristics of the species of the genus *Haemophilus*

Characteristics	1. <i>H. influenzae</i>	2. <i>H. aegyptius</i>	3. <i>H. aphrophilus</i>	4. <i>H. avium</i>	5. <i>H. ducreyi</i>	6. <i>H. haemoglobinophilus</i>	7. <i>H. haemolyticus</i>	8. <i>H. paracuniculus</i>	9. <i>H. paragaillinarum</i>	10. <i>H. parahaemolyticus</i>	11. <i>H. parainfluenzae</i>	12. <i>H. paraphrohaemolyticus</i>	13. <i>H. paraphrophilus</i>	14. <i>H. parvus</i>	15. <i>H. segnis</i>	a. " <i>H. somnus</i> "	b. " <i>H. agni</i> "
V-factor requirement	+	+	-	-	-	-	-	-	-	+	-	-	-	-	-	-	-
ALA→porphyrins	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Indole production ^a	p	-	-	-	-	-	p	+	+	+	+	+	+	+	+	+	p
Urease ^a	p	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ornithine decarboxylase ^a	p	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Arginine dihydrolase	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Hemolysis	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
D-Glucose, acid production	+	W	+	+	W	+	p	+	+	p	p	+	+	+	+	+	+
D-Glucose, gas production	-	-	+	-	-	-	W	+	+	+	+	+	+	+	+	+	+
Acid from:																	
D-Fructose	-	-	+	+	-	+	+	+	+	+	+	+	+	+	+	+	+
Sucrose	-	-	+	+	-	+	+	+	+	+	+	+	+	+	+	+	+
Lactose	-	-	+	+	-	+	-	-	-	-	-	-	-	-	-	-	-
D-Xylose	+	-	d	-	-	p	d	-	p	-	-	-	-	p	-	+	W
D-Ribose	+	W	+	-	-	+	+	-	+	-	-	-	+	+	-	+	+
D-Mannose	-	-	+	+	-	+	-	-	+	-	+	-	+	+	-	+	-
D-Mannitol	-	-	-	+	-	+	-	-	p	-	-	-	+	+	-	+	W
D-Sorbitol	-	-	-	-	-	-	-	-	+	-	-	-	-	-	-	+	-
β-Galactosidase (ONPG test) ^b	-	-	+	+	-	p	-	+	+	-	p	p	-	-	p	p	-
α-Fucosidase	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Catalase	+	+	-	-	-	-	+	+	-	p	p	+	+	+	p	+	+
CO ₂ enhances growth	-	-	+	+	-	-	-	+	-	-	-	+	+	+	-	+	-
Alkaline phosphatase	+	+	+	+	-	-	-	+	+	-	-	+	+	+	-	-	-
IgA1 protease	+	+	+	+	-	-	-	+	+	+	+	+	+	+	+	+	-

^aFor variations within *H. influenzae* and *H. parainfluenzae* see definition of biovars (Table BXII.γ.296).^bSome strains of *H. parainfluenzae* biovar II show weak (w) hemolysis.

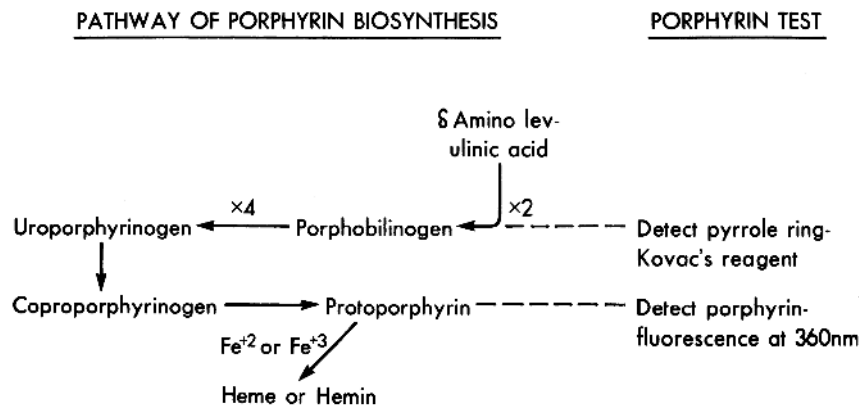


FIGURE BXII.γ.217. Steps in the biosynthesis of heme, with indications of compounds that are demonstrated in the Porphyrin test for X-factor requirement.

factor dependency in fresh isolates of *H. aphrophilus* is not due to the enzyme deficiencies in the porphyrin biosynthetic pathway described for *H. influenzae*. Indeed, White and Granick (1963) reported that one of Khairat's (1940) original isolates had all the enzymes of the biosynthetic pathway characteristic for hemin-independent species.

The V factor is minimally equivalent to nicotinamide mononucleotide (NMN) or nicotinamide riboside (NR), but is usually described as nicotinamide adenine dinucleotide (NAD). It is questionable if NAD phosphate (NADP) can serve as a general substitute for NAD (O'Reilly and Niven, 1986). Pyridine compounds that can serve as V factor can be characterized by possession of an intact pyridine-ribose bond in the β -configuration and a pyridine carboxamide bond at position 3. When NAD is supplied, it is cleaved by extracytoplasmic enzymes to yield NMN or NR, which is taken up and used in the biosynthesis of NAD (Wheat and Pittman, 1960; O'Reilly and Niven, 1986; Cynamon et al., 1988). It appears that none of the members of the family *Pasteurellaceae* are capable of *de novo* synthesis of NAD. V-factor dependency in some species can be explained by their inability to catalyze the formation of the nicotinamide-ribose bond, while all members of the family are incapable of amidating the carbonyl group on a nicotinate residue. Thus, the difference between V-factor dependence and independence within the family is much more subtle (the presence of nicotinamide phosphoribosyltransferase) than hitherto believed (reviewed by Niven and O'Reilly, 1990). Occurrence of V-factor-independent variants of *H. parainfluenzae* from human infections and of poultry isolates of *H. paragallinarum* has been detected in South Africa (Gromkova and Koornhof, 1990; Windsor et al., 1991; Mouahid et al., 1992; Mifflin et al., 1995). The property in *H. parainfluenzae* isolates is associated with a 5.25 kb plasmid that can be transferred to *H. influenzae* and confer independence of the V factor (Windsor et al., 1991). Recent studies demonstrate that this plasmid is normally present in strains of *H. ducreyi* (Martin et al., 2001).

The V factor is inactivated by autoclaving and is therefore absent in traditional agar or broth media. Although present in blood, V factor is unavailable in blood agar owing in part to its intracellular location and in part to the presence of NADase in the blood of many species (Krumwiede and Kuttner, 1938). In heated blood agar ("chocolate agar"), in which the blood is added to the basic medium at a temperature of approximately 70°C, the NAD is liberated from the blood cells and the NADase activity destroyed by the high temperature. Traditional sources

of V factor, apart from blood, have been yeast derivatives (Thjötta and Avery, 1921), which must be added after autoclaving of the medium. When met by crystalline NAD, V factor requirements of *H. influenzae* were found to range from 0.2–1 $\mu\text{g/ml}$; those of *H. parainfluenzae*, from 1–5 $\mu\text{g/ml}$, with some strains requiring as much as 25 $\mu\text{g/ml}$ for optimum growth (Evans et al., 1974).

When inoculated on media deficient in V factor but containing the X factor (e.g., blood agar) *Haemophilus* colonies cluster around contaminant colonies of other bacteria which produce the critical factors in excess (Grassberger, 1897). This phenomenon is called satellitism (Fig. BXII.γ.218) and is sometimes utilized for propagation and characterization of *Haemophilus* spp. It may be demonstrated by inoculating a medium such as blood agar with a strain requiring the V factor. The inoculated area is crossed with a single streak of *Staphylococcus aureus*. Early growth of the *Haemophilus* strain is confined to the area immediately adjacent to the line of staphylococcal growth. It gradually spreads peripherally but remains heaviest in the area nearest the "feeder" streak, reflecting the diffusion gradient of the limiting growth factor. Satellitism on blood agar is strongly suggestive of a V-factor requirement. However, strains of strongly hemolytic strains, particularly *H. parahaemolyticus*, do not show satellitism, due to liberation of ample V factor from the lysed blood cells (Kilian and Poulsen, in preparation). For such strains, the V-factor requirement can be demonstrated on a blood-free agar medium cross-inoculated with *S. aureus* or supplied with a NAD-containing paper disk, or on a medium to which blood is added before autoclaving.

The optimum growth temperature is 35–37°C and the minimum temperature is 20–25°C. Most *Haemophilus* strains are killed by heating at 55°C for 30 min. *H. influenzae* grows better under aerobic than anaerobic conditions. Raised carbon dioxide tension is beneficial, and may be required for surface growth of a number of species, including *H. paragallinarum*, *H. aphrophilus*, *H. paraphrophilus*, and *H. paraphrohaemolyticus*. In the case of *H. paragallinarum* propagated in liquid media, CO₂ has been found to replace a 5% serum requirement observed for some strains under fully aerobic conditions (Rimler et al., 1976; Blackall and Reid, 1982). A serum requirement also exists for *H. parasuis* and *H. ducreyi*. The precise role of serum in these situations is not known. Detoxification of media constituents has been suggested (Page, 1962). Sodium chloride at a concentration of 1–1.5% is essential for the growth of *Haemophilus paragallinarum* (Rimler et al., 1977). Completely synthetic media have been devised re-

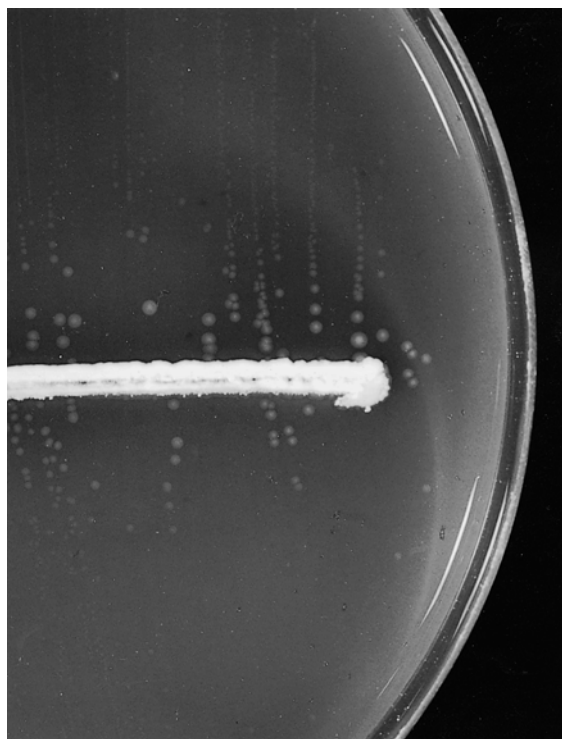


FIGURE BXII.7.218. Satellitic growth of *H. influenzae* colonies in the vicinity of a streak of *Staphylococcus aureus* that provides the V factor.

peatedly, particularly for *H. influenzae* and *H. parainfluenzae* (Herbst and Snell, 1949; Talmadge and Herriott, 1960; Butler, 1962; Wolin, 1963; Herriott et al., 1970; Klein and Luginbuhl, 1979). In all instances, the adequacy of the medium was tested either with a very limited number of isolates, or was found to support the growth of only a portion of the strains tested. The simplest (Klein and Luginbuhl, 1979) and one of the most complex (Herriott et al., 1970), when tested extensively in parallel, performed exactly alike with regard to their growth-supporting properties.

Metabolism Enzymes of the Embden–Meyerhof–Parnas, hexose monophosphate, Entner–Doudoroff, and tricarboxylic acid cycle pathways have been demonstrated in several *Haemophilus* species (Klein, 1940; White, 1966; Holländer, 1976). Glycolytic growth in the “classical” (White and Sinclair, 1971) sense, however (i.e., in the absence of a functional electron transport chain), has been observed only in X factor-requiring members of the genus, e.g., *H. influenzae*, *H. aegyptius*, and *H. haemoglobinophilus* (White, 1963). In the *H. parainfluenzae* strain studied, glycolysis (as well as oxidative catabolism of glucose) occurred only in the presence of a functional electron transport system and a terminal electron acceptor, viz. oxygen, nitrate, or fumarate (White, 1966; White and Sinclair, 1971). The electron transport chain in the species examined proceeds from flavoproteins (dehydrogenases) via cytochromes *b*, *c*, *d*, and *a* to demethylmenaquinone (DMK) to cytochrome *o* and/or oxygen or nitrate (White and Sinclair, 1971; Holländer, 1976). Strains of *Haemophilus* spp. contain mainly DMK-6 (mean value 78%) with smaller amounts (~10%) of menaquinone-8 (MK-8) and DMK-7. Only *H. aphrophilus* and *H. segnis* lack MK-8. *H. haemoglobinophilus* is unique among the present *Haemophilus* spp. in that it contains exclusively MK-7 (Mutters et al., 1993).

A comprehensive overview of the metabolism of *H. influenzae* as compared with *Escherichia coli* deduced from the genome sequences has been presented by Tatusov et al. (1996).

Genetics *H. influenzae* is the first free-living organism whose genome has been fully sequenced. The 1.83-Mb sequence of the strain Rd (rough variant of a capsular serovar d strain) contains a total of 1743 genes, of which more than 40% have no apparent counterparts of known function in other procaryotes that have been analyzed (Fleischmann et al., 1995). Genetic maps of a strain of *H. parainfluenzae* (genome size 2.34 Mb) and *H. influenzae* serovar b strain Eagan have been constructed by the use of pulsed field gel electrophoresis (Kauc and Goodgal, 1989; Butler and Moxon, 1990). Numerous genes and operons encoding virulence and other properties of *H. influenzae* and other *Haemophilus* species have been cloned and sequenced (available from the National Center for Biotechnology Information).¹

The genome sizes of *Haemophilus* species range between 1.8 and 2.8 Mb, with *H. influenzae* being the smallest and *H. ducreyi* the largest among species that have humans as their natural habitat (Burbach, 1987). The overall genetic diversity of the genus *Haemophilus*, as revealed by DNA–DNA hybridization studies, is considerable. The most comprehensive strain materials have been examined by Pohl (1979, 1981a) and Burbach (1987), who observed the following binding ratios to DNA of the type species *H. influenzae*: *H. aegyptius*, 70%; *H. haemolyticus* 48–52%; *H. parainfluenzae*, 15–33%; *H. aphrophilus*, 17%; *H. segnis*, 12%; *H. parahaemolyticus*, 14%; *H. paraphrohaemolyticus*, 16%; *H. parasuis* 26%; *H. paracuniculus*, 10%; and *H. haemoglobinophilus*, 22%. Relatively higher binding values, but with the same range, were reported in several other studies. Albritton et al. (1984) and Casin et al. (1986) found more than 90% homology between *H. influenzae* and *H. aegyptius*, whereas the lowest binding ratio (18%) in the study by Albritton et al. (1984) was observed between *H. influenzae* and *H. ducreyi*.

DNA–DNA hybridization studies also have revealed considerable heterogeneity within species. Burbach (1987) found that intra-species homologies among the species subjected to detailed analysis may be as low as 50% within *H. influenzae* and *H. parainfluenzae* and that certain strains of the two species show higher inter-species homology. This problem was solved by creating a new species designated “*H. intermedius*” (see discussion below) (Burbach, 1987).

Available data on 16S rRNA sequences of *Haemophilus* species are presented in Fig. BXII.7.215. of the chapter on *Pasteurellaceae* by Olsen et al. (this *Manual*). It is notable that relative relationships revealed by DNA–DNA hybridization data are not always reflected in the relationships suggested by the 16S rRNA sequence homologies.

Although many gene sequences from *H. influenzae* reveal mosaic-like patterns and thus evidence of recombination, the population structure of at least encapsulated forms is largely clonal. Multilocus enzyme electrophoresis (MLEE) analysis of over 2000 capsulated *H. influenzae* strains from 30 countries has revealed a limited number of distinct evolutionary lineages, with the major branches corresponding closely to the six capsular serovars (Musser et al., 1990). However, two distinct branches are observed among strains expressing a serovar b capsule. Only one of these branches (division I) contains strains regularly isolated from in-

1. At the time this manual went to press, this information could be accessed at www.ncbi.nlm.nih.gov/htbin-post/Taxonomy/wgetorg?id=12&lvl=3.

vative infections, and only 8–9 clones from this branch appear to be responsible for the majority of *H. influenzae* serovar b disease worldwide (Musser et al., 1988, 1990). The population of nonencapsulated *H. influenzae*, which is distinct from that containing capsulated strains, appears to be very large and heterogeneous (Musser et al., 1986; Porras et al., 1986). Whether this significant clonal diversity among non-encapsulated *H. influenzae* is due to more extensive recombination affecting the population structure or merely reflects a larger population size has not been properly analyzed.

Studies of mainly Australian isolates of *H. parausis* using MLEE analysis also reveal a clonal population structure with two major divisions that partly correlate with serovars (Blackall et al., 1997).

Several mechanisms of horizontal gene transfer have been detected in *Haemophilus* species. Alexander and Leidy (1951) utilized capsulation as a genetic marker to demonstrate the occurrence of transformation in *H. influenzae*. Noncapsulated R variants derived from each of the six capsular serovars a–f were examined for the ability to be transformed by DNA extracted from capsulated donor strains. Strain Rd proved to be the most dependable recipient and could be transformed to capsular types a, b, c, d, e, or f (or combinations of these), according to the capsular type of the DNA donor (Alexander et al., 1954). Catlin and Tartagni (1969) have detected serovar b antigen by immunofluorescence microscopy 40 min after adding transforming DNA to a population of strain Rd. Antibiotic resistance genes located in the chromosome, as well as on small plasmids, have been transferred between *H. influenzae* strains by genetic transformation (Stuy, 1979).

The transformation frequency for *H. influenzae* strain Rd in a rich medium increases from 10^{-7} in early exponential growth phase to about 10^{-4} in late exponential phase. The addition of cAMP to early-phase cultures increases the frequency of transformation to 10^{-4} (Wise et al., 1973; Redfield, 1991). Furthermore, under transient anaerobiosis or transfer to starvation medium, *H. influenzae* transformation frequencies are optimized to approximately 10^{-2} (Herriott et al., 1970). The increased competence under such conditions is paralleled by an altered envelope composition (Zoon and Scocca, 1975). A specific binding protein for homologous double-stranded DNA has been detected in the cell envelope of *H. influenzae* (strain Rd) (Deich and Smith, 1980). This membrane protein recognizes a specific 11-nt sequence that appears with much higher frequency in *Haemophilus* than in other genera (Sisco and Smith, 1979; Danner et al., 1980). Both intra- and interspecific transformation have been demonstrated *in vitro* in several other *Haemophilus* species, and the transformation efficiencies under experimental conditions reflect the overall genetic relationships (Leidy et al., 1956, 1959, 1965; Schaeffer, 1958; Steinhart and Herriott, 1968; Beattie and Setlow, 1970; Albritton et al., 1984).

Screening of *H. influenzae* strain Rd by various random insertional mutagenesis protocols have demonstrated that at least 5 gene loci are required for natural transformation (Larson and Goodgal, 1991; Redfield, 1991; Tomb et al., 1991; Chandler, 1992; Zulty and Barcak, 1995; Gwinn et al., 1997, 1998). Among the identified proteins are the type IV pilin-like protein and a drug-efflux transporter (Dougherty and Smith, 1999).

Plasmids of several sizes have been demonstrated in strains of *H. influenzae* and other *Haemophilus* species. Plasmids coding for resistance to a number of antibiotics have been demonstrated in *H. influenzae*, *H. parainfluenzae*, *H. parahaemolyticus*, and *H. ducreyi*. Two types of R plasmids occur in *H. influenzae*: the large 30–40

MDa conjugative plasmids and the small 2.5–4.4 MDa non-conjugative plasmids (De Graaf et al., 1976; Elwell et al., 1977; van Klingeren et al., 1977; Kaulfers et al., 1978; Laufs et al., 1979; Albritton, 1984). The large self-transferable plasmids carry the complete TnA or Tn10-like transposons in addition to antibiotic resistance genes and are often integrated into the chromosome (Stuy 1980; Levy et al., 1993). They have highly homologous core regions regardless of geographic origin and the antibiotic resistance markers carried (Elwell et al., 1977; Laufs and Kaulfers, 1977), and are closely related to plasmids coding for β -lactamase in *Enterobacteriaceae* (Laufs et al., 1978, 1979). Transfer of large plasmids coding for resistance to one to several antibiotics has been shown among strains of *H. influenzae* and between *H. influenzae* and *Escherichia coli* by a mechanism that requires cell-to-cell contact (Thorne and Farrar, 1975; van Klingeren et al., 1977; Stuy, 1979). The genetic transfer is likely mediated by a conjugation-like process. The same transfer mechanism has been demonstrated in *H. ducreyi* (Brunton et al., 1979; Deneer et al., 1982). Isolates of *H. ducreyi* may contain multiple plasmids of different sizes (1.8, 2.6, 2.8, 3.2, 5.4, 5.7, and 7.0 MDa) (Sarafian et al., 1991; Maclean et al., 1992).

The small R plasmids are highly related to gonococcal plasmids and usually carry only a fraction of the TnA sequence (de Graaf et al., 1976; Laufs et al., 1979). However, two sizes of plasmids (7.0 and 5.7 MDa) encoding β -lactamase with physically complete and functional TnA sequences have been identified in isolates of *H. ducreyi* (Brunton et al., 1982). The complete nucleotide sequence of a small (4.8 kb) yet broad-host-range plasmid encoding resistance to sulfonamides, streptomycin, and kanamycin in *H. ducreyi* has been reported (Dixon et al., 1994). Small cryptic plasmids have been detected in several *Haemophilus* species (Elwell et al., 1977; Stuy, 1979; Albritton et al., 1982b). A small 5.25 kb plasmid has been found to confer independence of V factor in occasional isolates of *H. parainfluenzae* (Windsor et al., 1991) and in *H. ducreyi* (Martin et al., 2001). Plasmids have not been detected in 75 *H. paragallinarum* isolates, including 20 antibiotic-resistant strains (Blackall, 1988), but more recently, a 6 kb plasmid was detected in one isolate (Blackall et al., 1991).

A ~24 kb cryptic plasmid (3031) has been associated with isolates of *H. influenzae* from Brazilian purpuric fever (Brenner et al., 1988a), although not all isolates possess it (Tondella et al., 1995).

Four different bacteriophages of *H. influenzae* have been reported: HP1 (and the mutants cl and c2), HP3, S2, and N3 (Boling et al., 1973; Stuy, 1978). The latter is morphologically distinct, with a longer tail and a contractile sheath. Differences in phage sensitivity and lysogeny have been demonstrated among *H. influenzae* serovars and a number of *Haemophilus* species (Stuy, 1978).

A variety of restriction endonucleases are produced by members of the genus *Haemophilus*. More than 20 different enzymes have been isolated and characterized.

Strains of *H. influenzae* serovar b have been shown to produce a bacteriocin to which other serovars and non-encapsulated *H. influenzae*, *H. parahaemolyticus*, and some strains of *H. parainfluenzae* and *H. haemolyticus* are sensitive (Venezia and Robertson, 1975; Stuy, 1978).

Antigenic structure There is a great deal of antigenic diversity in the genus and even within several species: *H. influenzae* (Pittman, 1931), *H. parausis* (Bakos, 1955), *H. paragallinarum* (Page, 1962; Kume et al., 1980; Blackall et al., 1990a), *H. paraphrophilus*, and *H. haemoglobinophilus* (Frazer et al., 1975). In spe-

cies in which isolates may possess a capsule, the capsular antigens furnish the basis for serovar specificity. *H. influenzae* isolates that lack a capsule are often referred to as "nontypeable," which is a misnomer that should be substituted by "non-encapsulated." All the capsular antigens studied in sufficient detail have proven to be polysaccharides. They are serologically quite specific, although more than one antigenic determinant may be present in a capsule (Williamson and Zinnemann, 1951, 1954; Branefors-Helander, 1972; Lucas, 1988). Capsular structure and composition have been studied most thoroughly for *H. influenzae*, for which Pittman (1931 and unpublished results) originally described six serologically distinguishable types a–f. The structure of the serovar a–f capsular polysaccharides (and a variant of serovar e, designated e') is shown in Fig. BXII.γ.219. All six polysaccharides, except those of serovars d and e, contain phosphate and may be characterized as teichoic acids (Crisel et al., 1975; Branefors-Helander, 1977; Branefors-Helander et al., 1979; Egan et al., 1980a, b; Branefors-Helander et al., 1980). Little is known of the physicochemical properties of the capsules of other *Haemophilus* species, apart from *H. parvus*, in which some, but not all, capsules appear to be composed of an acidic polysaccharide (Morozumi and Nicolet, 1986).

Capsule production in *H. influenzae* depends on a cluster of genes in an 18 kb chromosomal locus termed *cap*, which consists of three major regions. *Cap* region 1 contains four genes (*bex A–D*) necessary for capsular export and, in combination with *cap* region 3, forms the *cap* chassis. *Cap* region 2, located in the middle of the chassis, is the serovar-specific locus, which contains four genes involved in capsule biosynthesis. The *cap* chassis is flanked by the insertion sequence IS1016, forming a compound transposon. In most clinical isolates of *H. influenzae* serovar b (division I isolates), the *cap* locus is composed of multiple (usually two) tandemly repeating *cap* units, and is unique in having a partial deletion of most of the IS1016-*bex A* region, which allows

amplification, as well as irreversible loss of capsule expression (Kroll et al., 1993; Roche and Moxon, 1995). As many as 20% of cells in late-exponential liquid cultures of these strains have lost the ability to produce a capsule and will grow on agar medium as small colonies with an irregular surface, containing pleomorphic, often filamentous bacteria with cytoplasmic vacuoles (Roche and Moxon, 1995). A significant proportion of clinical isolates of *H. influenzae* serovar b contain more than 2 copies of the *cap* locus and produce large amounts of capsular polysaccharide (Hoiseth et al., 1992). In contrast, serovar b isolates belonging to phylogenetic division 2 (Musser et al., 1988) have only one copy, like strains of serovars c and d. Serovar a strains have a tandem duplication of intact *cap* loci. PCR-based capsular genotyping has been shown to accurately determine the capsular states of *H. influenzae* isolates, including strains that have lost expression of capsular polysaccharide (Leaves et al., 1995). Occasional clinical isolates of serovar a from Africa have revealed the same deletion in the *cap* locus as found in serovar b isolates. This deletion may be associated with enhanced virulence of some serovar a clones (Kroll et al., 1994). While the majority of non-encapsulated *H. influenzae* isolates are not the progeny of capsule ancestors, some studies have demonstrated isolates that contain *cap*-locus-specific sequences (St. Geme et al., 1994). The reported hybridization of the genome of Brazilian purpuric fever-associated isolates of *H. influenzae* with a *cap* locus probe (Carlone et al., 1989) is, however, due to the presence of IS1016 elements rather than capsule genes (Dobson et al., 1992).

Bacterial antigens cross-reactive with the *H. influenzae* serovar b capsule have been detected across a wide and unrelated variety of bacteria, including *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Streptococcus pyogenes*, *Streptococcus pneumoniae* (serovars 6, 15a, 29, 35a), *E. coli* (K 100), *Lactobacillus plantarum*, *Enterococcus faecium*, *Bacillus alvei*, and *Bacillus pumilus* (Alexander, 1958; Bradshaw et al., 1971; Argaman et al., 1974).

Serovar	Structure
a	4)-β-D-Glc-(1→4)-D-ribitol-5-(PO ₄ →
b	3)-β-D-Rib-(1→1)-D-ribitol-5-(PO ₄ →
c	4)-β-D-GlcNac-(1→3)-α-D-Gal-1-(PO ₄ →
	3 ↑ R
d	4)-β-D-GlcNac-(1→3)-β-D-ManANAc-(1→
	6 ↑ R
e	3)-β-D-GlcNac-(1→4)-β-D-ManANAc-(1→
e'	3)-β-D-GlcNac-(1→4)-β-D-ManANAc-(1→
	3 ↑ 2 β-D-fructose
f	3)-β-D-GalNac-(1→4)-α-D-GalNac-1-(PO ₄ →
	3 ↑ OAc

R = OAc (0.8)
H (0.8)

R = L-serine (0.41)
L-threonine (0.14)
L-alanine (0.41)

FIGURE BXII.γ.219. Structures of *H. influenzae* capsular polysaccharides. Compiled from data published by Crisel et al., 1975; Branefors-Helander, 1977; Branefors-Helander et al., 1979, 1981; Egan et al., 1980a, b; Tsui et al., 1981a, b; Zon and Robbins, 1983; Byrd et al., 1987. Ribose and fructose are in the furanose ring form. Glc, Gal, GlcNac, and ManANAc are in the pyranose ring form.

Among somatic antigens, obtained by various extraction procedures and sonic disruption, cross-reactions often occur between serovars within species (Branefors-Helander, 1979; Blackall, 1989) and between species within the genus (Tunevall, 1953; Omland, 1964; Branefors-Helander, 1979; Schiøtz et al., 1979). Among the surface proteins expressed by *H. influenzae* strains, some (P4, P6, lipoprotein D, D15) are immunologically highly conserved, while many others (e.g., P1, P2, P5, pili) show significant antigenic diversity. Antigenic diversity has also been demonstrated for cell wall lipooligosaccharide and the secreted IgA1 protease. The genetic mechanisms creating this diversity include point mutations, phase variation (e.g., alteration of promoter structure and translational frameshift due to slipped strand mispairing), and horizontal gene transfer and homologous recombination (reviewed by Gilsdorf, 1998).

Based on heat-stable somatic antigen extracts, a total of 15 serovars (1 through 15) have been detected among isolates of *H. parvus* (Kielstein and Rapp-Gabrielson, 1992). Subsequent studies have suggested additional antigenic diversity (Blackall et al., 1997). Several serotyping schemes for *H. paragallinarum* have been reported. The first serological typing system based on whole cell plate agglutination was developed by Page (1962), who recognized three serovars (A, B, and C). This and other similar typing schemes detect several distinct antigens, of which some are common to all members of the species (Blackall, 1989). Other typing systems for *H. paragallinarum* are based on somatic heat-stable antigens and a hemagglutinin (Hinz, 1980; Kume et al., 1983). The Kume scheme, which is based on a heat-labile, trypsin-sensitive hemagglutinin (HA-L), originally recognized three serogroups, termed I, II, and III, and a total of seven serovars, termed HA-1 to HA-7. The serovars are based on minor differences that can be detected with absorbed antisera. Two additional hemagglutinin serovars have been described by Blackhall and his co-workers (Eaves et al., 1989; Blackall et al., 1990a), who also proposed a rationalized nomenclature that allows easy addition of new serovars and reconciles the schemes of Page and Kume. Under this altered scheme, the three recognized serogroups I, II, and III are renamed A, C, and B, respectively, and correspond to Page's serovars. Within each of the serogroups, the hemagglutinin serovars are numbered sequentially, allowing new serovars to be added in numerical order. The nine currently recognized serovars are termed A-1, A-2, A-3, A-4, B-1, C-1, C-2, C-3, and C-4 (Blackall et al., 1990a).

Several *H. influenzae* LPS oligosaccharide epitopes are similar to those in pathogenic *Neisseria* species and cross-react with human glycosphingolipid epitopes (Roche and Moxon, 1995). Studies with monoclonal antibodies have demonstrated species-specific epitopes in the LPS of *H. ducreyi* (Borrelli et al., 1995).

Comprehensive proteomic maps of primarily *H. influenzae*, but also including *H. parainfluenzae*, *H. haemolyticus*, and *H. parahaemolyticus*, have been reported by several research groups (Cash et al., 1997; Fountoulakis et al., 1997, 1998a, 1998b).

Antibiotic sensitivity Wild-type strains of the *Haemophilus* species are susceptible to therapeutically achievable concentrations of a wide range of antibiotics. Representative minimum inhibitory concentrations (MICs) of common antibiotics for *H. influenzae* are ($\mu\text{g/ml}$): benzyl penicillin, 1–2; ampicillin/amoxycillin, 0.5; cefuroxime, 0.5; cefotaxime, 0.06; ceftriaxone, 0.03; cefaclor, 4; imipenem, 1; chloramphenicol, 0.5; tetracycline, 1; rifampicin, 1; erythromycin, 0.5–8; ciprofloxacin, 0.015; gentamicin, 1; sulfamethoxazole, 4; and trimethoprim, 0.5 (Butt et al., 1997b; Slack and Jordens, 1998).

The macrolide (erythromycin, oleandomycin) and lincosamide antibiotics (lincomycin, clindamycin) are generally less active, and bacitracin has so little effect on *H. influenzae* and other *Haemophilus* species that it is used, in a concentration of 5–19 U/ml, in media for the selective isolation of haemophili (Hovig and Aandahl, 1969; Ederer and Schurr, 1971; see below).

Since 1973, ampicillin resistance among *Haemophilus* species has become an increasing problem. Markedly differing rates of resistance among clinical isolates have been reported in various parts of the world, in some areas exceeding 60% (Campos et al., 1984, 1986). Ampicillin resistance is predominantly due to plasmid-encoded production of β -lactamases, which belong most often to the TEM-1-type, less often to the ROB-1-type, and occasionally to the VAT-1 type (Williams et al., 1974; Elwell et al., 1975; Bell and Plowman, 1980; Markowitz, 1980; Rubin et al., 1981; Maclean et al., 1992; Scriver et al., 1994; Vali et al., 1994; Shanahan et al., 1996). It has been suggested that the ROB-1 β -lactamase has an animal reservoir, as it has been detected in isolates of *Actinobacillus pleuropneumoniae* (Medeiros et al., 1986). Occasional isolates of *H. influenzae* show resistance to the β -lactam agents as a result of alterations in penicillin-binding proteins (Powell, 1988; Mendelman et al., 1990). Plasmid-encoded resistance to other antimicrobics has been observed. These include chloramphenicol (Manten et al., 1976; van Klingeren et al., 1977), tetracycline (Dang Van et al., 1975a; Bryan, 1978), and kanamycin (Dang Van et al., 1975b). Plasmids that simultaneously transfer ampicillin, chloramphenicol, and tetracycline resistance have been found in *H. influenzae* (Bryan, 1978). While tetracycline resistance appears to occur at rates comparable to penicillin-ampicillin resistance in some populations (Piot et al., 1977; Green et al., 1979), chloramphenicol resistance shows significant variation in prevalence (Brotherton et al., 1976; Ward et al., 1978; Butt et al., 1997b; Slack and Jordens, 1998). Resistance to trimethoprim and rifampicin has been observed (Green et al., 1979; Powell et al., 1992; Murphy et al., 1981). Chloramphenicol resistance is almost always due to the production of a plasmid-encoded chloramphenicol acetyl transferase (CAT) type II (Roberts et al., 1980). Non-CAT-mediated resistance is chromosomally encoded and due to a permeability barrier associated with the loss of an outer membrane protein (Burns et al., 1985).

H. parainfluenzae generally shows resistance levels higher than those of nonresistant *H. influenzae* to the antimicrobial agents commonly used on *Haemophilus* species (Kamme, 1969; Mayo and McCarthy, 1977). Both β -lactamase and non- β -lactamase-mediated resistance to ampicillin have been observed in *H. parainfluenzae*, usually at a higher rate than in *H. influenzae* (Green et al., 1979; Kauffman et al., 1979; Walker and Smith, 1980). Transmissible chloramphenicol resistance has been described in *H. parainfluenzae* and credited to CAT-II (Cavanagh et al., 1975; Shaw et al., 1978), while plasmid-encoded aminoglycoside resistance in the same species has been found to be mediated by a phosphotransferase (Le Goffic et al., 1977). Similar observations have been made for isolates of *H. parahaemolyticus* and *H. paraphrophilus* (Jones et al., 1976; Green et al., 1979).

The following minimum inhibitory concentrations ($\mu\text{g/ml}$) were observed on 19 isolates of *H. ducreyi* (Hammond et al., 1978b): vancomycin, 8–128; polymyxin, 32–128; penicillin G or ampicillin, 4.0 (unless β -lactamase-positive); cloxacillin, 32–64; cephalothin, 4.0–8.0; tetracycline, 0.4–32; doxycycline, 0.25–8.0; chloramphenicol, ≤ 4.0 ; rifampin, ≤ 4.0 ; sulfisoxazole, ≤ 8.0 ; and nalidixic acid, ≤ 8.0 . Plasmid-mediated resistance to sulfonamides, aminoglycosides, tetracyclines, chloramphenicol, and β -

lactam antibiotics have been demonstrated in *H. ducreyi*. It is not unusual for a single *H. ducreyi* isolate to contain multiple resistance plasmids, including more than one plasmid that confers resistance to β -lactam antibiotics (Trees and Morse, 1995).

Resistance to streptomycin is common in strains of *H. paragallinarum*, and strains resistant to tetracycline and sulfonamides also occur. There is no specific knowledge about resistance mechanisms in *H. paragallinarum*. Plasmids are not regularly associated with antibiotic resistance in this species (Blackall, 1988; Blackall et al., 1991).

Pathogenicity Among the *Haemophilus* species that colonize man, *H. influenzae* is clearly the most important from a clinical point of view. Although not responsible for epidemic influenza, as was originally believed, it is involved in a variety of numerically important and severe infections (for review see Turk and May, 1967). These infections can be divided into two groups: (a) acute, pyogenic, and usually invasive infections in which *H. influenzae* (almost exclusively serovar b) is the primary pathogen, and (b) noninvasive infections (often chronic) in which primarily non-encapsulated *H. influenzae* play an important, though in many cases probably a secondary, part.

The acute infections include meningitis in children, of which *H. influenzae* is one of the three leading causes, and other septicemic conditions with local implications, such as epiglottitis, cellulitis, arthritis, and osteomyelitis. More than 95% of *H. influenzae* isolates from these conditions produce a serovar b capsule, and belong to a limited number of clones (Musser et al., 1990), the majority of which are of biovar I (Kilian, 1976; Albritton et al., 1978; Kilian et al., 1979; Oberhofer and Bach, 1979). The introduction in many countries of a conjugated vaccine based on the *H. influenzae* serotype b capsular polysaccharide and one of several protein carriers has virtually eliminated invasive infections due to this serovar (Eskola et al., 1987; Jordens and Slack, 1995; Peltola, 2000). A relatively high prevalence of serovar a isolates have been observed from invasive infection in Papua New Guinea and in White Mountain Apache Indian children (Losonsky et al., 1984; Gratten et al., 1985). Capsulated strains of several serovars may also cause pneumonia. Noncapsulated strains of *H. influenzae* are often implicated in chronic bronchitis, sinusitis, conjunctivitis, and otitis media and have recently been recognized as a major cause of lower respiratory disease in children in the developing world (Foxwell et al., 1998). They occur frequently in the lower respiratory tract of patients with cystic fibrosis during acute exacerbations. Isolates from such conditions are genetically very diverse (Porras et al., 1986; van Alphen et al., 1997), but usually belong to the biovars II and III, the same biovars that colonize the nasopharynx of healthy individuals (Kilian, 1976; Albritton et al., 1978; Oberhofer and Bach, 1979; Granato et al., 1983; Harper and Tilse, 1991). *H. parainfluenzae* biovars identical to those that predominate in the upper respiratory tract are also a relatively frequent finding in respiratory secretions from patients with chronic bronchitis, but their pathogenic role is not clear (Taylor et al., 1992). Occasionally, *H. influenzae* and *H. parainfluenzae* are isolated from genital, pelvic, and urinary tract infections (Quentin et al., 1989; Morgan and Hamilton-Miller, 1990). *H. parahaemolyticus* is rarely isolated from healthy individuals but may be a pathogenic factor in abscesses in the oral cavity and in upper and lower respiratory tract infections (Sims, 1970; Kilian and Poulsen, unpublished studies).

H. aegyptius is a frequent cause of acute and contagious conjunctivitis, mainly in hot climates. Due to difficulties in differentiating *H. aegyptius* from *H. influenzae*, which also causes con-

junctivitis, the natural history of *H. aegyptius* infections is poorly understood. There are indications, however, that *H. aegyptius* is associated with a more acute form of conjunctivitis and that the organism, in contrast to *H. influenzae*, can colonize eyes without predisposing conditions being present (see discussion of taxonomic relationship between the two species below). Particular clones of the *H. influenzae/H. aegyptius* complex are causing a disease called Brazilian purpuric fever (BPF), which is a fulminant and often fatal infection that starts as conjunctivitis and primarily affects young children. An outbreak of systemic infections due to *H. aegyptius*-like bacteria in sheep in Nigeria has been reported (Akpavie et al., 1994).

H. ducreyi is the causative agent of the venereal disease soft chancre, or chancroid, which is particularly common in Africa, Asia, and Latin America. There has been renewed interest in this pathogen and in chancroid owing to the association between genital ulcers and HIV infection (Trees and Morse, 1995).

H. parainfluenzae, *H. parahaemolyticus*, *H. paraphrohaemolyticus*, *H. segnis*, *H. aphrophilus*, and *H. paraphrophilus*, which occur in the mouth and oropharynx of healthy individuals, are opportunistic pathogens. The oral cavity is usually a likely source of the etiologic agents in infections caused by these organisms. Such infections include endocarditis, brain abscesses, dental abscesses, jaw infections, and infections following human bites or finger sucking (for review, see Frederiksen and Kilian, 1981; Frederiksen, 1993).

The clinical condition of swine in which *H. parasuis* is most frequently encountered is Glässer's disease (polyserositis), a systemic stress-related infection producing fibrinous inflammation of the membranes lining the large body cavities, joints, and meninges. While no antecedent virus infection has been invoked as a precipitant of this condition, stressful events like weaning, weather changes, or movement to new quarters commonly, but not invariably, precede it (for review see Bisgaard, 1993).

H. paragallinarum causes an infection of the upper respiratory tract of chickens called fowl coryza. It begins in the nasal passages and sinuses and spreads to the conjunctivae, and can extend to the air sacs and lungs (Blackall, 1989).

H. haemoglobinophilus is of low pathogenicity. On rare occasions, it has been implicated in urogenital inflammatory disease of dogs. *H. felis* has been isolated from the lower respiratory tract of a cat with chronic obstructive pulmonary disease. Its pathogenic potential probably resembles that of *H. parainfluenzae* in humans (Inzana et al., 1992). *H. paracuniculus* was isolated from the small intestine of rabbits, but its pathogenic potential is unknown (Targowski and Targowski, 1979).

Ecology *Haemophilus* species are obligate parasites and form part of the indigenous flora of the mucous membranes of the human upper respiratory tract, including the oral cavity, and may occasionally be isolated from the vagina and intestinal canal. Humans are the only natural host of the following species: *H. influenzae*, *H. aegyptius*, *H. haemolyticus*, *H. ducreyi*, *H. parainfluenzae*, *H. parahaemolyticus*, *H. paraphrohaemolyticus*, *H. aphrophilus*, *H. paraphrophilus*, and *H. segnis*. *Haemophilus* species constitute approximately 10% of the total bacterial flora in the human upper respiratory tract, and *H. parainfluenzae* accounts for the largest proportion of *Haemophilus* species in the pharynx and oral cavity at all ages. While *H. parainfluenzae* is ubiquitous in the human pharynx and oral cavity, *H. influenzae* is found exclusively behind the palatal arches (Sims 1970; Kilian and Schjøtt, 1975; Kuklinska and Kilian, 1984; Liljemark et al., 1984). Point prevalence studies of children have shown carriage rates for *H. in-*

fluensae of 25–84% in the pharynx, although in the absence of local infection, they usually amount to less than 1% of the total microbiota. In contrast to adults, who are less frequently colonized, children often carry multiple clones that show rapid turnover (Kuklinska and Kilian, 1984; Spinola et al., 1986; Trotter et al., 1989; Faden et al., 1995; Smith-Vaughan et al., 1997). Among *H. influenzae* isolates from the upper respiratory tract, only a minority (2–7%) were found by most studies conducted in open communities to be encapsulated. The relative frequencies of the six capsular serovars a–f in such studies were 15, 40, 4, 6, 19, and 15%, respectively (reviewed by Moxon, 1986). Significantly higher nasopharyngeal carriage rates of serovar b strains have been observed among children in institutions such as orphanages and day care centers (Turk and May, 1967). The general vaccination of infants with a conjugate vaccine that includes the serovar b capsular polysaccharide has resulted in a significant reduction of oropharyngeal carriage of *H. influenzae* serovar b in some countries (Takala et al., 1991).

H. parainfluenzae is carried by most individuals on the mucosal surfaces of the pharynx and oral cavity, and as members of the biofilm (dental plaque) forming on tooth surfaces. Dynamic colonization patterns similar to those observed for *H. influenzae* have been demonstrated for *H. parainfluenzae* in the pharynx of adults (Kerr et al., 1993). *H. aphrophilus*, *H. paraphrophilus*, and *H. segnis* have a particular predilection for the dental plaque of humans, and the two former species occur primarily in the bacterial deposits between teeth and in deepened gingival crevices in patients with periodontal disease (Kilian and Schiøtt, 1975; Liljemark et al., 1984). *H. haemolyticus* may also be isolated from subgingival dental plaque (Kilian, unpublished observations). The carriage rate of *Haemophilus* species in the anogenital area is approximately 10% or less (Sturm, 1986; Drouet et al., 1989; Martel et al., 1989), although *Haemophilus* spp. have been recovered from over a quarter of stool specimens (Palmer, 1981). Occurrence of *H. ducreyi* in healthy individuals has not been documented.

Haemophilus-like bacteria (V-factor dependent) are also isolated from a variety of mammalian and avian species, including laboratory rodents and rabbits (for review see Grebe and Hinz, 1975; Kilian and Frederiksen, 1981a; Rayan et al., 1987; Nicklas 1989; Busse et al., 1997), but it is questionable whether these belong in the genus *Haemophilus* as presently defined. The only species associated with animals that are still formally (though probably incorrectly) included in the genus *Haemophilus* are *H. parasuis*, *H. paragallinarum*, *H. paracuniculus*, *H. felis*, and *H. haemoglobinophilus*.

H. parasuis has its natural habitats in pigs and is part of the resident microbiota of the upper respiratory tract, mainly in the nasal cavity. The carriage rate may differ between herds (Smart and Miniats, 1989; Møller and Kilian, 1990). Using a selective culture medium, examination of 233 Danish slaughterhouse pigs representing different herds yielded growth of *H. parasuis* from the nasal cavity of 30% of animals (Møller et al., 1993).

H. paragallinarum is found in the respiratory tracts of poultry, but is also an important pathogen (Blackall, 1989p). *H. haemoglobinophilus* is a frequent commensal inhabitant of the lower genital tract of dogs. *H. felis* was isolated from the nasopharynxes of 6 of 28 apparently healthy cats (Inzana et al., 1992).

ENRICHMENT AND ISOLATION PROCEDURES

The traditional way of demonstrating *Haemophilus* organism in samples from mucosal membranes is to inoculate a blood agar plate and cross-inoculate it with a *Staphylococcus* strain to provide

the V factor. Most *Haemophilus* species will grow to sizeable colonies in the vicinity of the feeder strain after overnight incubation at 35–37°C in air supplemented with 5–10% extra CO₂. However, in mixed cultures haemophili may easily be overgrown. Thus, respiratory samples often give rise to heavy growth of haemophili on selective media in cases where no apparent *Haemophilus* colonies are detectable by routine cultivation methods.

Chocolate agar supplemented with 100–300 µg/ml (6.3–18.9 U/ml) of bacitracin is a very satisfactory medium for the selective isolation of species of *Haemophilus* (Hovig and Aandahl, 1969; Ederer and Schurr, 1971). Some pharyngeal and oral *Neisseria* species grow on this medium, as do *Actinobacillus actinomycetemcomitans* and *Eikenella corrodens*. An alternative selective medium that gives equally satisfactory results is chocolate agar supplemented with bacitracin (5 U/ml) and cloxacillin (5 µg/ml) (Sims, 1970). A selective agar medium that allows differentiation of *H. influenzae* and *H. parainfluenzae* (and other V-factor dependent species) based on the detection of sucrose fermentation by the latter has been described by Roberts et al. (1987) and in modified form by Taylor et al. (1990).

For the more fastidious haemophili (e.g., *H. ducreyi* and *H. aegyptius*), higher isolation rates are obtained with chocolate agar enriched with 1% IsovitaleX (BBL) (Hammond et al., 1978a; Vastine, et al., 1974). GC-HgS agar consisting of GC agar (GIBCO Laboratories) supplemented with 3 mg of vancomycin per liter, 1% hemoglobin, 5% fetal bovine serum, 1% IsoVitaleX (BD Diagnostics) or other comparable enrichment has a high sensitivity for isolation of *H. ducreyi* from clinical specimens; so has MH-HB agar, i.e., Mueller–Hinton agar (BBL) supplemented with 5% chocolated horse blood, 1% IsoVitaleX, and 3 mg/l of vancomycin. It is generally accepted that a combination of two media, e.g. GC-HgS and MH-HB, is optimum for the isolation of *H. ducreyi*, possibly because of differences in nutritional requirements between strains, but the sensitivity for *H. ducreyi* is never 100%. Some lots of fetal calf serum inhibit growth of *H. ducreyi*. Fetal calf serum can be replaced by either activated charcoal or bovine albumin, but not by newborn calf serum (for a review see Trees and Morse, 1995). Some strains of *H. paragallinarum* require chicken serum for growth, while other strains simply show improved growth in its presence (Hinz, 1973; Blackall and Reid, 1982). Sodium chloride at a concentration of 1.0–1.5% is essential for the growth of *H. paragallinarum* (Rimler et al., 1977).

MAINTENANCE PROCEDURES

Plate and tube cultures of haemophili usually survive no longer than 1 week without subcultivation. Survival is often better at room temperature than at 4°C. The most satisfactory way of conserving cultures is by lyophilization, e.g., in skim milk. Levinthal broth cultures remain viable for at least 2 years when frozen at –70°C in sealed glass ampules.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Test for porphyrin synthesis (X-factor requirement) Like many other bacteria, hemin-independent *Haemophilus* strains excrete porphobilinogen (PBG) and porphyrins, intermediates in the hemin biosynthetic pathway (Fig. BXII.γ.217), when supplied with δ-aminolevulinic acid (ALA). In contrast, hemin-requiring strains do not excrete these compounds because they lack the enzymes responsible for their synthesis (Biberstein et al., 1963). This is the rationale for the porphyrin test, which is the easiest and most reliable means of determining X-factor requirement in *Haemophilus* organisms (Kilian, 1974).

A substrate of the following composition is used: δ -aminolevulinic acid, (Sigma), 2 mM; and MgSO_4 , 0.08 mM; in 0.1 M phosphate buffer, pH 6.9. The substrate is distributed in 0.5-ml quantities in small glass tubes and is inoculated with a heavy loopful of bacteria from an agar plate culture. After incubation for 4 h at 37°C, the result is read under Wood's light (360 nm), preferably in a dark room. A red fluorescence from the bacterial cells and/or the fluid is indicative of porphyrins (Fig. BXII.γ.217), meaning that the strain is independent of the X factor.

An alternative method of reading is to add 0.5 ml of Kovacs' reagent, shake vigorously, and allow the phases to separate. A red color in the lower water phase, indicative of porphobilinogen (Fig. BXII.γ.217), means that the strain is independent of X factor. With this method of reading, an inoculated "substrate" without ALA must be included to avoid false positive reactions due to indole, which also gives a red color with Kovacs' reagent (Fig. BXII.γ.217).

The traditional use of paper disks impregnated with X factor on agar media cannot be recommended for the demonstration of an X-factor requirement. Even with care, use of this method results in misidentifications approaching 20% (M. Kilian and K.R. Eriksen, unpublished results). Such misidentifications undoubtedly account for most of the reported isolates of *H. parainfluenzae* from invasive infections.

V-factor requirement The satellite phenomenon mentioned above is widely used as a means of determining V-factor requirement. This phenomenon may be demonstrated on an agar medium lacking the V factor by cross-inoculating the plate with an appropriate feeder strain (e.g., *Staphylococcus* or *Pseudomonas*). More unequivocal results may be obtained by the application of a V-factor-impregnated paper disk. Media in which all ingredients have been autoclaved are consistently free of V factor, whereas ordinary blood agar contains varying amounts of available V factor. A convincing satellite phenomenon is, therefore, sometimes difficult to achieve on ordinary blood agar plates, particularly with strains of hemolytic species. The best results are obtained on a blood agar medium in which the blood (5–10%) has been added before autoclaving. This medium is completely devoid of V factor but otherwise satisfies all growth requirements of the *Haemophilus* species, including the X factor.

Fermentation of carbohydrates Fermentation of carbohydrates can be studied in phenol red broth base (Difco) supplemented with 1% of the respective carbohydrates and 10 mg/l each of NAD and hemin, the former added as a solution sterilized by filtration (Kilian, 1976). In this medium, most strains of *H. parainfluenzae*, *H. parahaemolyticus*, *H. aphrophilus*, *H. paraphrophilus*, and *H. haemolyticus* release hydrogen and CO_2 during fermentation of glucose (Kilian, 1976). This activity may be demonstrated by inserting an inverted Durham tube in the glucose broth. *H. paracuniculus* is reported to be unable to grow in phenol red broth base supplemented with V factor. Purple broth base supplemented with NAD (10 mg/l) is recommended for the examination of fermentation reactions in this organism (Targowski and Targowski, 1979).

Hemagglutination The hemagglutinating activity found in most fresh isolates of *H. influenzae* and *H. aegyptius*, and in some isolates from fowl and swine is demonstrated as described by Davis et al. (1950). The property is easily lost upon subcultivation of isolates.

Typing of capsules Commercial antisera are available for the differentiation of the six capsular serovars of *H. influenzae*, and several techniques may be applied. Slide agglutination is the traditional method and gives reliable results when performed with care. It is wise to consider as positive only those agglutinations that occur within a few seconds. Delayed reactions should be interpreted very cautiously, as commercial antisera also contain antibodies against somatic antigens. Although more time-consuming, the use of the Quellung reaction greatly strengthens the reliability of typing. An alternative serological method that gives reliable results is coagglutination using commercial reagents (Himmelreich et al., 1985). Typing may also be achieved by demonstration of serovar-specific gene sequences. A probe (pUO38) that recognizes the cap region of all six serovars reveals 14 distinct banding patterns in *EcoRI*-digested chromosomal DNA, which can be directly translated into serovars (Kroll et al., 1991). Capsular typing by PCR is another method that accurately determines the capsular states of *H. influenzae* isolates, including b^- strains. It is easier to perform than probe analysis, as there is no need for DNA extractions, Southern blotting, and hybridization. The method furthermore recognizes four cap b patterns not detected by pUO38 hybridization (Leaves et al., 1995). Details about probes and primers are provided by Herbert et al. (1998).

Micromethods for Examination of *Haemophilus* Strains Micromethods that do not require growth are recommended for the detection of indole production, urease, ornithine, and lysine decarboxylases, and arginine dihydrolase and for the demonstration of glycosidase activities. All these tests use 0.5-ml quantities of the substrate and are inoculated by suspension of a heavy loopful of bacteria in the substrate. The results can usually be read after incubation for 4 h at 37°C, but the incubation may be extended to 24 h.

Indole test The substrate is 0.1% L-tryptophan in 0.067 M phosphate buffer at pH 6.8. After incubation, 0.5 ml of Kovacs' reagent is added and the mixture shaken. A red color in the upper alcohol phase indicates the presence of indole (Clark and Cowan, 1952).

Urease test The basal medium consists of (per 100 ml of distilled water): KH_2PO_4 , 0.1 g; K_2HPO_4 , 0.1 g; NaCl, 0.5 g; and phenol red solution (1:500, prepared by dissolving 0.2 g in 92 ml distilled water + 8 ml 1 N NaOH), 0.5 ml. pH is adjusted to 7.0 with 5 N NaOH, autoclave, and add 10.4 ml of urea solution (20% aqueous, sterilized by filtration). A red color that develops within 4 h after inoculation indicates urease activity (Lautrop and Lacy, 1960).

Amino acid decarboxylases Ornithine and lysine decarboxylases and arginine dihydrolase can be demonstrated in Møller's medium (Møller, 1955). When this is heavily inoculated with positive strains, a purple color develops within 4 h.

Glycosidases Glycosidase activities can be demonstrated in 0.1% buffered solutions (w/v) of the respective chromogenic nitrophenol derivatives (Kilian, 1978). A yellow color developing within 4 h indicates glycosidase activity.

Most of the biochemical tests mentioned above can be performed with commercial micromethod kits, such as API 10E, API 20E, HNID (Quentin et al., 1992), the Minitek System (BBL) (Oberhofer and Bach, 1979), API NH (Barbé et al., 1994), Rapid NH and RIM-*Haemophilus* (Barbé et al., 1994).

DIFFERENTIATION OF THE GENUS *HAEMOPHILUS* FROM OTHER GENERA

According to the definition of the genus employed in the previous edition of this *Manual*, the demonstrable need of X or V factor on the part of a Gram-negative rod or coccobacillus would qualify that organism as a *Haemophilus* strain. Conversely, the absence of such need would exclude a bacterium from the genus (Kilian and Biberstein, 1984). However, in the intervening period several V-factor-dependent species have been transferred from the genus *Haemophilus* to other genera within and outside the family *Pasteurellaceae* based on genetic data. "*H. pleuropneumoniae*" has been transferred to the genus *Actinobacillus* (Pohl et al., 1983). "*Haemophilus avium*", which turned out to be a heterogeneous taxon, has been transferred to the genus *Pasteurella* as *P. avium*, *P. volantium*, and an additional, yet unnamed, taxon designated *Pasteurella* species A (Mutters et al., 1985b; Piechulla et al., 1985b). Finally, "*H. equigenitalis*" has been reclassified as *Taylorella equigenitalis* (Sugimoto et al., 1983).

As a result of the reclassification of these species, V-factor requirement is now a property of some species of both *Actinobacillus* and *Pasteurella*. In addition, three V-factor-dependent taxa of porcine origin have been proposed as new species within the genus *Actinobacillus*, i.e., *A. minor* (formerly *Haemophilus* taxon "minor group" of Kilian et al., 1978), *A. porcinus*, and *A. indolicus* (Møller et al., 1996a). Conversely, it has become clear that some of the remaining *Haemophilus* species (i.e., *H. parainfluenzae* and *H. paragallinarum*) may include occasional members that are independent of the V factor, at least in some cases as a result of a transferable plasmid (see above). Similarly, as discussed above, isolates of *H. aphrophilus* often lack a demonstrable requirement for X factor, and the apparent requirement in fresh isolates is not due to a lack of enzymes in the biosynthetic pathway of heme as in *H. influenzae*. As a result, the previous definition of the genus is no longer valid.

Despite the reallocation of some species, the definition of the genus *Haemophilus* and the delineation of the individual genera in the family *Pasteurellaceae* have not become any clearer. As discussed by Olsen et al. in the chapter describing the family *Pasteurellaceae* (this *Manual*), available data suggest that it will be difficult to divide the family *Pasteurellaceae* into genera that are phenotypically and phylogenetically coherent. However, there is no doubt that several of the species currently assigned to the genus *Haemophilus* belong elsewhere. Based primarily on DNA-DNA hybridization data of Burbach (1987), Mutters et al. (1989) have proposed restricting the genus *Haemophilus sensu stricto* to *H. influenzae*, *H. aegyptius*, *H. haemolyticus*, "*H. intermedius*", and *H. parainfluenzae*. In contrast, phylogenetic analyses by 16S rRNA sequence comparisons suggest that the genus should instead be restricted to *H. influenzae*, *H. aegyptius*, *H. haemolyticus*, and "*H. intermedius*"; whereas *H. parainfluenzae* is only distantly related. *H. aphrophilus*, *H. paraphrophilus*, and *H. segnis* belong to a separate cluster related to *Actinobacillus actinomycetemcomitans* (see Fig BXII.γ.215 of Olsen et al. in the chapter describing the family *Pasteurellaceae*, this *Manual*).

Allocation of isolates to one of the existing species of *Haemophilus* usually presents no problems. This can be achieved in the clinical microbiological laboratory by examination of phenotypic traits or, if sequencing facilities are available, by partial sequencing of 16S rRNA genes. However, a decision concerning the correct allocation of newly discovered taxa to any one of the existing genera within *Pasteurellaceae* is problematic. There are

at present no phenotypic traits known that correlate with the 16S rRNA sequence homology data and that would provide a workable basis for reconciling the genetic information with the practical needs for laboratory identification of genera and species. Thus, at present, the decision would be made based on genetic data, in particular, comparison of 16S rDNA sequences of an unknown isolate with available sequences for the respective type species, but even then the most natural allocation would not necessarily be clear-cut.

Of all the secondary criteria usually cited as characteristic of the genus, such as carbohydrate fermentation, alkaline phosphatase activity, and nitrate reduction, only the lattermost is without exception among the currently recognized species, and none are exclusive. Other chemotaxonomic criteria are discussed by Olsen, et al. (see the chapter on the family *Pasteurellaceae*, this *Manual*). There are no motile members of the genus, and the question of whether motility would exclude an organism meeting the other standards of the genus has never arisen.

Organisms with growth factor requirements comparable to those of *Haemophilus* spp. occur sporadically among other, unrelated taxa (Jensen and Thofern, 1953; Beljanski, 1955; Caldwell et al., 1965). The basis for excluding them from the genus *Haemophilus* is, in addition to genetic data such as significant differences from *H. influenzae* in 16S rRNA genes, based on morphological, tinctorial, and cultural characteristics identifying them with their parent taxa.

TAXONOMIC COMMENTS

In addition to the described problems related to the definition of the genus, several other taxonomic problems remain to be solved in the genus *Haemophilus*. These problems include species that seem to belong not in the genus, but rather elsewhere in the family *Pasteurellaceae*, as well as problems related to some of the "true" *Haemophilus* species. According to phylogenetic analyses based on 16S rDNA sequence comparisons, the species *H. ducreyi*, *H. parainfluenzae*, *H. parahaemolyticus*, *H. paraphrohaemolyticus*, *H. haemoglobinophilus*, *H. parasuis*, *H. paragallinarum*, *H. paracuniculus*, and *H. felis* do not cluster with the type species *H. influenzae* and seem to be misplaced in the genus *Haemophilus* (see Fig BXII.γ.215 of Olsen et al. in the chapter on the family *Pasteurellaceae*, this *Manual*).

The taxonomic position of *H. ducreyi* has been questioned for a number of years in spite of its requirement for X factor and a mol% G + C content of DNA within the accepted range for *Haemophilus* species. DNA hybridization data indicate that *H. ducreyi* is unrelated to *H. influenzae* (Casin et al., 1985). Sequence comparisons of 16S rDNA of selected isolates of *H. ducreyi*, including the type strain (CIP542), with representatives of a large number of species support its exclusion from the genus *Haemophilus* but confirm that it belongs in the family *Pasteurellaceae* (Dewhirst et al., 1992).

H. parainfluenzae is a genetically very diverse species. Although Mutters et al. (1993) included it in *Haemophilus sensu stricto*, 16S rRNA sequence data of a limited number of strains reveal that it is phylogenetically only distantly related to *H. influenzae* and belongs elsewhere in the family *Pasteurellaceae*. Further studies of a larger number of representative strains reflecting the diversity of the species are required to establish the natural affiliation of this species. Likewise, *H. parahaemolyticus* clusters with *A. lignieresii*, but the 16S rRNA sequence on which this position is based appears to be incorrect (Hedegaard et al., 2001). *H. parasuis* clusters with *Actinobacillus indolicus*; *H. paracuniculus*, with *Acti-*

nobacillus capsulatus, and *H. paragallinarum*, with a cluster of avian species including *Pasteurella gallinarum* (see discussion of the family *Pasteurellaceae* by Olsen et al. in the chapter on the family *Pasteurellaceae* in this *Manual*). Solutions to these problems must await clarification of the overall taxonomic structure of the family *Pasteurellaceae*. The fact that results of DNA reassociation and 16S rRNA sequencing are in some cases in conflict emphasizes the need for more comprehensive genetic approaches.

Numerous taxonomic questions need to be solved, even within the genus *Haemophilus sensu stricto*. Several of these problems are associated with *H. influenzae* and its close relative *H. aegyptius*. Based on cluster analyses of comprehensive DNA–DNA reassociation data, Burbach (1987) recognized two subgroupings within *H. influenzae* and proposed them as subspecies, designated “*H. influenzae* subsp. *influenzae*” and “*H. influenzae* subsp. *meningitidis*”, respectively. “*H. influenzae* subsp. *influenzae*” includes the type and consists of noncapsular strains and encapsulated strains of serovars a, c, and f, all belonging to biovars II, III, and VII. “*H. influenzae* subsp. *meningitidis*” includes encapsulated strains of serovars a, b, d, and e, as well as noncapsular strains. Biovars I, II, IV, and V are represented among these strains. Excepting a single biovar II strain included in “subsp. *meningitidis*”, a positive ornithine decarboxylase test can distinguish “subsp. *meningitidis*” from “*H. influenzae* subsp. *influenzae*”. However, with regard to serovars and biovars, these subgroups are not consistent with those observed by cluster analysis of multilocus enzyme electrophoresis data (Musser et al., 1986, 1988).

Among strains previously identified as *H. influenzae*, Burbach (1987) furthermore observed a distinct cluster of three strains with biochemical properties as described for biovar VIII (Sottnek and Albritton, 1984). The cluster is only distantly related to *H. influenzae* but is close to *H. haemolyticus*. The designation “*H. intermedius* subsp. *gazogenes*” was applied to this taxon, as the strains produce gas in glucose broth. Three other strains previously identified as *H. parainfluenzae* biovars III and IV are closely related to “*H. intermedius* subsp. *gazogenes*”, in spite of differences in X-factor requirement, and these were designated “*H. intermedius* subsp. *intermedius*”. Analysis of 16S rRNA sequences of two representative strains confirm that these taxa are distinct from *H. influenzae* and *H. parainfluenzae*, respectively, and are closely related to *H. haemolyticus* (see Fig BXII.γ.215. of Olsen et al. in the chapter on the family *Pasteurellaceae*, this *Manual*). However, the species “*H. intermedius*” has not been formally validly named and, phenotypically, appears difficult to handle in practice.

Studies from North America and France have revealed a relatively high prevalence of noncapsular *H. influenzae* biovar IV among isolates from cases of neonatal bacteremia and meningitis and obstetric infections (Albritton et al., 1982a; Wallace et al., 1983; Quentin et al., 1990). Phylogenetic analyses performed by multilocus enzyme electrophoresis, DNA–DNA hybridization, and rRNA gene polymorphism strongly suggest that such strains form a separate species related to *H. influenzae*, *H. aegyptius*, and *H. haemolyticus* (Quentin et al., 1990, 1993, 1996). It is not clear if all isolates with the characteristics of biovar IV belong in this taxon. This taxon is provisionally referred to as “*Haemophilus quentini*” in Fig. BXII.γ.215 of Olsen et al. (see chapter on the family *Pasteurellaceae*, this *Manual*).

Several studies have addressed the relationship between *H. influenzae* and *H. aegyptius* and the question of whether both deserve specific recognition. Phenotypically, *H. aegyptius* is similar, but not identical to *H. influenzae* (Mazloum et al., 1982). In the three biochemical tests used to distinguish the biovars of *H.*

influenzae, *H. aegyptius* is identical to biovar III (Kilian, 1976; Mazloum et al., 1982), for which reason many researchers have identified all conjunctival isolates with those characteristics as *H. aegyptius*. This may explain some of the conflicting results that have been published on the degree of phenotypic relatedness of the two taxa. Clearly, “true” *H. influenzae* (biovars II and III) are frequently also isolated from cases of conjunctivitis in temperate climates (Kilian et al., 1976). DNA–DNA hybridization data, transformation experiments, and 16S rRNA sequence comparisons all confirm that the two species are closely related (Leidy et al., 1959; Leidy et al., 1965; Pohl, 1981a; Albritton et al., 1984; Casin et al., 1986; Burbach, 1987). While Casin et al. (1986) concluded that the two taxa are indistinguishable, Burbach (1987) found that *H. aegyptius* strains form a distinct entity. The latter conclusion is supported by phenotypic differences and by studies of outer membrane profiles, which have revealed a distinct pattern in strains of *H. aegyptius* (Carlone et al., 1985). Although these differences indicate that *H. aegyptius* is a distinct subpopulation, they do not necessarily show that both *H. influenzae* and *H. aegyptius* warrant status as separate species.

Based on their DNA reassociation data, Casin et al. (1986) argued that *H. aegyptius* be considered a junior subjective synonym of *H. influenzae*. However, this is incorrect. While *H. influenzae* is the type species, the epithet *aegyptius* Trevisan 1889b has formal priority over the epithet *influenzae* Lehmann and Neumann 1896. “To avoid confusion” Brenner and coworkers (1988a) used the informal designation “*H. influenzae* biovar *aegyptius*” to denote strains formerly called *H. aegyptius*. The term was also applied to isolates from the disease Brazilian purpuric fever (BPF) and subsequently has been used frequently in the literature on this pathogen. Most isolates from the original outbreak and subsequent cases of BPF in Brazil belong to at least three different clones (“BPF clones”), although minor differences in ribotype and plasmid content have been observed within clones. Isolates from occasional cases in Australia are not closely related to the Brazilian clone (Musser and Selander, 1990). The Brazilian BPF clone fails to ferment D-xylose (Brenner et al., 1988a), like strains of *H. aegyptius* and occasional strains of *H. influenzae* (Mazloum et al., 1982). Recent phylogenetic and other analyses demonstrate that BPF clone isolates and *H. aegyptius* are two recently diverged but distinct populations that are separate from *H. influenzae* (Kilian, Poulsen, and Lomholt, unpublished). Thus, the specific status and nomenclature of the two present taxa *H. influenzae* and *H. aegyptius* remain an unsolved problem.

Collectively, the observations described above emphasize that there is a need for comprehensive population genetic analyses of *H. influenzae* and *H. parainfluenzae* strains that fully reflect the genetic diversity within these species in order to clearly define them, as well as neighboring species.

The two species *H. aphrophilus* and *H. paraphrophilus* are differentiated by the requirement for V factor by the latter (Table BXII.γ.294). *H. aphrophilus* has an apparent X-factor requirement when freshly isolated, but contains all the enzymes of the heme biosynthetic pathway (White and Granick, 1963). On subculture, variant colonies consisting of X factor-independent cells arise and apparently supplant the original population (Boyce et al., 1969). Strains carried by type culture collections usually do not have the X-factor requirement. In practice, *H. aphrophilus* is identified by criteria other than the growth factor requirement. Numerous genetic and chemotaxonomic analyses have been applied to strains of the two species to clarify their taxonomic relationship. Results of DNA–DNA hybridization, genetic transformation

(Pohl, 1981a; Tanner et al., 1982; Potts et al., 1985; Tønnum et al., 1990), 16S rRNA sequencing (Dewhirst et al., 1992), ribotyping (Sedláček et al., 1993), multilocus enzyme electrophoresis (Caugant et al., 1990), and whole cell protein mapping by two-dimensional gel electrophoresis (Olsen et al., 1987) fail to identify two separate groups. In contrast, according to Brondz and Olsen (1990) and Myhrvold et al. (1992), multivariate statistical analyses of LPS carbohydrates and enzyme data support their recognition as separate taxa. No formal action has so far been taken, and the two species are maintained as separate species in this chapter.

A proposal was made (Potts et al., 1985) to transfer the species *Actinobacillus actinomycetemcomitans*, which is associated with periodontal disease in humans, to the genus *Haemophilus*, based on its close genetic relationship to *H. aphrophilus* (Tanner et al., 1982; Potts et al., 1985). However, due to the rather distant relationship with the type species *H. influenzae*, as judged by DNA reassociation studies, this proposal has not been generally supported. Comprehensive phylogenetic analyses based on 16S rDNA sequence comparisons reveal that the species clusters neither with *H. influenzae* nor with *Actinobacillus lignieresii* but do confirm its close association with *H. aphrophilus*, *H. paraphrophilus*, and *H. segnis* (see Fig. BXII.γ.215 of Olsen et al. in the chapter on the family Pasteurellaceae, this *Manual*).

The previous edition of *Bergey's Manual of Systematic Bacteriology* listed three taxa as *species incertae sedis* under the genus *Haemophilus*: "*Haemophilus somnus*", "*Haemophilus agni*", and "*Haemophilus equigenitalis*" (Kilian and Biberstein, 1984). The lattermost

was subsequently transferred to a new genus *Taylorella* (Sugimoto et al., 1983). "*Haemophilus somnus*", which is associated with cattle as both a commensal and a recognized pathogen, and the phenotypically and genetically related "*Haemophilus agni*" are not genetically related to the genus *Haemophilus*, as demonstrated by DNA-DNA hybridization (Piechulla et al., 1986), DNA-rRNA hybridization (de Ley et al., 1990), and the degree of 16S rRNA sequence homology (Dewhirst et al., 1993). They may represent a new genus within the family Pasteurellaceae (Bisgaard, 1995). De Ley et al., (1990) suggested that both be transferred to genus "*Histophilus*", "*Histophilus ovis*" being a senior synonym for "*Haemophilus agni*". However, none of these names are valid, and no types have been designated. As the taxa are still often referred to as "*Haemophilus somnus*" and "*Haemophilus agni*", the descriptions are maintained here under Other Organisms.

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In the previous edition of this *Manual*, the chapter on the genus *Haemophilus* was co-authored with Dr. E.L. Biberstein. I am grateful for his many contributions and for what remains in this version.

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DIFFERENTIATION OF THE SPECIES OF THE GENUS *HAEMOPHILUS*

The differential characteristics of the species of *Haemophilus* are presented in Table BXII.γ.294. Other characteristics of the species are presented in Table BXII.γ.295.

List of species of the genus *Haemophilus*

1. ***Haemophilus influenzae*** (Lehmann and Neumann 1896) Winslow, Broadhurst, Buchanan, Krumwiede, Rogers and Smith 1917, 561^{AL}. (*Bacterium influenzae* Lehmann and Neumann 1896, 187.) *in.flu.en'zae*. Italian n. *influenza* influenza; M.L. gen. n. *influenzae* of influenza.

Coccobacilli or small regular rods 0.3–0.5 × 0.5–3.0 μm. Colonies on chocolate agar are smooth, low, convex, grayish, and translucent and attain a diameter of 0.5–1.0 mm in 24 h. Capsulated strains usually produce larger and more mucoid colonies (1–3 mm), which show a tendency to coalesce with no visible line of demarcation. On transparent agar media, colonies of capsulated strains show iridescence when examined under obliquely transmitted light.

Among encapsulated isolates, six serovars (a-f) have been identified (Pittman, 1931; M. Pittman, unpublished results). The structure of the serovar a-f capsular polysaccharides is shown in Fig. BXII.γ.219. A total of 41 capsular and somatic antigenic determinants have been detected in a strain of *H. influenzae* serovar b by crossed immunoelectrophoresis. A considerable number of these determinants are related to antigenic determinants of strains of *H. haemolyticus*, *H. parainfluenzae*, and various enterobacteria (Schjötz et al., 1979).

H. influenzae strains may be assigned to any of seven biovars (I-VII) based on three biochemical characteristics, indole production, and urease and ornithine decarboxylase activities (Tables BXII.γ.294 and BXII.γ.296) (Kilian, 1976; Oberhofer and Bach, 1979; Gratten 1983). Strains described as *H. influenzae* biovar VIII (Sottnek and Albritton, 1984) were subsequently shown to be distinct from *H. influenzae* and were assigned to the not validly published taxon "*H. intermedius* subsp. *gazogenes*" (see above). The majority of strains with a serovar b capsule are of biovar I, an observation that first suggested that serovar b strains are distinct from non-capsulated strains, most of which belong to biovars II and III. All isolates so far examined with a serovar d and e capsule are of biovar IV (Kilian, 1976).

H. influenzae possesses neuraminidase activity (Müller and Hinz, 1977). Virtually all strains produce an extracellular endopeptidase (IgA1 protease) capable of inducing specific cleavage of a proline-serine or a proline-threonine peptide bond in the hinge region of human immunoglobulin A1 (Kilian et al. 1996).

H. influenzae is present in the nasopharynxes of a majority of healthy children. The carriage rate in adults is somewhat lower (Kilian and Frederiksen, 1981a). It is rarely encountered in the human oral cavity and has not been

TABLE BXII-γ.295. Further characteristics of the species of the genus *Haemophilus*

Characteristics	1. <i>H. influenzae</i>	2. <i>H. aegyptius</i>	3. <i>H. aphrophilus</i>	4. <i>H. avium</i>	5. <i>H. ducreyi</i>	6. <i>H. haemoglobinophilus</i>	7. <i>H. haemolyticus</i>	8. <i>H. parvauniculus</i>	9. <i>H. paragaillinarum</i>	10. <i>H. parahaemolyticus</i>	11. <i>H. parainfluenzae</i>	12. <i>H. paraphrohaemolyticus</i>	13. <i>H. paraphrophilus</i>	14. <i>H. parasuis</i>	15. <i>H. segnis</i>	a. " <i>H. somnus</i> "	a. " <i>H. somnus</i> "	+
Lysine decarboxylase	p	+	+	+	+	p	p	+	+	+	+	+	+	+	+	+	+	+
Oxidase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
H ₂ S (lead acetate)	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Acid production from:																		
L-Arabinose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
L-Rhamnose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
D-Galactose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Sorbitose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Cellobiose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Maltose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Melibiose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Trehalose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Melzitose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Raffinose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Inulin	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Dulcitol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Glycerol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
meso-Erythritol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Inositol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Xylitol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Esculin, salicin, adonitol	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
α-Galactosidase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
α-Glucosidase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
β-Glucosidase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
α-Mannosidase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
β-Xylosidase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
β-Gluconidase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Nitrate reduction	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Nitrite reduction	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

*W, weak reaction.

TABLE BXII.γ.296. Key to the differentiation of the biovars of *H. influenzae* and *H. parainfluenzae*

Characteristic	<i>H. influenzae</i> biovars							<i>H. parainfluenzae</i> biovars							
	I	II	III	IV	V	VI	VII	I	II	III	IV	V	VI	VII	VIII
Indole	+	+	−	−	+	−	+	−	−	−	+	−	+	+	+
Urease	+	+	+	+	−	−	−	−	+	+	+	−	−	+	−
Ornithine decarboxylase	+	−	−	+	+	+	−	+	+	−	+	−	+	−	−

detected in any animal species apart from chimpanzees. Capsulated strains are harbored intermittently in the nasopharynx of a minority of healthy individuals (3–7%). Serovars b and f are the serovars most frequently encountered, whereas serovar c strains are rare (Moxon, 1986).

H. influenzae was isolated originally from cases of endemic influenza and was regarded as its causative agent at the time. It is frequently isolated from chronic infections of the upper and lower human respiratory tract, paranasal sinuses, middle ears, and conjunctivae, in which conditions *H. influenzae* plays an important, though often probably a secondary, part. Strains implicated in such conditions are usually noncapsulated and belong to the biovars II or III like the majority of isolates from healthy upper respiratory tracts (Kilian and Frederiksen, 1981a). Encapsulated strains of serovar b are among the three most common causes of bacterial meningitis in children and occasionally cause acute epiglottitis (obstructive laryngitis) cellulitis, osteomyelitis, and joint infections (Turk and May, 1967).

The mol% G + C of the DNA is: 39 (*T_m*).

Type strain: 680 of Pittman (noncapsulated, biovar II) ATCC 33391, NCTC 8143.

GenBank accession number (16S rRNA): M35019.

Additional Remarks: Reference strains: NCTC 8466 (serovar a, biovar I); NCTC 7279 (serovar b, biovar I); NCTC 8469 (serovar c, biovar II); NCTC 8470 (serovar d, biovar IV); NCTC 10479 (serovar e, biovar IV); NCTC 8473 (serovar f, biovar I); NCTC 4560 (noncapsulated, biovar III); NCTC 11394 (noncapsulated, biovar V).

2. **Haemophilus aegyptius** (Trevisan 1889b) Pittman and Davis 1950, 413^{AL} (*Bacillus aegyptius* Trevisan 1889b, 13.) *ae.gyp' ti.us.* L. adj. *aegyptius* Aegyptian.

The common name of the organism is Koch–Weeks bacillus. Long slender rods 0.2–0.3 × 2.0–3.0 μm. Growth of freshly isolated strains on chocolate agar is slow. After 48 h, colonies attain a diameter of about 0.5 mm and are smooth, low, convex, grayish, and translucent. The species does not grow on Tryptic soy agar (Difco) with X and V factors added, in contrast to *H. influenzae*. Forms comet-like colonies in semisolid agar media (Pittman and Davis, 1950).

Capsules have not been demonstrated. Bacteria of this species produce an extracellular IgA1-cleaving endopeptidase (Lomholt and Kilian, 1995; Kilian et al., 1996) and have neuraminidase activity (Müller and Hinz, 1977).

Features which may be of use in distinguishing *H. aegyptius* from *H. influenzae* include poorer growth on most media, a lack of indole production, and D-xylose fermentation, the distinct bacillary morphology, a hemagglutinating activity, the comet-like growth in semisolid media, and the susceptibility to troleandomycin. However, none of these features unequivocally differentiate the two species.

Some strains lack the enzyme ferrochelatase, which in-

serts iron into protoporphyrin IX, and thus require protoheme as X-factor (White and Granick, 1963; Kilian et al., 1976; Ruckelshausen and Holländer, 1978). The organism causes acute or subacute infectious conjunctivitis in hot climates. It has not been demonstrated in healthy individuals. Particular clones (“BPF clones”) closely related to *H. aegyptius* have been associated with Brazilian purpuric fever. The clones have been referred to as *H. influenzae* biogroup *aegyptius* (see comments above). They are characterized by particular patterns of outer membrane proteins and alleles of genes encoding house-keeping enzymes, rRNA gene restriction pattern, resistance to serum bactericidal activity, a 24-MDa plasmid (not present in all isolates), a unique antigenic variant of IgA1 protease, and the lack of D-xylose fermentation (Brenner et al., 1988a; Irino et al., 1988; Porto et al., 1989; Musser and Selander, 1990; Lomholt and Kilian, 1995). The taxonomic questions associated with *H. aegyptius* and its close relative *H. influenzae* are discussed above.

The mol% G + C of the DNA is: 39 (*T_m*).

Type strain: 180a of Pittman, ATCC 11116.

GenBank accession number (16S rRNA): M75044.

3. **Haemophilus aphrophilus** Khairat 1940, 505^{AL} *a.phro' phi.us.* Gr. n. *aphros* foam; Gr. adj. *philos* loving; M.L. adj. *aphrophilus* foam loving.

Short regular rods 0.45–0.55 × 1.5–1.7 μm with only occasional filamentous forms. Colonies on chocolate agar incubated in air supplemented with 10% extra CO₂ are high convex, granular, yellowish, and opaque and reach a diameter of 1.0–1.5 mm within 24 h. When plates are incubated without extra CO₂, the growth is characteristically stunted, with very small colonies interspersed with a few larger colonies. Grows on blood agar incubated in air plus CO₂ without feeder strain. Growth in broth media is granular, with a heavy sediment on the bottom of the tube and adhering colonies on the walls, which are difficult to remove.

H. aphrophilus possesses all the enzymes of the hemin biosynthetic pathway characteristic for hemin-independent species (White and Granick, 1963). Accordingly, it gives a positive, though usually weak, reaction for both porphobilinogen and porphyrins in the porphyrin test. The reason for an apparent need for hemin-containing media at primary isolation remains to be elucidated.

H. aphrophilus is a frequent member of the microflora of human dental plaque, particularly between the teeth and in the gingival pockets (Kraut et al., 1972; Kilian and Schiøtt, 1975). Occasionally causes endocarditis and brain abscesses in humans. Has been isolated from spinal fluid, wounds, and jaw infections (King and Tatum, 1962).

The mol% G + C of the DNA is: 42 (*T_m*).

Type strain: PM1, ATCC 33389, NCTC 5906.

GenBank accession number (16S rRNA): M75041.

4. **Haemophilus ducreyi** (Neveu-Lemaire 1921) Bergey, Harrison, Breed, Hammer and Huntoon 1923, 271^{AL} (*Coccobacillus ducreyi* Neveu-Lemaire 1921, 20.)
du.crey'i. M.L. gen. n. *ducreyi* of Ducrey; named after Ducrey, the bacteriologist who first isolated this organism.

Slender rods in pairs or chains, measuring 0.5×1.5 – $2.0 \mu\text{m}$, often seen in “school of fish”-like arrangements. Capsules have not been detected. Growth is poor on most laboratory media. Colonies on chocolate agar after 72 h are predominantly small (~ 0.5 mm in diameter), flat, smooth, grayish, and translucent, but often with a few interspersed larger colonies having an otherwise identical appearance. Colonies may be slid intact across the surface of agar plates. Growth on blood agar is very sparse, and there is no satellite growth around *Staphylococcus* colonies. Produces a cell-associated hemolysin with cytotoxic activity (Alfa et al., 1996) and a soluble cytotoxin (Cope et al., 1997). Either one or both of these toxins results in weak β hemolysis on blood agar plates.

Clinical isolates usually appear to be asaccharolytic. However, under favorable growth conditions, some strains show a late positive reaction for glucose fermentation. The species is inert in most of the traditional biochemical tests (Tables BXII.γ.294 and BXII.γ.295), but produces a wide array of peptidases. At least seven variants have been detected among 105 clinical isolates based on cell protein profiles (Odumeru et al., 1983). Causes the human venereal disease known as soft chancre or chancroid. A carrier state in healthy individuals has not been detected.

The mol% G + C of the DNA is: 38 (T_m).

Type strain: CIP 542, DSM 8925.

GenBank accession number (16S rRNA): M63900, M75078.

5. **Haemophilus felis** Inzana, Johnson, Shell, Møller and Kilian 1999, 341^{VP} (Effective publication: Inzana, Johnson, Shell, Møller and Kilian 1992, 2111.)
fe'lis. L. n. *felis* a cat; L. gen. n. *felis* of a cat.

Coccobacilli or small regular rods, 0.45 – 0.55×1.5 – $1.7 \mu\text{m}$, with occasional filamentous forms. Colonies on chocolate agar incubated in air plus 10% extra CO_2 are raised, smooth, and rounded and reach a diameter of 0.5–1.0 mm after 24 h. Colonies on brain–heart infusion agar supplemented with NAD are very adherent and show yellow pigmentation when collected on a loop. On blood agar, satellite growth is observed around a feeder strain of *Staphylococcus* or an NAD-impregnated paper disk. Weak hemolysis is observed after incubation for 3 d. Requires increased concentration of CO_2 in incubation atmosphere for primary isolation. This requirement is lost upon subcultivation. Has been isolated from the lower respiratory tract of a cat with chronic obstructive pulmonary disease and from the nasopharynxes of 6 of 28 apparently normal cats.

The mol% G + C of the DNA is: 38 (T_m).

Type strain: TI189, ATCC 49733.

GenBank accession number (16S rRNA): AF224292.

6. **Haemophilus haemoglobinophilus** (Lehmann and Neumann 1907) Murray 1939, 309^{AL} (*Bacterium haemoglobinophilus* (Lehmann and Neumann 1907, 270.)
hae.mo.glo.bi.no'phi.us. M.L. n. *haemoglobinum* hemoglobin; Gr. adj. *philos* loving; M.L. adj. *haemoglobinophilus* hemoglobin-loving.

Small, slightly pleomorphic, noncapsulated rods. Colonies on chocolate agar are smooth, convex, and translucent with a small granular area on top, and reach a diameter of 1–2 mm within 24 h. In contrast to other members of the genus, they grow well on blood agar and show no satellite growth around a streak of *Staphylococcus*.

Differs from all other *Haemophilus* species in lacking alkaline phosphatase. Belongs to the normal flora of the preputial sac of dogs, and is probably of low pathogenicity.

The mol% G + C of the DNA is: 38 (T_m).

Type strain: XIII of Kristensen, NCTC 1659.

GenBank accession number (16S rRNA): M75064.

7. **Haemophilus haemolyticus** Bergey, Harrison, Breed, Hammer and Huntoon 1923, 269^{AL}
hae.mo.ly'ti.cus. Gr. n. *haema* blood; Gr. adj. *lyticus* loosening, dissolving; M.L. adj. *haemolyticus* blood dissolving.

Small, regular, noncapsulated coccobacilli or short rods, with occasional filamentous forms. Colonies on chocolate agar are smooth, convex, grayish, and translucent and reach a diameter of 0.5–1.5 mm within 24 h. Produce clear hemolytic zones on bovine or sheep blood agar. However, the hemolytic activity may be lost upon subcultivation. Is found in the nasopharynx of a minority of the healthy human population and often in the bacterial deposits on human teeth below the gingival margin (subgingival dental plaque). No pathogenic potential has ever been demonstrated.

The mol% G + C of the DNA is: 39 (T_m).

Type strain: AQ/3273 of McIves, NCTC 10659.

GenBank accession number (16S rRNA): M75045.

8. **Haemophilus paracuniculus** Targowski and Targowski 1984, 355^{VP} (Effective publication: Targowski and Targowski 1979, 36.)

pa.ra.cu.ni'cu.us. Gr. prep. *para* alongside of, resembling; L. n. *cuniculus* rabbit, or possibly the specific epithet of “*Haemophilus cuniculus*”, a species that has never been described; M.L. adj. *paracuniculus* like “*Haemophilus cuniculus*”.

Small coccobacilli or pleomorphic rods with occasional filamentous forms. In broth media supplemented with V factor, the organism forms chains of rods. Colonies on chocolate agar are smooth, high convex, grayish, and opaque. Capsules have not been described. Does not grow in phenol red broth base supplemented with V factor. Has been isolated from the gastrointestinal tract of rabbits with mucoid enteritis. The pathogenic significance of the organism has not been established.

The mol% G + C of the DNA is: 40 (Bd).

Type strain: RF15888, ATCC 29986.

GenBank accession number (16S rRNA): M75061.

9. **Haemophilus paragallinarum** Biberstein and White 1969, 77^{AL}

pa.ra.gal.li.na'rum. Gr. prep. *para* alongside of, resembling; M.L. gen. pl. n. *gallinarum* specific epithet; M.L. adj. *paragallinarum* resembling (*Haemophilus*) *gallinarum*.

Coccobacilli to pleomorphic rods with occasional filamentous forms. Most strains are capsulated. Colonies on chocolate agar are smooth, convex, grayish, and semi-opaque and attain a diameter of 0.5–1.0 mm within 48 h in air supplemented with 10% CO_2 . Growth is feeble in air without extra CO_2 . Young cultures (8- to 24-h-old) of cap-

sulated strains produce iridescent colonies on transparent agar media. Growth is enhanced by serum added to cultivation media (Hinz, 1973; Rimler et al., 1976). Occasional V-factor-independent strains have been isolated in South Africa (Bisgaard Taxon 31) (Mouahid et al., 1992; Miflin et al., 1995).

Several serotyping schemes have been proposed. Three serovars (A–C) have been defined based on heat-labile surface antigens (Page, 1962). Kato and Tsubahara (1962) have used the designations I, II, and III for three serovars, of which serovar III may be a noncapsulated variant of serovar I (Kume et al., 1980). Hinz (1980) has established six serovars based on heat-stable somatic antigens. Nine serovars based on a hemagglutinin are currently recognized according to the proposal by Blackall et al. (1990a), which combines some of the former schemes in one, with the designations A-1, A-2, A-3, A-4, B-1, C-1, C-2, C-3, and C-4.

Produces neuraminidase and N-acetylneuraminase pyruvate lyase (Hinz and Müller, 1977). Present in the respiratory tract of poultry. Causes respiratory tract disease in chickens known as infectious coryza. The species includes organisms previously labeled "*Haemophilus gallinarum*".

The mol% G + C of the DNA is: 42 (T_m).

Type strain: IPDH 2403, serovar A, ATCC 29545.

GenBank accession number (16S rRNA): M75057.

10. *Haemophilus parahaemolyticus* Pittman 1953, 750^{AL}

pa.ra.hae.mo.ly'ti.cus. Gr. prep. *para* alongside of, resembling; M.L. n. *haemolyticus* specific epithet; M.L. adj. *para-haemolyticus* resembling (*Haemophilus*) *haemolyticus*.

Small, pleomorphic rods, usually with long, filamentous forms. Growth on chocolate agar is similar to that of *H. parainfluenzae*. A distinct β -hemolytic zone is produced on blood agar plates, most clearly detectable on bovine or sheep blood agar. Growth on blood agar usually is not restricted to the area around a feeder strain, presumably due to the release of ample V factor from blood cells that lyse as a result of the hemolytic activity.

Has been associated with acute pharyngitis, purulent oral infections, occasional cases of endocarditis, and exacerbation of chronic lower respiratory tract infections. Strains of this species produce an extracellular IgA1 protease closely related to the *H. influenzae* endopeptidase, which is capable of cleaving human IgA1 (Kilian and Poulsen, unpublished studies). Tuyau and Sims (1974) observed neuraminidase activity in members of this species. However, this cannot be confirmed with well-defined chromogenic substrates (Kilian and Poulsen, unpublished studies). Hemolytic porcine isolates that were previously assigned to this species belong in the species *Actinobacillus pleuropneumoniae*.

The mol% G + C of the DNA is: 40–41 (T_m).

Type strain: 536, NCTC 8479.

GenBank accession number (16S rRNA): AJ295746.

11. *Haemophilus parainfluenzae* Rivers 1922, 431^{AL}

pa.ra.in.flu.en'zae. Gr. prep. *para* alongside of, resembling; M.L. n. *influenzae* specific epithet; M.L. gen. n. *parainfluenzae* intended to mean like the species *H. influenzae*.

Small, pleomorphic rods, usually with long, filamentous forms. Occasional strains possessing a capsule have been described (Sims, 1970). Colonies on chocolate agar are grayish white or yellowish opaque and reach a diameter of

1–2 mm after 24 h. Some strains produce flat, smooth colonies with an entire edge, others show a serrated edge, and yet others grow as very rough wrinkled colonies. The irregular forms of colonies are usually coherent in texture and can be slid intact across the surface of the agar plate. Strains growing as rough-type colonies often convert into the smooth type after some *in vitro* transfers. Some isolates show weak β -hemolysis on blood agar, but the property is often lost after several subcultivations. Growth in broth media may or may not be granulated.

Eight biovars have been identified based on three biochemical reactions: indole formation, and urease and ornithine decarboxylase activities (Table BXII.γ.294) (Kilian, 1976; Oberhofer and Bach, 1979; Bruun et al., 1984a; Doern and Chapin, 1987). It is not clear if strains with the characteristics of biovar V do indeed belong in the species *H. parainfluenzae*. They are phenotypically similar to *H. segnis* and *H. paraphrophilus*. The majority of human isolates can be assigned to biovars I–III.

H. parainfluenzae is ubiquitous in the human oral cavity and pharynx and may be present in the normal vaginal flora (Sims, 1970; Tuyau and Sims, 1975; Kilian and Schiøtt, 1975; Liljemark et al., 1984). Organisms closely resembling *H. parainfluenzae* have been isolated from monkeys, swine, rabbits, and rats (Kilian and Frederiksen, 1981a). Tuyau and Sims (1974) reported that the species has neuraminidase activity. However, this cannot be confirmed with well-defined chromogenic substrates. Is of low pathogenicity, but is occasionally implicated in endocarditis and lower respiratory tract infections in humans.

The mol% G + C of the DNA is: 40–41 (T_m).

Type strain: ATCC 33392, NCTC 7857 (biovar I).

GenBank accession number (16S rRNA): M75081.

Additional Remarks: Reference strains include NCTC 10665 (biovar II), NCTC 11607 (biovar III).

12. *Haemophilus paraphrohaemolyticus* Zinnemann, Rogers, Frazer and Deveraj 1971, 143^{AL}

par.aph.ro.hae.mo.ly'ti.cus. M.L. adj. *paraphro* resembling (*Haemophilus*) *aphrophilus*; M.L. adj. *haemolyticus* blood dissolving; M.L. adj. *paraphrohaemolyticus* like *H. aphrophilus*, but hemolytic.

When grown in air with 10% extra CO₂, it forms short to medium-length rods, 0.75–2.5 × 0.4–0.5 μ m, with occasional short filaments; in air without added CO₂, it forms short to long, coarse rods with involution forms and twisted filaments. Requires increased CO₂ tension in incubation atmosphere for satisfactory growth and good β -hemolysis on blood agar. Apart from the CO₂ requirement, the biochemical, physiological, and ecological characteristics are similar to those of β -hemolytic isolates of *H. parainfluenzae*. Has been isolated from human sore throats, ulcers of the mouth, sputum, and urethral discharge of adult males, but pathogenic significance is unknown.

The mol% G + C of the DNA is: 40–41 (T_m).

Type strain: L1, NCTC 10670.

GenBank accession number (16S rRNA): M75076.

13. *Haemophilus paraphrophilus* Zinnemann, Rogers, Frazer and Boyce 1968, 418^{AL}

pa.ra.phro'phi.lus. Gr. prep. *para* alongside of, resembling; M.L. adj. *aphrophilus* specific epithet; M.L. adj. *paraphrophilus* resembling (*Haemophilus*) *aphrophilus*.

Short, regular rods with occasional filamentous forms. Involution forms occur under fully aerobic incubation. Growth characteristics are identical to those described for *H. aphrophilus*. In contrast to *H. aphrophilus*, *H. paraphrophilus* requires V factor and is not dependent on X factor on primary isolation. The two species have otherwise almost identical properties (Tables BXII.γ.294 and BXII.γ.295) and are genetically closely related (see discussion under Taxonomic comments).

Found as a member of the normal flora of the human oral cavity and pharynx (Kilian and Frederiksen, 1981a). May cause subacute endocarditis, paronychia, and brain abscesses and has been isolated from osteomyelitis of the jaw, an inflamed appendix, urine of children with congenital malformation of urogenital tract, and the vaginas of mature women.

The mol% G + C of the DNA is: 42 (T_m).

Type strain: Reece, ATCC 29241, NCTC 10557.

GenBank accession number (16S rRNA): M75042.

14. ***Haemophilus parasuis*** Biberstein and White 1969, 77^{AL}
pa.ra.su'is. Gr. prep. *para* alongside of, resembling; M.L. n. *suis* specific epithet; M.L. adj. *parasuis* (*Haemophilus*) *suis*-like.

Thin, pleomorphic rods of varying length. Growth on chocolate agar is very feeble after 48–72 h. The colonies are smooth, grayish, and translucent and reach a diameter of about 0.5 mm. A marked enhancement of growth on chocolate agar, with colonies attaining a diameter of 1–2 mm, is seen around a streak of a *Staphylococcus* (Biberstein et al., 1977).

Some isolates form capsules, but capsule production does not appear to be associated with virulence. Some, but not all, capsules precipitate with Cetavlon and appear to be acidic polysaccharides (Morozumi and Nicolet, 1986). Four

serovars (A–D) are designated, based on capsular polysaccharides (Bakos et al., 1952). Noncapsulated strains are antigenically heterogeneous. Based on heat-stable antigen extracts, a total of 15 serovars (1–15) have been described, and subsequent studies have suggested additional antigenic diversity (Kielstein and Rapp-Gabrielson, 1992; Blackall et al., 1997). Two variants (PAGE patterns I and II) have been identified based on differences in patterns of cellular proteins observed by polyacrylamide gel electrophoresis (Nicolet and Krawinkler, 1981; Morozumi and Nicolet, 1986).

Member of the normal flora of the upper respiratory tract of swine. Causes respiratory tract infections and polyserositis (Glässer's disease). Strains in culture collections labeled "*Haemophilus suis*" require only V factor and belong to *H. parasuis* according to the proposal of Biberstein and White (1969).

The mol% G + C of the DNA is: 41–42 (T_m).

Type strain: 1374 of Shope, NCTC 4557.

GenBank accession number (16S rRNA): M75065.

15. ***Haemophilus segnis*** Kilian 1976, 35^{AL}
seg'nis. L. adj. *segnis* slow, sluggish.

Pleomorphic rods, often showing a predominance of irregular, filamentous forms. Colonies on chocolate agar are smooth or granular, convex, grayish white, and opaque and reach a diameter of about 0.5 mm after incubation for 48 h. Growth in fermentation media is slow, and reactions are negative or weakly positive. Fermentation of sucrose is usually stronger than fermentation of glucose. Is a regular member of the human oral flora, particularly in dental plaque, and can be isolated from the pharynx. Has been isolated in pure culture from a pancreatic abscess.

The mol% G + C of the DNA is: 43–44 (T_m).

Type strain: HK 316, ATCC 33393, NCTC 10977.

GenBank accession number (16S rRNA): M75043.

Other Organisms

1. "*Haemophilus somnus*"* Bailie 1966, 64 (*Haemophilus*-like organism, Kennedy, Biberstein, Howarth, Frazier and Dungworth 1960, 403; *Actinobacillus actinoides*-like organism, Bailie, Anthony and Weide 1966, 165; *Actinobacillus* sp., Gossling 1966, 18; *Haemophilus somnifer* Miles, Anthony and Dennis 1972, 431.)

som' nus. L. masc. *somnus* sleep.

Gram-negative short rods, coccobacilli, or filaments. Most strains noncapsulated. Nonmotile. Not acid-fast. Colonial characteristics: 0.2–0.6 mm in diameter, raised, circular, smooth, entire colonies on beef blood agar after 24 h (Kennedy et al., 1960); 0.5–1.5 mm on PPLO–*Haemophilus* agar (Nicolet, 1971; Corboz and Nicolet, 1975); 1–2 mm after 3 d on brain–heart infusion calf blood–yeast extract plates (Garcia-Delgado et al., 1976). Prolonged incubation on blood agar produces colonies with a slightly granular appearance, papillate centers, and flattened peripheries. Most authors report no or weak hemolytic activity.

There is usually no growth of fresh isolates on infusion, tryptose, trypticase soy, serum, or hemoglobin agar under any atmospheric condition. Growth occurs on blood and

chocolate agars, and on serum–yeast extract PPLO–agar under 5–20% carbon dioxide. Poor or no growth in ambient air, or anaerobically on any solid medium (Kennedy et al., 1960; Garcia-Delgado et al., 1976). Satellitic growth around *S. aureus* occurs in an atmosphere of increased CO₂, but the growth factor supplied by *S. aureus* has not been identified (Kennedy et al., 1960). Occasional strains lack the special atmospheric and nutritional requirements described, and all cultures apparently adapt gradually to fully aerobic growth. No growth response to X and V factors. Biochemical activities are shown in Tables BXII.γ.294 and BXII.γ.295.

All isolates examined have antigens in common, regardless of geographic and anatomic origin (Shigidi and Hoerlein, 1970; Corboz and Nicolet, 1975; Garcia-Delgado et al., 1976; Corboz and Wild, 1981). Cross-reactions at low titer with a number of other mostly Gram-negative bacteria have been reported (Miller et al., 1975). Significant diversity has been detected among isolates based on biochemical activities, plasmid profiles, and DNA fingerprinting (Fussing and Wegener, 1993).

Antimicrobial susceptibility: Only rough, qualitative data exist on antimicrobial susceptibility; these are based on non-standardized disk diffusion tests. These suggest susceptibility to ampicillin, bacitracin, cephaloridine, chlorampheni-

*Editorial Note: "*Haemophilus somnus*" has been described as a member of the genus *Haemophilus*. By current standards it does not qualify for inclusion in the genus.

col, dihydrostreptomycin, erythromycin, novobiocin, penicillin G, polymyxin B, and tetracycline, and resistance to lincomycin, neomycin, and sulfonamides (Garcia-Delgado et al., 1976).

Pathogenicity: "*H. somnus*" causes septicemia and meningoencephalomyelitis in cattle (Kennedy et al., 1960) and is involved in respiratory (Brown et al., 1970) and genital infections (Waldhalm et al., 1974a), including abortions (Chladek, 1975), in that species. Experimental infections of mice, guinea pigs, rabbits, and sheep have been reported (Kennedy et al., 1960; Panciera et al., 1968; Miles et al., 1972).

Ecology: The reservoir of "*H. somnus*" appears to be the mucous membranes of the normal bovine respiratory and genital tracts (Corstvet et al., 1973; Waldhalm et al., 1974a).

The mol% G + C of the DNA is: 37.3 ± 0.2 (T_m).

GenBank accession number (16S rRNA): M75046.

2. "*Haemophilus agni*"* Kennedy, Frazier, Theilen and Biberstein 1958, 645.

agn' ni. L. gen. n. *agri* of the lamb.

Morphology: Gram-negative, nonmotile, non-acid-fast, pleomorphic rods or coccobacilli, $0.3\text{--}0.7 \times 0.5\text{--}2.0$ μm . Long, thread-like forms containing spherical and thickened fusiform bodies occur. Pleomorphism diminishes on passage. Presence of capsules has been suggested.

Colonial characteristics: Colonial characteristics in 5–10% CO₂: the colonies are convex and translucent and range from $<0.5\text{--}1.5$ mm. Colonies become flattened peripherally and acquire more sharply contoured edges but not significantly larger size on further incubation.

Nutrition and growth conditions: Very slight growth on blood agar in air. Colonial growth on blood agar under 5–10% CO₂, as described above. Satisfactory growth occurs on hemoglobin cystine agar and chocolate agar. No growth response to X and V factors. No satellitic growth with staphylococci. No growth on MacConkey, tryptose, 10% equine serum (with and without 1% yeast hydrolysate), mycoplasma agar, coagulated blood serum, or gelatin.

Metabolism: Metabolism and biochemical activities are shown in Tables BXII.γ.294 and BXII.γ.295.

Antigenic structure: This organism cross-reacts consistently with "*H. somnus*".

Antimicrobial susceptibility: Antimicrobial susceptibility has not been reported.

Pathogenicity: This organism is associated with septicemia, meningitis, polyarthritis myositis, pneumonia, and mastitis of sheep. Of experimental animals, only suckling mice are susceptible.

Ecology: The reservoir of "*H. agni*" has not been determined.

An organism occupying a position close to both "*H. agni*" and "*H. somnus*" has been encountered in a number of inflammatory diseases of sheep in Australia (Roberts 1956; Claxton and Everett 1966; Rahaley and White 1977) and New Zealand (Kater et al., 1962) and named "*Histophilus ovis*" (Roberts, 1956). De Ley et al., (1990) consider "*Histophilus ovis*" to be a senior synonym of "*H. agni*".

The mol% G + C of the DNA is: 36.8 ± 0.3 (T_m).

Genus IV. *Lonepinella* Osawa, Rainey, Fujisawa, Lang, Busse, Walsh and Stackebrandt 1996, 362^{VP} (Effective publication: Osawa, Rainey, Fujisawa, Lang, Busse, Walsh and Stackebrandt 1995b, 372)

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Lone.pin.el' la. M.L. fem. n. *Lonepinella* named after Lone Pine Koala Sanctuary, a private zoo in Australia.

Cells coccoidal, or short to extremely filamentous and rod-shaped. Facultatively anaerobic. Nonmotile. **Neither X nor V factor is required for growth.** No growth under 15°C or over 40°C; optimum, 37°C. Indole produced. Urea not hydrolyzed. **Produces gallic acid from tannic acid (tannase positive), and decarboxylates gallic acid to pyrogallol. Oxidase and catalase reactions are negative.** Acid, but not gas, is produced from D-glucose and some other carbohydrates. **Occur in the normal intestinal flora of the koala.** Analysis of 16 S rDNA places the genus into the family Pasteurellaceae, in the Gammaproteobacteria.

The mol% G + C of the DNA is: 37.5.

Type species: *Lonepinella koalarum* Osawa, Rainey, Fujisawa, Lang, Busse, Walsh and Stackebrandt 1996, 362 (Effective publication: Osawa, Rainey, Fujisawa, Lang, Busse, Walsh and Stackebrandt 1995b, 372.)

FURTHER DESCRIPTIVE INFORMATION

Based upon 16S rDNA analysis, isolates from the arboreal marsupial *Phascolarctos* (koala) form a separate branch within the

radiation of members of the genera *Pasteurella*, *Haemophilus*, and *Actinobacillus*. Cells of *Lonepinella* strains are pleomorphic, including coccoidal forms ($0.5\text{--}0.6 \times 1.0\text{--}1.2$ μm), short rods ($0.5 \times 3\text{--}5$ μm), and extremely filamentous rods ($0.5 \times 50\text{--}100$ μm), especially after prolonged incubation on plate media (Fig. BXII.γ.220). The organisms are nonmotile.

Polyamine composition of *Lonepinella* strains is characterized by the co-occurrence of putrescine and small amounts of diaminopropane, while cadaverine, tyramine, sym-norspermidine, spermidine, and spermine are lacking (Osawa et al., 1995b). This is in contrast to the other members of the family Pasteurellaceae, which have different polyamines (Busse et al., 1997). In addition, *Lonepinella* strains contain an unique, unidentified substance not reported in any other bacteria.

Lonepinella strains grow in nutrient broth supplemented with 0.1% yeast extract with uniform turbidity. On horse-blood supplemented Wilkins–Chalgren anaerobe agar or brain–heart infusion agar, smooth, opaque, more or less convex, gray to white colonies 1.0–3.0 mm in diameter without hemolysis are formed after 3 d at 37°C. No pigment is produced. Although *Lonepinella* strains are facultatively anaerobic, anaerobic conditions retard or suppress growth. Colony formation is also markedly suppressed on MacConkey agar. Optimum growth occurs at 37°C

*Editorial Note: "*Haemophilus agni*" has been described as a member of the genus *Haemophilus*. By current standards it does not qualify for inclusion in the genus.

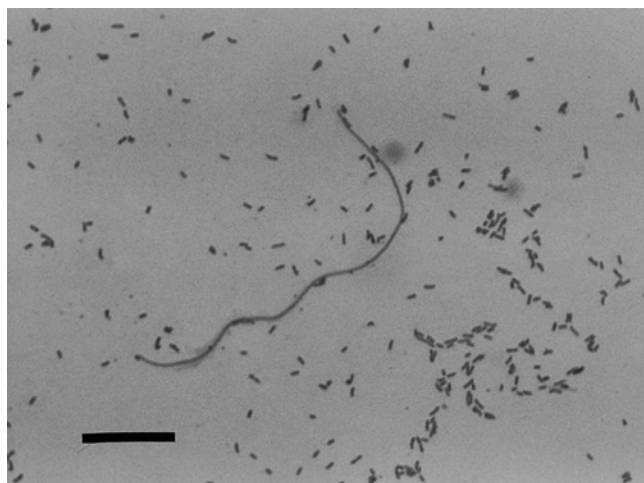


FIGURE BXII.γ.220. Gram-stained cells of *Lonepinella koalarum*, including an extremely long cell, from a colony formed on Wilkins–Chalgren anaerobe agar supplemented with 5% defibrinated horse blood after incubation at 37°C for 48 h. Bar = 20 μm.

when cultured in brain heart infusion. No growth occurs at 15°C or 40°C. Neither X nor V factor is required for growth.

After 3–4 d microaerophilic incubation at 37°C, *Lonepinella* strains form clear zones around colonies on a nonselective plate medium, the surface of which is treated with filter-sterilized tannic acid solution (Fig. BXII.γ.221). This indicates the ability to digest the tannin–protein complex formed on surface of the medium (Osawa, 1992).

Glucose, melibiose, rhamnose, sucrose, amygdalin, and L-arabinose are fermented without gas. D-mannitol, myo-inositol, and glucitol are not utilized. The Voges–Proskauer test and the β-galactosidase reaction are positive. The following enzyme activities are negative: oxidase, catalase, urease, arginine dihydrolase,

lysine decarboxylase, ornithine decarboxylase, tryptophan deaminase, and gelatinase. H₂S and indole not produced.

Lonepinella is unique by virtue of its “tannase” (tannin acylhydrolase) activity, which hydrolyzes gallotannin to yield gallic acid. It was long believed that tannase could be produced only by fungal strains belonging to the genera *Aspergillus* (Pourrat et al., 1987) and *Candida* (Aoki et al., 1976), but recent studies have revealed an almost identical enzymatic activity in a bacterial species, *Streptococcus bovis* biotype I (currently proposed as a new species, “*S. gallolyticus*”; Osawa et al., 1995a) (Osawa, 1990; Osawa et al., 1993c). *L. koalarum* is considered to be yet another bacterial species with tannase activity. This activity enables the organism to degrade the tannin–protein complex. Furthermore, the organism decarboxylates the released gallic acid to pyrogallol (Osawa et al., 1993c).

16S rDNA analysis revealed that *Lonepinella* does not belong to the family *Enterobacteriaceae*, but instead forms a phylogenetic cluster within the family *Pasteurellaceae*. Similarity values for strains of *Lonepinella* and members of the genera *Pasteurella*, *Haemophilus*, and *Actinobacillus* range from 93 to 94.5%. A similar range is found for the intra- and intergeneric relationships of the three genera. EMBL accession numbers are Y17189 (ACM 3666^T), Y17190 (strain ACM 3896), and Y17191 (strain ACM 3496). The mol% G + C content of the DNA, as determined by the thermal melting point method (Marmur and Doty, 1962), is 37.5.

Antibiotic sensitivity testing of two *L. koalarum* strains has indicated their resistance to vancomycin, but sensitivity to the following antimicrobials: kanamycin, colistin, chloramphenicol, erythromycin, oxytetracycline, penicillin, gentamicin, rifampin, and ampicillin (Osawa et al., 1995b).

Research on the ecology of *L. koalarum* has been stimulated by frequent isolation of the organism from fecal microflora of koala, an arboreal marsupial animal feeding exclusively on eucalyptus leaves (Osawa, 1992). The organism makes up more than 60% of the facultative anaerobic fecal microflora, and colonizes

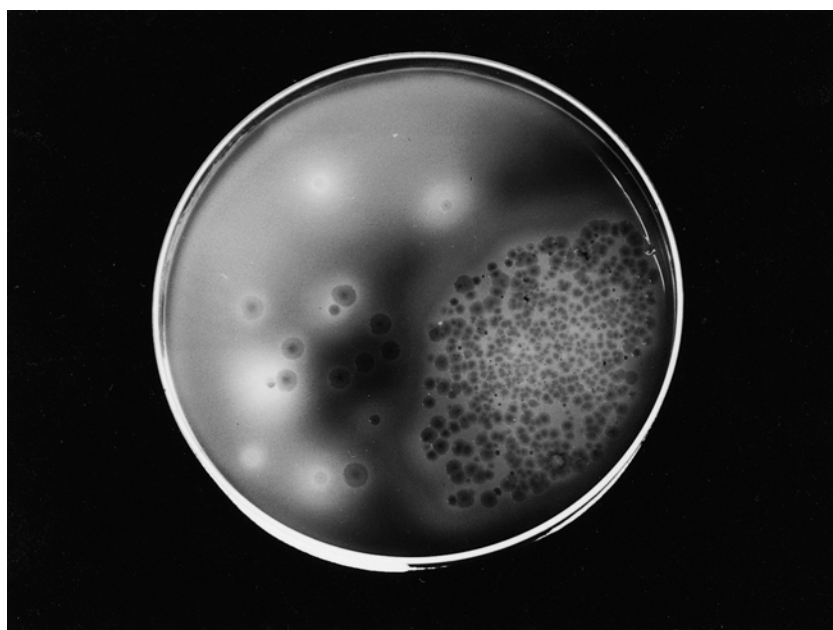


FIGURE BXII.γ.221. Growth of *Lonepinella koalarum*, with clear zones formed around colonies on vancomycin- and tannin-treated Wilkins–Chalgren anaerobe agar.

on the wall of the animal's well-developed caecum and proximal colon (Osawa et al., 1993a). Furthermore, a quantitative study on the microflora in "pap," a special maternal feces consumed by the juvenile around the time of dietary transition from maternal milk to eucalyptus leaves, revealed that "pap" has markedly higher viable counts of *L. koalarum* than the normal feces (Osawa et al., 1993b). The evidence strongly suggests that the organism plays a significant role in the digestion of a tannin-rich diet in the alimentary tract of the host animal.

ENRICHMENT AND ISOLATION PROCEDURES

The best way of isolating *Lonepinella* from fecal samples is to inoculate a plate containing a selective agar, vancomycin- and tannin-treated Wilkins-Chalgren anaerobe agar (Osawa, 1992), and incubate it microaerophilically (5–15% O₂) at 37°C for 3 d. After incubation, *L. koalarum* colonies are easily distinguished from others in mixed cultures, because they are generally about 1–2 mm in diameter with distinct clear zones around them (due to degradation of the tannin–protein complex formed on the surface of the plate). However, the plate allows growth of some enterobacterial species, including *Escherichia coli* and *Klebsiella*; thus, detection of the target colony might be difficult in fecal samples containing a large number of enterobacteria. The organism is readily cultured in conventional enrichment liquid media, such as nutrient broth and heart infusion broth, incubated at 37°C.

MAINTENANCE PROCEDURES

Plate and broth cultures of *Lonepinella* species usually survive no longer than 1 week without subcultivation. Survival is better at room temperature than at 4°C. The most satisfactory way of conserving cultures is by lyophilization, e.g., in skim milk. Wilkins-Chalgren anaerobe agar cultures remain viable for at least 1 year when frozen at –70°C in sealed glass ampoules.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Test for tannase activity Tannase activity may be directly detected by the clear-zone formation of colonies grown on the selective tannin-treated plate medium (Osawa, 1992). The specific clear zone is, however, sometimes difficult to recognize due to overgrowth of other bacteria in the samples. For a more definite and reliable detection of tannase activity, a visual reading method developed by Osawa and Walsh (1993) should be performed. This method is based on two phenomena: (i) the ability of tannase to hydrolyze methyl gallate, which has a molecular structure similar to gallotannin, to release free gallic acid and (ii) the green coloration of gallic acid after prolonged exposure to oxygen under alkaline conditions. Briefly, the isolate is cul-

tured on Wilkins-Chalgren anaerobe agar aerobically at 37°C for 3 d. After incubation, fresh culture on the plate is suspended in substrate medium containing NaH₃PO₄ (33 mM) and methyl gallate (20 mM) to prepare a dense suspension (equivalent to a no. 3 McFarland turbidity standard). The suspension is then incubated aerobically at 37°C for 24 h. After incubation, the suspension is alkalinized with an equal amount of saturated NaHCO₃ solution and left in the atmosphere at room temperature for at least 15 min. The development of any green coloration in the medium is an indication of tannase activity.

Test for gallate decarboxylase activity In addition to tannase activity, *Lonepinella* strains are capable of decarboxylating gallic acid to pyrogallol. The most rapid and convenient method to detect this activity was developed by Osawa and Walsh (1995). Briefly, the isolate is inoculated into Wilkins-Chalgren anaerobe broth or nutrient broth supplemented with gallic acid (10 mM), and incubated anaerobically at 37°C for 3 d. After incubation, the culture is centrifuged, and the supernatant added to an equal amount of saturated NaHCO₃ solution. Any orange coloration of the medium that develops is an indication of positive gallate decarboxylating activity.

DIFFERENTIATION OF THE GENUS *LONEPINELLA* FROM OTHER GENERA

See Table BXII.γ.297 for major phenotypic characteristics that differentiate *Lonepinella* from *Pasteurella*, *Haemophilus*, and *Actinobacillus*.

TAXONOMIC COMMENTS

Lonepinella strains were initially placed within the family *Enterobacteriaceae*, since they are facultatively anaerobic, oxidase-negative, Gram-negative bacilli and reduce nitrates to nitrites, thus phenotypically resembling *Enterobacter agglomerans* (Osawa, 1992). However, further comparison with representatives of the *Gammaproteobacteria* has shown them to group within the radiation of organisms presently assigned to the family *Pasteurellaceae* (Osawa et al., 1995b). The phylogenetic placement of these *Phascolarctos* isolates within members of the family *Pasteurellaceae* was unexpected, as none of the species of the family known at the time, *Pasteurella*, *Haemophilus*, or *Actinobacillus*, had been reported to produce tannase and gallate decarboxylase (Mannheim, 1984). The rationale for describing the isolates as a species of a new genus, *Lonepinella*, was based on these physiological properties, their origin from feces of the koala, and the phylogenetic distinctness from the type species of the *Pasteurellaceae* genera. As demonstrated by the studies of Dewhirst et al. (1992, 1993), the phylogenetic structure of the family does not reflect the taxo-

TABLE BXII.γ.297. Differential phenotypic characteristics of the genus *Lonepinella* and other genera^a

Characteristics	<i>Lonepinella</i>	<i>Actinobacillus</i>	<i>Haemophilus</i>	<i>Pasteurella</i>
V-factor requirement	–	–	+	–
Growth on MacConkey agar	–	+	–	D
Production of:				
Tannase	+	–	–	–
Gallate decarboxylase	+	–	–	–
Catalase	–	D	D	D
Indole	–	–	D	D
Ornithine decarboxylase	–	–	D	D
Acetoin (VP reaction)	+	D	–	–

^aFor symbols, see standard definitions.

nomic affiliation of species into genera to which they presently belong (Fig. BXII.γ.222). On the other hand, the branching points of deeply rooting lineages have no high statistical significance, as is demonstrated by low (<80%) bootstrap values (Fig. BXII.γ.222). Thus, the branching point of the species *Lonepinella*

koalarum changes with the number and selection of reference species. However, the distinctness of the *Lonepinella* lineage has not been changed by either the composition of strains included in the phylogenetic trees or by the algorithm used for their generation.

List of species of the genus *Lonepinella*

1. ***Lonepinella koalarum*** Osawa, Rainey, Fujisawa, Lang, Busse, Walsh and Stackebrandt 1996, 362^{VP} (Effective publication: Osawa, Rainey, Fujisawa, Lang, Busse, Walsh and Stackebrandt 1995b, 372.)

koala' rum. Eng. n. *koala* an arboreal marsupial, *Phascolarctus cinereus*, called "koala"; L. gen. pl. *koalarum* of koalas.

The description is as given for the genus. Cells coccoidal, $0.5\text{--}0.6 \times 1.0\text{--}1.2 \mu\text{m}$, or short to extremely filamentous and rod-shaped, $0.5\text{--}0.6 \mu\text{m} \times 3\text{--}100 \mu\text{m}$. D-glucose, sucrose, and amygdalin are utilized at 37°C within 48 h, with the production of acid, but not gas. D-mannitol, *myo*-inositol, and glucitol not utilized. Tannase and gallate decarboxylase positive.

Occurs in the normal intestinal flora of koalas; probably providing the animals with digests of tannin-rich eucalyptus leaves in their hind-gut.

The mol% G + C of the DNA is: 37.5 (T_m).

Type strain: ACM 3666, ATCC 700131, DSMZ 10053.

GenBank accession number (16S rRNA): Y17189.

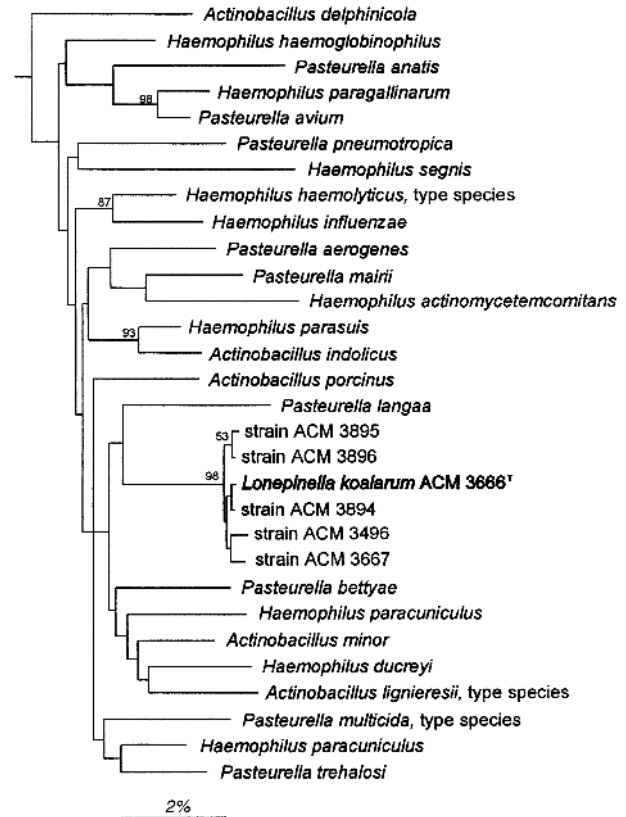


FIGURE BXII.γ.222. Dendrogram of 16S rDNA relationships of strains of the species *Lonepinella koalarum* and some selected members of the family *Pasteurellaceae* that represent individual lines of descent (Dewhirst et al., 1992, 1993; Osawa et al., 1995b). The type species of *Pasteurella*, *Haemophilus*, and *Actinobacillus* are indicated. The dendrogram was generated using the algorithm of Saitou and Nei (1987) on dissimilarity values, as calculated by Jukes and Cantor (1969). The sequences of some members of the *Enterobacteriaceae* were used as the root. Numbers refer to bootstrap values (300 trees). Only values higher than 80% are presented. Bar = 2% sequence divergence.

Genus V. *Mannheimia* Angen, Mutters, Caugant, Olsen and Bisgaard 1999a, 82^{VP}

REINIER MUTTERS, ØYSTEIN ANGEN AND MAGNE BISGAARD

Mann.hei' mi.a. L. n. fem. *Mannheimia* named in tribute to the German microbiologist Walter Mannheim.

Gram-negative, nonmotile rods or coccobacilli. Cell diameters of $\sim 0.5 \times 1.2 \mu\text{m}$ are similar to those of *Pasteurella* species. **Some species exhibit bipolar staining.** Endospores are not formed. Growth is mesophilic and facultatively anaerobic or microaerophilic. Neither X nor V factor is required for growth. Glucose is fermented without gas production. **Oxidase reaction is normally**

positive, but may be variable. **Alkaline phosphatase test is positive and nitrate is reduced.** Simmon's citrate, arginine dihydrolase, and Voges-Proskauer tests are negative, and there is no fermentation of D-adonitol or L-sorbose. All strains ferment D-mannitol. Urease reaction is negative. Trehalose and D-mannose are not fermented. Cultural characteristics are typical of those of the

family *Pasteurellaceae*. A zone of **β -hemolysis is commonly observed**, depending on media, type of blood, and pH.

The mol% G + C of the DNA is: 39.2–43.9.

Type species: *Mannheimia haemolytica* (Newsom and Cross 1932) Angen, Mutters, Caugant, Olsen and Bisgaard 1999a, 82 (*Pasteurella haemolytica* Newsom and Cross 1932, 715.)

FURTHER DESCRIPTIVE INFORMATION

Maximum likelihood analysis of 16S rDNA sequences of *Mannheimia* strains results in five distinct but affiliated clusters (Angen et al., 1999a). The mean sequence differences within each cluster range from 0.9 to 2.5%, while the mean sequence differences between the five clusters range from 2.8 to 4.4%. Bootstrap analysis identifies clusters II–V as monophyletic in more than 80% of the repeats. Strains of cluster I, however, cluster together in only 39% of the repeats.

Cluster I includes the type strain of the genus, *M. haemolytica* (NCTC 9380, GenBank accession no. AF060699). Differences in only two nucleotide positions separate this strain from strain U57066 which has the same sequence as strains of serovars 1, 5–9, 12–14, 16, and UG1 of the former *P. haemolytica* (Davies et al., 1996). These two sequences cluster together in 71% of the bootstrap trees. Cluster I also includes *M. glucosida*, the sequences of which show considerable heterogeneity and do not appear as a monophyletic group in the phylogenetic analysis (Angen et al., 1999a). Clusters II–IV contain *M. ruminalis*, *M. granulomatis*, and *M. varigena*, respectively. Cluster V currently includes only unnamed taxa. The genus *Mannheimia* is most closely related to the genus *Actinobacillus*, as evaluated by distances to the type species of *Pasteurella*, *Actinobacillus*, and *Haemophilus*.

The polyamine patterns of the five named species of *Mannheimia* can be obtained from the publication of Busse et al. (1997). Members of the genus are characterized by the presence of 1,3-diaminopropane as the predominant compound (77.6–95.2%), followed by spermine (2.2–12.8%), spermidine (1.7–11.6%), putrescine (0.8–3.8%), and cadaverine (0.2–1.1%). The variations observed are on the same order as those observed for the type strains of the three subspecies of *P. multocida*.

Surface colonies on bovine blood agar are generally circular, slightly raised, and regular, with an entire margin. The surface of the colonies is regular, smooth, shiny, and non-transparent, with a grayish tinge. A diameter of ~1–2 mm is observed after 24 h aerobic incubation at 37°C. A narrow zone of β -hemolysis normally surrounds the colonies, which have a butyrous consistency, but weakly hemolytic strains are often isolated. The colony size of strains isolated from pigs is generally smaller. These strains, however, often produce a very distinct zone of β -hemolysis.

Organisms in Gram-stained preparations from the above cultures are rather pleomorphic without a distinctive cellular morphology. All strains grow well in commonly used broths and in conventional media used for phenotypic characterization. Growth occurs between 25°C and 40°C, with an optimum temperature of 37°C.

As with most *Pasteurella sensu stricto* and actinobacilli, the recovery of *Mannheimia* spp. from clinical specimens does not normally present a problem unless the material is contaminated. When plates are overgrown by *Proteus* spp., the specimen should be replated on blood agar containing an increased amount of agar or a detergent. Media such as trypticase soy blood agar base (BBL), Difco–tryptose blood agar base, or Oxoid–Columbia blood agar base containing 5% bovine or ovine blood are generally preferred for initial isolation. The degree of hemolysis is dependent on the medium used and on the animal species from which the blood originates. A selective medium for isolation of *M. haemolytica* from specimens containing mixed populations of bacteria has been described by Morris (1958).

DNA–DNA hybridization data are available for 102 pairs of strains (Angen et al., 1999a). DNA binding of 91% is observed between the type strain of *M. haemolytica* and the most divergent strain within the ribotype cluster containing the type strain. The different biovars of *M. glucosida* cluster above 85% DNA binding by single linkage. The group of the *M. glucosida* biovars clusters with the type strain of *M. haemolytica* at a DNA binding level of 71%. Strains classified as *M. ruminalis* cluster at 88% DNA binding, while strains identified as *M. granulomatis* cluster at a DNA binding level of 86%. A maximum level of 85% DNA binding is observed for strains identified as *M. varigena*.

Several candidates for new species of *Mannheimia* have been identified by Angen et al. (1999a). However, several taxa allocated to this genus remain to be characterized by DNA–DNA hybridization. Further DNA–DNA hybridizations are necessary before conclusions as to the taxonomic interrelationships between these taxa can be drawn.

Information as to the use of ribotyping, random amplified polymorphic DNA analysis, and pulsed-field gel electrophoresis for discrimination of the former *P. haemolytica* complex may be obtained from the publication of Kodjo et al. (1999). For review of the structure, function, and properties of cellular and extracellular components of *P. haemolytica* see Adlam (1989), while the molecular approach for isolating genes from *M. haemolytica* A1 that code for secreted and soluble antigens has been reviewed by Lo (1995). Major outer membrane proteins of *M. haemolytica* serovars 1–15 have been investigated by Morton et al. (1996).

Since previous publications based upon a few phenotypic tests and subsequent serotyping might be questioned, reinvestigations are needed to obtain solid data on the ecology, antibiotic sensitivity, and pathogenicity of species of the genus *Mannheimia*.

MAINTENANCE PROCEDURES

Proper maintenance of cultures of members of the genus *Mannheimia* includes subculturing on complex media. Cultures remain viable for no longer than a week under refrigeration. Plates should be kept in plastic bags to avoid drying. For long term preservation, lyophilization or storage at –80°C in nutrient broth containing glycerol or a mixture of glucose and serum is recommended.

List of species of the genus *Mannheimia*

1. *Mannheimia haemolytica* (Newsom and Cross 1932) Angen, Mutters, Caugant, Olsen and Bisgaard 1999a, 82^{VP} (*Pasteurella haemolytica* Newsom and Cross 1932, 715.)
hae.mo.ly'ti.ca. Gr. n. *haima* blood; Gr. n. *lysis* solution; adj. from *lyticos*, Latinized as M.L. fem. adj. *lytica* dissolving, referring to the hemolysis seen on blood agar.

Corresponds to *P. haemolytica* biogroup 1 (Bisgaard and Mutters, 1986b). Cells are Gram-negative, nonmotile, small rods and coccobacilli. Colonies are regular, smooth, and grayish on blood agar and are 1–2 mm in diameter after 24 h of incubation. Most strains show a characteristic β -hemolysis on bovine blood agar. Biochemical characteristics

important for characterization are listed in Tables BXII.γ.298 and BXII.γ.299 and under the description of genus *Mannheimia*. D-sorbitol, D-xylose, maltose, and dextrin are fermented. No strains ferment L-arabinose or glucosides. Strains are negative in ornithine decarboxylase and NPG (β-glucosidase) and positive in ONPF (α-fucosidase). Isolated from pneumonia in cattle and sheep and from septicemia in lambs and mastitis in ewes. Some of the serotypes are probably part of the resident microflora of the upper respiratory tract of ruminants (Frank, 1989; Gilmour and Gilmour, 1989).

A total of 17 serovars have been described within *M. haemolytica*, among which serovars T3, 4, 10, and 15 now are classified as "*Pasteurella trehalosi*". Representatives of serovars A1, 2, 5, 6, 7, 8, 9, 12, 13, 14, and 16 are found in *Mannheimia haemolytica*. Serotype 11 has been reclassified as *Mannheimia glucosida*.

Virulence factors and pathogenesis of *M. haemolytica* in cattle have been reviewed by Confer et al. (1995). Most studies on pathogenesis and virulence factors have been done with strains classified as serotype A1, which should represent *M. haemolytica* (Bisgaard and Mutters, 1986b; Davies et al., 1996; Angen et al., 1999a). Several potential virulence factors have been described, including the polysaccharide capsule (Chae et al., 1990), LPS (Brogden, 1992), neuraminidase (Straus et al., 1993), transferrin binding proteins (Ogunnariwo and Schryvers, 1990), and sialoglycoprotease (Abdullah et al., 1991). Production of a leucotoxin has been observed in all serotypes, as well as in several untypable strains (Chang et al., 1987; Burrows et al., 1993). It has been shown to be specific for bovine macrophages, neutrophils, lymphocytes, and platelets (Clinkenbeard and Upton, 1991), and several studies have indicated that it plays a major role in the pathogenesis of pneumonic pasteurellosis (Henricks et al., 1992; Hughes et al., 1994; Tatum et al., 1998).

The type strain was isolated in 1956 from sheep in the UK. The type strain represents serovar A2, is ONPX (β-xylosidase) and ONPG negative and shows a late *meso*-inositol fermentation.

The mol% G + C of the DNA is: 43.6 (T_m).

Type strain: J.A.Watt 1266A&B, CCUG 12392, NCTC 9380.

GenBank accession number (16S rRNA): AF060699.

2. ***Mannheimia glucosida*** Angen, Mutters, Caugant, Olsen and Bisgaard 1999a, 83^{VP}

glu.co'si.da. Gr. adj. *glykos* sweet; L. adj.-forming suff. *-id (-us)*; fem. *-ida*; M.L. adj. *glucosida* pertaining to glucosides, which are fermented by almost all biovars.

Corresponds to Frederiksen's third taxon of *M. haemolytica* (Frederiksen, 1973), classified as *P. haemolytica* biogroups 3 and 5 by Bisgaard and Mutters (1986b) and reclassified as *M. haemolytica* biogroups 3A–H by Angen et al. (1997). The species also includes NPG and *meso*-inositol positive strains of *M. haemolytica* biogroup 9 (Bisgaard and Mutters, 1986b). *M. glucosida* can be divided into 9 biovars, A–I (Angen et al., 1999a). Cells are Gram-negative, non-motile, small rods. Colonies are regular, smooth, and grayish on blood agar and are 1–2 mm in diameter after 24 h of incubation. All strains are hemolytic on bovine blood agar. All strains ferment D-sorbitol, D-xylose, maltose, and dextrin. For the ONPG and NPG tests, all biovars are positive. The biovars are separated by differences in ornithine decarboxylase, ONPX, ONPF, and fermentation of L-arabinose and glucosides. Further biochemical characteristics are listed in Tables BXII.γ.298 and BXII.γ.299 and under the description of genus *Mannheimia*. Typeable strains within *M. glucosida* all belong to serovar A11 (Angen et al., 1999b). Most strains investigated have been isolated from sheep, although isolates of serotype 11 have been reported from cattle (Quirie et al., 1986). This group of bacteria is normally not associated with disease and probably represents part of the resident microflora in the upper respiratory tract (Biberstein and Gillis, 1962). Murray et al. (1992) have shown that the iron binding proteins in serotype A11 differed from other serotypes. Sialoglycoprotease and neuraminidase activity have not been observed within serotype A11 (Lee et al., 1994). Serotype A11 produces a leucotoxin that is serologically and functionally similar to leucotoxin from *M. haemolytica* (A1), but immunoblots have shown that strains of serotype A11 seem to produce more variants of these toxins than do the other serotypes (Burrows et al., 1993).

The type strain was originally received as B558 from Moredun Research Institute. It was isolated from ovine lung in the UK, and belongs to biovar B and serotype 11 (Angen et al., 1997).

The mol% G + C of the DNA is: 41.6 (T_m).

TABLE BXII.γ.298. Phenotypic characters separating existing species of *Mannheimia*^a

Characteristics	1. <i>M. haemolytica</i>	2. <i>M. glucosida</i>	3. <i>M. granulomatis</i>	4. <i>M. ruminalis</i>	5. <i>M. varigena</i>
β-Hemolysis (bovine blood)	+	+	–	–	+
Ornithine decarboxylase	–	d	–	–	d
L(+)Arabinose	–	d	–	–	+
D(+)Xylose	+	+	d	–	+
<i>meso</i> -Inositol	d	(+)	d	–	d
D(–)Sorbitol	+	+	+	d	–
L(+)Rhamnose	–	–	–	–	d ^b
β-Glucosidase (NPG)	–	+	+	–	– ^b
Glycosides ^c	–	d	d	–	– ^b
α-Fucosidase (ONPF)	+	+	–	d	– ^b
β-Xylosidase (ONPX)	–	d	d	–	–

^aSymbols: +, 90% or more of the strains positive within 1–2 days; (+), 90% or more of the strains positive within 3–14 days; –, 10% or less of the strains are positive within 14 days; d, 11–89% of the strains are positive; w, weak positive.

^bStrains of Bisgaard Taxon 36 are positive.

^cGlycosides: cellobiose, esculin, amygdalin, arbutin, gentiobiose, and salicin.

TABLE BXII.γ.299. Additional phenotypic characters of existing species of *Mannheimia*^{a,b}

Characteristics	1. <i>M. haemolytica</i>	2. <i>M. glucosida</i>	3. <i>M. granulomatis</i>	4. <i>M. ruminantis</i>	5. <i>M. varigena</i>
Gram stain	—	—	—	—	—
Motility at 22°C and 37°C	—	—	—	—	—
Catalase	+	+	d	+	+
Oxidase	+	+	d	d	+
Glucose (Hugh and Leifson)	F	F	F	F	F
Symbiotic growth	—	—	—	—	—
Porphyrim test	+	+	+	+	+
Citrate (Simmons)	—	—	—	—	—
Mucate, acid	—	—	—	—	—
Malonate, base	—	—	—	—	—
H ₂ S/TSI	—	—	—	—	—
KCN, growth	—	—	—	—	—
Methyl-red, 37°C	—	—	—	—	—
Voges-Proskauer, 37°C	—	—	—	—	—
Nitrate, reduction	+	+	+	+	+
Nitrate, gas	—	—	—	—	—
Urease	—	—	—	—	—
Alanine aminopeptidase	+	+	+	+	+
Arginine dehydrolase	—	—	—	—	—
Lysine decarboxylase	—	—	—	—	—
Phenylalanine deaminase	—	—	—	—	—
Indole	—	—	—	—	—
Phosphatase	+	+	+	+	+
Gelatinase	—	—	—	—	—
Tween 20	—	—	—	—	—
Tween 80	—	—	—	—	—
Growth on MacConkey agar	d	d	—	d	—
Pigment	—	—	—	—	—
Glycerol	d	(+)	(+)	d	(+)
<i>meso</i> -Erythritol	—	—	—	—	—
Adonitol	—	—	—	—	—
D(+)-Arabitol	—	—	—	—	—
Xylitol	—	—	—	—	—
D(–)-Arabinose	d	w	—	d	d
D(–)-Ribose	+	+	+	+	+
L(–)-Xylose	—	—	—	—	—
Dulcitol	—	—	—	—	—
D(–)-Mannitol	+	+	+	+	+
D(–)-Fructose	+	+	+	+	+
D(+)-Fucose	—	—	—	—	—
L(–)-Fucose	+	+/(+)	d	d	(+)
D(+)-Galactose	+	+/(+)	+/(+)	+	+/(+)
D(+)-Glucose, acid	+	+	+	+	+
D(+)-Glucose, gas	—	—	—	—	—
D(+)-Mannose	—	—	—	—	—
L(–)-Sorbitol	—	—	—	—	—
Lactose	+	+/(+)	d	(+)	d
<i>o</i> -Nitrophenyl-β-D-galactopyranoside (ONPG)	+	+	d	+	+
Maltose	+	+	+/(+)	d	d
D(+)-Melibiose	—	—	—	—	— ^c
Sucrose	+	+	+	+	+
Trehalose	—	—	—	—	—
D(+)-Melizitose	—	—	—	—	—
Raffinose	(+)	d	d	d	d
Dextrin	+	+/(+)	d	d	d
D(+)-Glycogen	—	—	—	—	—
Inulin	—	—	—	—	—

(continued)

Type strain: P925, CCUG 38457.*GenBank accession number (16S rRNA)*: AF053889.

3. *Mannheimia granulomatis* (Ribeiro, Carter, Frederiksen and Riet-Correa 1990) Angen, Mutters, Cagant, Olsen and Bisgaard 1999a, 82^{VP} (*Pasteurella granulomatis* Ribeiro, Carter, Frederiksen, and Riet-Correa 1990, 105.) *gran.nu.lo' ma.tis*. L. dim. n. *granulum* a small grain; Gr. suff. -oma a swelling or tumor; M.L. n. *granuloma* a granuloma; M.L. n. *granulomatis* of a granuloma.

Cells are Gram negative, nonmotile, small rods and coccobacilli. Colonies are regular, smooth, and grayish on blood agar and are 1–2 mm in diameter after 24 h of incubation. No strains were hemolytic on bovine blood agar, but they have been reported as hemolytic using ovine blood (Ribeiro et al., 1989). All strains are negative for ornithine decarboxylase and ONPG and positive for NPG and fermentation of D-sorbitol. Strains are variable in fermentation of D-xylose, maltose, dextrin, glucosides, and *meso*-inositol.

TABLE BXII.γ.299. (cont.)

Characteristics	1. <i>M. haemolytica</i>	2. <i>M. glucosida</i>	3. <i>M. granulomatis</i>	4. <i>M. ruminalis</i>	5. <i>M. varigena</i>
D(+) Turanose	—	—	—	—	—
β-N-CH ₃ -Glucosamide	—	—	—	—	—
α-Galactosidase	—	—	—	—	— ^c
α-Glucosidase (PNPG)	—	—	—	—	—
β-Glucuronidase (PGUA)	—	—	—	d	—
α-Mannosidase	—	—	—	—	—

^aSymbols: +, 90% or more of the strains positive within 1–2 days; (+), 90% or more of the strains positive within 3–14 days; —, 10% or less of the strains are positive within 14 days; d, 11–89% of the strains are positive; w, weak positive; +/(+), positive or delayed positive reaction later than 48 h; F, fermentation.

^bIncubation temperature 37°C.

^cstrains of Bisgaard Taxon 36 are positive.

Further biochemical characteristics are listed in Tables BXII.γ.298 and BXII.γ.299 and under the description of genus *Mannheimia*. The species includes the following taxa: *Pasteurella granulomatis*, isolated from bovine panniculitis (Ribeiro et al., 1989; 1990; Riet-Correa et al., 1992); Bisgaard Taxon 20 biovars 1 and 2, isolated from bronchopneumonia and conjunctivitis in leprine species and deer (Devriese et al., 1991); and *M. haemolytica* biogroup 3J, isolated from deer and the oral cavity of cattle (Angen et al., 1997). No virulence factors have yet been described.

The type strain was isolated from bovine panniculitis in Brazil and was previously classified as *Pasteurella granulomatis* (Ribeiro et al., 1989). The type strain is ONPG positive and ferments maltose, dextrin, glucosides, and *meso*-inositol, but not D-xylose.

The mol% G + C of the DNA is: 39.2 (*T_m*).

Type strain: 26, ATCC 49244.

GenBank accession number (16S rRNA): AF053902.

4. ***Mannheimia ruminalis*** Angen, Mutters, Caugant, Olsen and Bisgaard 1999a, 83.^{VP}

ru.mi.na'lis. L. n. *rumen* stomach, especially first stomach of ruminants; L. adj.-forming suff. *-alis*; M.L. adj. *ruminalis* referring to ruminants, from which the bacteria have been isolated.

Cells are Gram-negative, nonmotile, nonhemolytic, small rods or coccobacilli. Colonies are regular, smooth, and grayish on blood agar and are 1–2 mm in diameter after 24 h of incubation. All strains are ornithine decarboxylase, NPG, ONPG, and ONPX negative and do not ferment L-arabinose, *meso*-inositol, or glucosides (Table BXII.γ.298). Further biochemical characteristics are given in the description of genus *Mannheimia*. *M. ruminalis* consists of two distinct phenotypic groups. Biovar 1 consists of strains that do not ferment maltose or dextrin, show variable reactions for fermentation of D-sorbitol and D-xylose, and correspond to Bisgaard Taxon 18 biovars 1 and 3 (Bisgaard et al., 1986). Biovar 2 strains ferment maltose and dextrin and correspond to *M. haemolytica* biogroup 8D (Angen et al., 1997). All strains have been isolated from the rumen of sheep or cattle and are not associated with disease. Virulence factors have not been described. The strains have previously been called *Actinobacillus lignieresii* (Phillips, 1961, 1964, 1966).

The type strain was originally isolated as strain S1/15/6 from a sheep rumen in the UK by J.E. Phillips, and previously was classified as Bisgaard Taxon 18 biovar 1 (Bisgaard et al., 1986). The type strain belongs to biovar 1 and does not ferment D-sorbitol and D-xylose.

The mol% G + C of the DNA is: not known.

Type strain: HPA 92, CCUG 38470.

Additional Remarks: AF053900.

5. ***Mannheimia varigena*** Angen, Mutters, Caugant, Olsen and Bisgaard 1999a, 83.^{VP}

va.ri'gena. L. fem. adj. *varia* varied; L. adj.-forming element *-gena* originating from; L. adj. *varigena*, of different origin, referring to the fact that strains of this species have been isolated both from ruminants and pigs.

Cells are Gram-negative, nonmotile, small rods or coccobacilli. Colonies are regular, smooth, and grayish on blood agar and are 1–2 mm in diameter after 24 h of incubation. Most strains are hemolytic on bovine blood agar. All strains ferment L-arabinose and D-xylose, but not D-sorbitol. Most strains do not ferment glucosides; those fermenting glucosides also ferment D-melibiose. Most strains are ornithine decarboxylase positive (biovar 1) and correspond to *M. haemolytica* biogroup 6, Bisgaard Taxon 36, and Bisgaard Taxon 15 biovar 1. Ornithine-decarboxylase-negative strains (biovar 2) were isolated from pigs and correspond to Bisgaard Taxon 15 biovar 2. Strains of *M. varigena* have been isolated from pneumonia, mastitis, and septicemia in cattle, as well as from the oral cavity, rumen, and intestines (Angen et al., 1997). Isolates from pigs have been associated with septicemia, enteritis, and pneumonia and have also been isolated from the upper respiratory tract (Bisgaard, 1984; McLaughlin et al., 1991). Some of these strains have been shown to produce a leucotoxin toxic against leucocytes from both cattle and pigs (Chang et al., 1993).

The type strain was isolated from a bovine with pneumonia in West Germany, and was previously classified as *M. haemolytica* biogroup 6 (Bisgaard and Mutters, 1986b). This strain is indole negative, ONPG and ONPF positive, and NPG negative and ferments maltose and *meso*-inositol, but not dextrin (Tables BXII.γ.298 and BXII.γ.299).

The mol% G + C of the DNA is: 41.7 (*T_m*).

Type strain: 177, CCUG 38462.

GenBank accession number (16S rRNA): AF053893.

Genus VI. **Phocoenobacter** Foster, Ross, Malnick, Willems, Hutson, Reid and Collins 2000, 138^{VP}

GEOFFREY FOSTER AND MATTHEW D. COLLINS

Pho.coe.no.bac' ter. M.L. n. *phocoena* derived from Gr. n. *phokaina* porpoise; M.L. masc. n. *bacter* rod; M.L. masc. n. *phocoenobacter* a rod from a porpoise.

Cells spherical, ovoid or rod shaped, 0.2–0.5 × 0.5–1.0 μm. Occasional longer forms occur. **Some cells demonstrate bipolar staining.** Gram negative. **Nonmotile. Facultative anaerobes.** Optimum temperature for growth, 37°C. Do not require X or V factors for growth. No growth occurs on MacConkey agar. **Catalase negative, oxidase positive. Nitrates are reduced to nitrites. Voges–Proskauer positive.** Arginine dihydrolase, ornithine decarboxylase, lysine decarboxylase, urease and indole are not produced. Acid produced from glucose. Isolated from a porpoise.

The mol% G + C of the DNA is: 41.5.

Type species: Phocoenobacter uteri Foster, Ross, Malnick, Willems, Hutson, Reid and Collins 2000, 139.

FURTHER DESCRIPTIVE INFORMATION

Cells are pleomorphic after isolation, with coccal, coccobacillary, and bacillary forms all present. After 48 h incubation on sheep blood agar, colonies are 0.5 mm in diameter, circular, entire, low convex, smooth, and gray. Colonies are not sticky on primary isolation. Nonhemolytic on sheep blood agar initially, although alpha hemolysis is evident in older cultures. Growth on Columbia agar without serum or blood is absent but a scanty growth may be obtained on tryptic soy agar without additives. Grows at 37°C and 22°C, but not at 42°C.

Because the genus contains only one species, all the characteristics given for the genus also describe the species *Phocoenobacter uteri*.

ENRICHMENT AND ISOLATION PROCEDURES

Phocoenobacter strains can be isolated on blood-containing media. Growth is poor or absent on media without blood. Increased CO₂ does not enhance the number or size of the colonies.

MAINTENANCE PROCEDURES

Surface cultures survive up to a week on Columbia sheep blood agar and slightly longer on blood-enriched nutrient agar slopes. Both lyophilization and storage on cryobeads at –80°C are effective for long-term maintenance.

DIFFERENTIATION OF THE GENUS *PHOCOENOBACTER* FROM OTHER GENERA

Phocoenobacter can be differentiated from all other members of the *Pasteurellaceae* by the following tests. It can be separated from *Pasteurella* by its negative catalase, indole, and D-mannose reactions; from *Haemophilus* by its positive Voges–Proskauer reaction and lack of X or V requirement; from *Actinobacillus* by not producing urease; from *Mannheimia* by its negative catalase and mannitol fermentation reactions; and from *Lonepinella* by its positive oxidase reaction.

TAXONOMIC COMMENTS

The genus *Phocoenobacter* was described to accommodate a Gram-negative pleomorphic rod originating from the uterus of a porpoise. The genus is phylogenetically related to the *Pasteurellaceae*.

The family *Pasteurellaceae* is phylogenetically very complex and there is extensive intermixing of species of some of its constituent genera. The genus *Phocoenobacter* forms a distinct subline within the *Pasteurellaceae* and does not display a particularly close affinity with any described species of the family. The nearest named relative of *Phocoenobacter* in terms of 16S rDNA sequence similarity is *Haemophilus ducreyi*, but treeing analysis shows this association is statistically not significant.

List of species of the genus *Phocoenobacter*

1. ***Phocoenobacter uteri*** Foster, Ross, Malnick, Willems, Hutson, Reid and Collins 2000, 139^{VP}
u' te.ri. L. masc. n. *uteris* uterus.

The characteristics are as described for the genus and as listed in Table BXII.γ.300. Acid is produced from glucose but not from adonitol, dulcitol, galactose, inositol, inulin, lactose, maltose, mannitol, mannose, melibiose, raffinose, rhamnose, salicin, sorbitol, sucrose, trehalose, or xylose. β-galactosidase positive. Alkaline phosphatase, esterase lipase, acid phosphatase, and naphthol- AS-BI-phosphohydrolase are positive. Leucine arylamidase is weakly positive.

The mol% G + C of the DNA is: 41.5 (*T_m*).

Type strain: NCTC 12872, M1063U/93.

GenBank accession number (16S rRNA): X89379.

TABLE BXII.γ.300. Characteristics of *Phocoenobacter uteri*^a

Characteristics	Reaction or result
<i>Production of:</i>	
Catalase	–
Oxidase	+
Indole	–
Arginine dihydrolase	–
Lysine decarboxylase	–
Ornithine decarboxylase	–
Acetoin	+
β-Galactosidase (ONPG)	+
<i>Fermentation of:</i>	
Glucose	+
Adonitol, dulcitol, galactose, inositol, inulin, lactose, maltose, mannitol, mannose, melibiose, raffinose, rhamnose, salicin, sorbitol, sucrose, trehalose, xylose	–
Growth on MacConkey agar	–
X or V factor requirement	–

^aSymbols: +, positive; –, negative.

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